

Esperimenti

Anomalous Transport in Soaring Flights

Matteo Di Sante

Working draft — September 22, 2026

Scope

This volume collects the experimental and modelling chapters following Chapter 2 of the thesis. The introduction, dataset description, and Appendices 2.A and 2.B are in the companion thesis volume, *Anomalous Transport in Soaring Flights* ([main.pdf](#)). Cross-references to that material point to the thesis PDF. The original chapter numbering, 3–7, is retained.

Material incorporated into the thesis. Passages printed in **red** have been incorporated, with revised estimators and newly measured results, into Chapter 3 of the thesis. Their original wording and numerical results are retained here as a record of the exploratory analysis; the current definitions and estimates are in the thesis. Unmarked material, including the anisotropy and PCA analyses, has not been transferred.

Historical calculations not regenerated. The fixed-origin scaling extension of the exploratory analysis has not been regenerated. Its affected passages in Chapters 4 and 7 remain marked as pending. The current fixed-segment analysis is complete in thesis Chapter 3 and uses a separate numerical report. Earlier available measurements remain here.

Contents

3	Global observables	1
3.1	Mean-square displacement and growth exponents	1
3.1.1	Effective and local growth exponents	4
3.2	Directional structure, terrain and equipment	5
3.2.1	East–north balance in the retained archive	5
3.2.2	Regional principal axes	5
3.2.3	Terrain, equipment and wind	15
3.2.4	Regional changes of velocity and direction	20
3.2.5	Third differences and the distribution of changes	24
4	Preliminary global transport	28
4.1	Drift removal and route closure	28
4.1.1	Removing a constant velocity	28
4.1.2	Open and closed routes	31
4.2	Quantiles and self-similarity	32
4.2.1	The scaling prediction	32
4.2.2	A fixed population across lags	33
4.2.3	Relative spread	35
4.3	Vector scaling and dependence	37
4.3.1	Testing east, north and radial scaling	37
4.3.2	Moment spectrum	40
4.3.3	Distribution collapse	41
4.3.4	Signed joint distributions	46
4.3.5	Testing approximate scaling beyond one-lag magnitudes	47
4.4	Time-averaged transport by region and initial altitude	51
4.4.1	Flight weighting and the fixed cohort	51
4.4.2	Regional contrasts in the fixed cohort	52
4.4.3	Initial-altitude bands and their regional composition	52
4.4.4	Paraglider scaling by altitude and circuit type	55
4.5	Flight duration and equipment composition	58
4.6	Constraints for the phase analysis	62
5	Flight phases	63
5.1	What the three phases mean	63
5.2	From a Markov chain to hidden-state inference	63
5.2.1	Constructing the probability law	64
5.2.2	Likelihood and uncertainty about the states	65
5.2.3	Decoding a whole sequence	65
5.2.4	Learning from unlabelled observations	66
5.3	From the cleaned flight to observations	66
5.4	A probabilistic segmentation, and its assumptions	68
5.4.1	The fitted hidden Markov model	68
5.4.2	Numerical estimation in the fitted model	69
5.4.3	Why the current decoder differs from the original fit	69
5.5	What has been checked	71
5.5.1	Archive support and reproducible examples	71

5.5.2	The manual check that is still needed	72
5.6	What the segmentation will let us measure	76
5.6.1	Keeping model assumptions separate from measured physics	77
5.7	Chapter summary	77
6	The Vilpellet flight-phase segmentation	78
6.1	Input, fitted parameters and admissible trajectories	78
6.2	From geometry to three binary observations	79
6.3	A categorical HMM and its decoded labels	80
6.4	Retained runs, saved products and an example	81
6.5	How to compare the two methods and where to study the code	83
7	Comparison with stochastic models	84
7.1	The moment spectrum and the Lévy-walk benchmark	84
7.2	Centred kurtosis and heterogeneity	85
7.3	Velocity memory across resolutions	87
7.4	Model predictions and remaining tests	89
4.A	Squared-displacement distributions	93

Chapter 3

Global observables

Circling, gliding and searching for lift combine into the displacement of a whole flight. We measure its growth, state how effective growth exponents are fitted, and ask how directional structure varies with take-off region and equipment. The analysis precedes the phase labels introduced in Chapter 5; Chapter 7 then compares these measurements with explicit stochastic-process predictions.

The archive contains 156,406 paraglider and 6,094 hang-glider flights in the retained transport snapshot. A complete scan of the cleaned archive supplies measurements spanning 10s to 10,000s. All retained segments meeting the common-grid cadence and lag-support requirements enter; there is no cap on the number of flights.

Table 3.1. Population support in this and the following chapter. The common-grid analysis includes every eligible flight. The fixed long-flight population supplies an additional control with identical flights and origins across lags (Sec. 4.2.2).

Population	Paragliders	Hang gliders
Retained cleaned archive	156,406	6,094
Eligible on the 10-s grid	155,085	6,060
Fixed long-flight control	14,360	563

This chapter defines displacement growth for the complete archive, states how effective exponents are fitted, and measures the directional structure of the retained flights across regions and equipment groups. Chapter 4 then removes constant drift and examines distributional structure, using the same populations.

Reworked in thesis Section 3.1. Original analysis retained below.

3.1 Mean-square displacement and growth exponents

Let $\mathbf{r}_f(t) = (E_f(t), N_f(t))$ be the horizontal position of flight f . Its retained segments form the set $\mathcal{S}_f$, with fs denoting segment s of flight f .

A displacement is

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f(t, \tau) = |\mathbf{X}_f(t, \tau)|. \quad (3.1)$$

For segment fs , of duration T_{fs} , the time-averaged MSD is

$$\overline{\delta_{fs}^2}(\tau) = \frac{1}{T_{fs} - \tau} \int_0^{T_{fs} - \tau} R_f(t, \tau)^2 dt, \quad 0 < \tau < T_{fs}, \quad (3.2)$$

with t measured from the start of that segment. Only windows lying entirely within a retained segment are admitted, so an increment never crosses a data gap. Let $n_{fs}(\tau)$ count the admissible windows of segment fs and $n_f(\tau) = \sum_{s \in \mathcal{S}_f} n_{fs}(\tau)$ their total over the flight. Combining a flight's segment sums with those counts gives the flight-level mean

$$\overline{\delta_f^2}(\tau) = \frac{\sum_{s \in \mathcal{S}_f} n_{fs}(\tau) \overline{\delta_{fs}^2}(\tau)}{n_f(\tau)}, \quad (3.3)$$

a window-count-weighted average of the segment TA-MSDs contributing to that flight. Averaging further over the population requires a choice of sampling unit:

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_{\text{fl}}(\tau)} \sum_f \overline{\delta_f^2}(\tau), \quad M_2^{\text{window}}(\tau) = \frac{\sum_f n_f(\tau) \overline{\delta_f^2}(\tau)}{\sum_f n_f(\tau)}, \quad (3.4)$$

where $N_{\text{fl}}(\tau)$ is the number of flights with at least one admissible window. The first gives each contributing flight equal weight, whereas the second gives each admissible window equal weight and consequently weights longer flights more strongly. Substituting Eq. (3.3) into the second cancels the flight denominators,

$$M_2^{\text{window}}(\tau) = \frac{\sum_f \sum_{s \in \mathcal{S}_f} n_{fs}(\tau) \overline{\delta_{fs}^2}(\tau)}{\sum_f \sum_{s \in \mathcal{S}_f} n_{fs}(\tau)}, \quad (3.5)$$

so the window average does not depend on how segments are grouped into flights. Both averages still combine flights made under different meteorological and geographical conditions.

Figure 3.1 uses a third convention, equal weight per segment:

$$M_2^{\text{segment}}(\tau) = \frac{1}{N_{\text{seg}}(\tau)} \sum_f \sum_{s \in \mathcal{S}_f} \overline{\delta_{fs}^2}(\tau), \quad (3.6)$$

with $N_{\text{seg}}(\tau)$ the number of segments answering τ . Only segments with at least eight fixes and support at τ enter. A split flight can contribute several segments, and the archive holds 1.24 paraglider and 1.26 hang-glider segments per contributing flight.

In discrete time, a segment fs of N_{fs} fixes $\mathbf{r}_0, \dots, \mathbf{r}_{N_{fs}-1}$ at interval Δt_{fs} contributes

$$\widehat{\delta_{fs}^2}(k\Delta t_{fs}) = \frac{1}{N_{fs} - k} \sum_{j=0}^{N_{fs}-k-1} |\mathbf{r}_{j+k} - \mathbf{r}_j|^2, \quad 1 \leq k < N_{fs}. \quad (3.7)$$

The common-grid diagnostics below use a 10 s grid, excluding segments with slower native cadence or fewer than two grid points. Starting times are $0, \tau, 2\tau, \dots$ within each segment, giving non-overlapping first differences. Segment sums and origin counts are combined within flights. The duration analysis in Sec. 4.5 instead uses the native-grid segment TA-MSDs, with its own support and weighting stated there.

The common-grid second moments then receive equal flight weights; baseline quantiles and higher moments pool windows.

All three averages share one property that limits how far they can be read: the population contributing to τ is the set of segments long enough to support it, so it shrinks as τ grows. Panel (d) of Fig. 3.1 shows that loss directly. A slope measured along such a curve combines the growth of displacement with a change of the flights being averaged, and the two cannot be separated after the fact. The curves of this section are therefore archive descriptions. Section 4.2.2 fixes flights, weights and origins once, and every exponent and shape comparison in this chapter is taken from that fixed cohort.

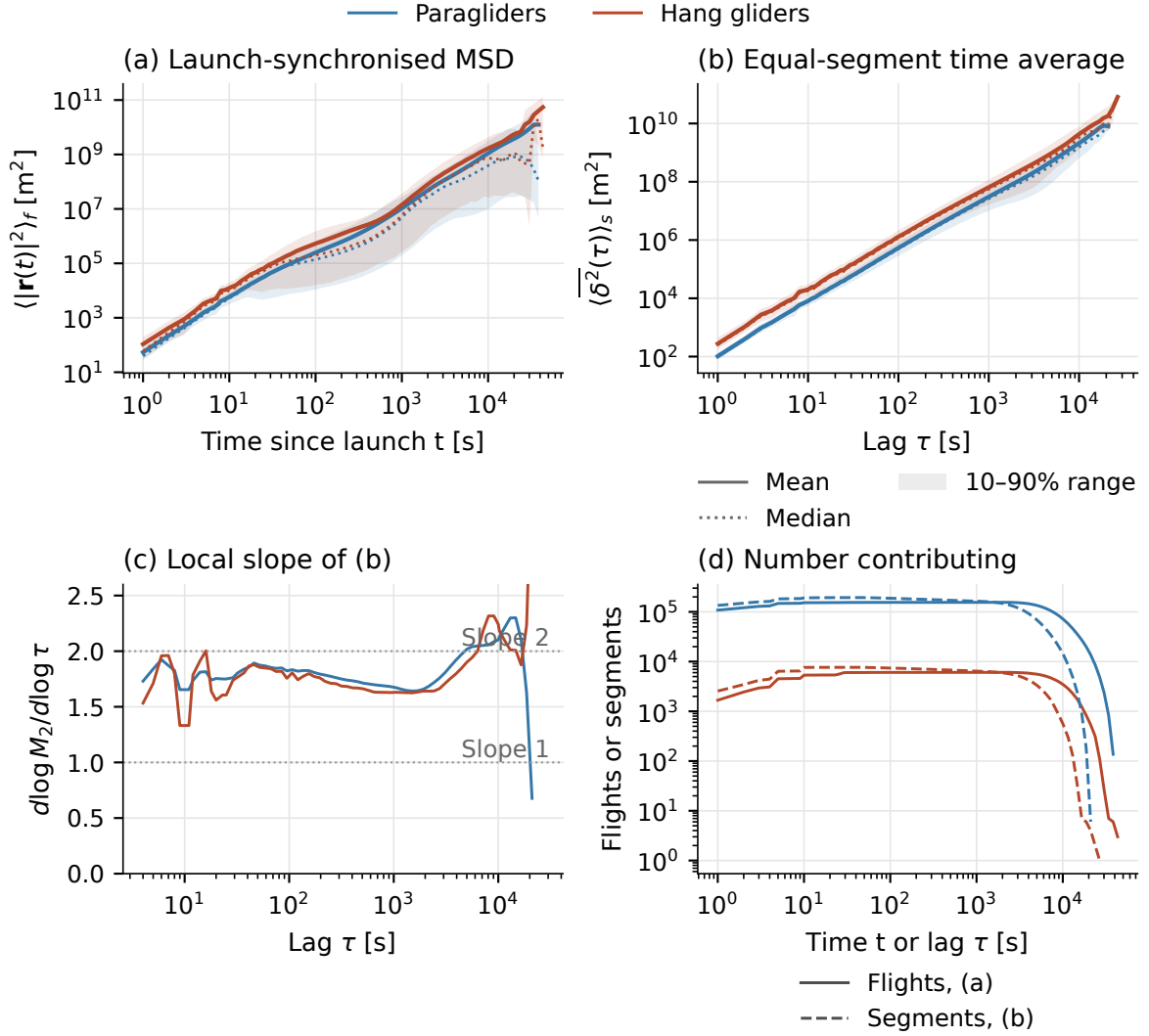


Figure 3.1. Archive displacement summaries. (a) Launch-synchronised average, retained as a description of the heterogeneous collection. (b) Time-averaged MSD, averaged equally over retained segments. Solid lines are means, dotted lines medians, and shading the 10–90% range across contributing flights in (a), or segments in (b); shading is not an uncertainty band on the mean. (c) Local slope of (b). (d) Contributing flights for (a) and segments for (b), which fall as the lag grows: the averaged population changes along every curve in this figure. Panel (a) uses time since launch; panels (b,c) use within-segment lag.

Figure 3.1a also gives the launch-synchronised average, $\text{MSD}_{\text{launch}}(t) = \langle |\mathbf{r}_f(t) - \mathbf{r}_f(0)|^2 \rangle_f$. Flights occur in different places, seasons and years; this curve describes the archive without assigning it a common process exponent.

Figure 3.2 evaluates the three conventions on one set of segment curves, with the same lag support and the same population, so their weights are the only difference between them. Over 10 s to 10,000 s the fitted exponents span 0.019 for paragliders and 0.015 for hang gliders. The amplitudes separate further. Here M_2^{window} exceeds M_2^{segment} at every lag, by up to 21.7% and 28.4% respectively, with the largest departure near 3,000 s; M_2^{flight} stays below it by at most 1.8% and 3.9%. Window weighting lets the longest flights, which supply most of the admissible origins, set the level. The equal-flight and equal-segment curves nearly coincide because the archive averages close to one segment per contributing flight. Departures of this size between two averages of the same displacement data follow from the weighting convention

alone. They do not establish nonergodicity in a heterogeneous collection.

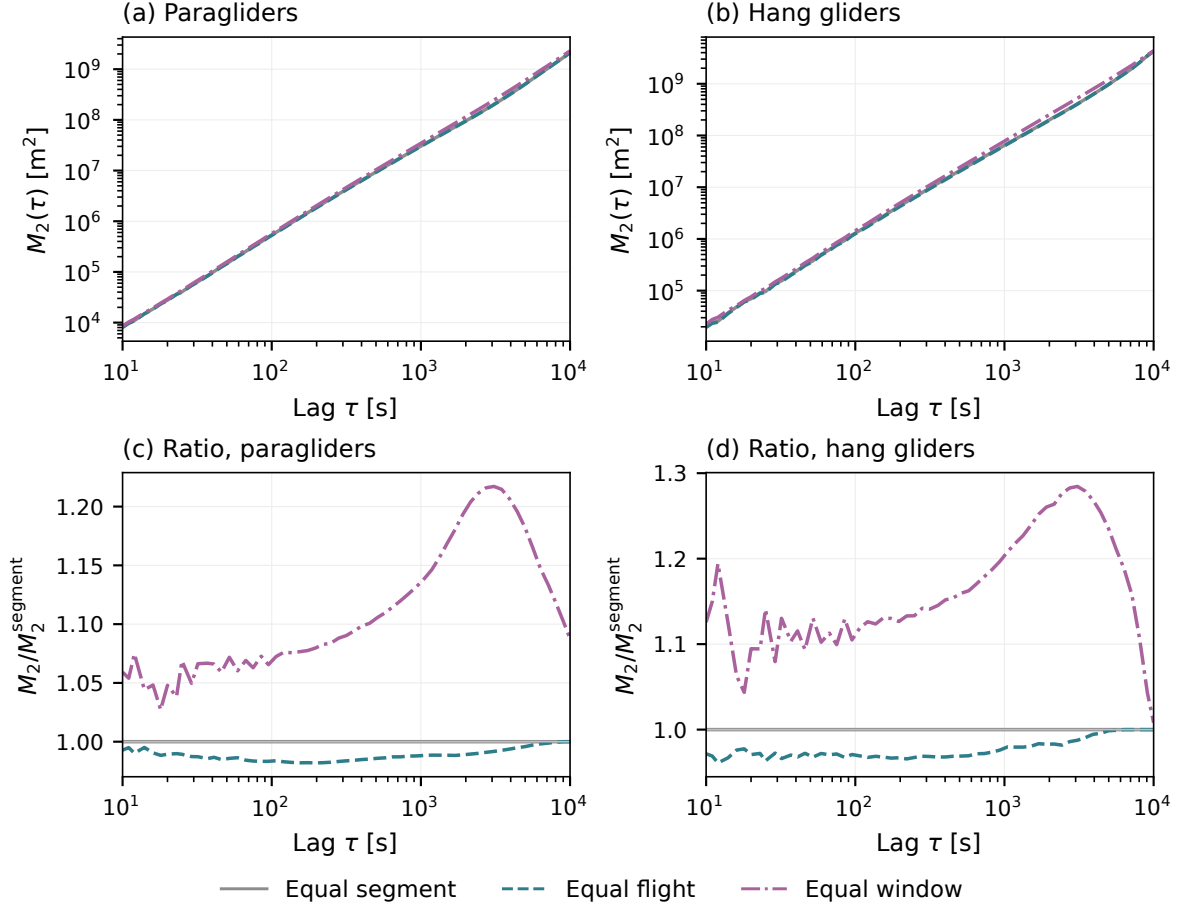


Figure 3.2. The three population weightings of one set of segment TA-MSDs. (a,b) M_2^{segment} of Eq. (3.6), with M_2^{flight} and M_2^{window} of Eq. (3.4), against within-segment lag τ . (c,d) The same curves divided by M_2^{segment} at the same lag, which is what separates them on these axes. All three read the same 194,048 paraglider and 7,662 hang-glider segment curves, with the lag support of Eq. (3.7). The comparison isolates the weighting convention; the contributing population still changes with lag, identically for all three curves.

3.1.1 Effective and local growth exponents

The comparison interval is 10s to 10,000s. For a positive observable $Y(\tau)$ we distinguish its fitted slope and its local slope,

$$\log Y(\tau) \simeq a + b \log(\tau/\tau_0), \quad b_{\text{loc}}(\tau) = \frac{d \log Y}{d \log \tau}, \quad (3.8)$$

Here logarithms use the stated units, equivalently $\log(Y/Y_0)$, and $\tau_0 = 1$ s. The common-grid and duration reports fit ordinary least squares in $\log_{10} Y$, with equal weight per admitted lag. A changing local slope makes b an effective summary of that interval.

Cadence and smoothing can still affect the 10s end; only long flights contribute near 10,000s. Coverage and local slopes accompany each fit. Changing the fitting interval measures sensitivity to curvature and selection.

Sampling uncertainty needs a separate calculation. Windows from one flight, and flights sharing a site and day, may be dependent. Resampling whole site-day groups with their

complete curves applies the bootstrap [1] at that level, assuming exchangeable groups. The baseline curves and slopes below are descriptive; where intervals are reported, their resampling unit and assumptions are stated.

The drift-removing second difference, and the route geometry that no drift removal can cancel, enter with the preliminary transport measurements of Chapter 4. The rest of this chapter stays with displacement itself and asks how its direction depends on where a flight starts and what it flies.

3.2 Directional structure, terrain and equipment

We next ask how directional structure varies across take-off regions and equipment groups, and which environmental interpretations these contrasts can support. The quantities differ from the fixed-population collapse test of Chapter 4: launch-time second moments describe mean motion and spread together, whereas regional increment covariances describe centred spread over a lag. Their clocks, weighting and contributing flights must therefore be specified before comparing them.

3.2.1 East–north balance in the retained archive

Isotropy requires the joint displacement law to remain unchanged under rotation. Equality of east and north second moments is one necessary condition. For positions measured from launch, define

$$\rho_{EN}(t) = \frac{\langle E_f(t)^2 \rangle_f}{\langle N_f(t)^2 \rangle_f}, \quad \langle E_f(t)^2 \rangle_f = \text{Var}_f[E_f(t)] + \langle E_f(t) \rangle_f^2. \quad (3.9)$$

This uncentred ratio combines mean motion and spread. Here t is time since launch, and its contributing flights change with t . Unity checks one moment condition for isotropy.

Over 10 s to 10,000 s, the unresampled ratio ranges from 0.72 to 0.87 for paragliders and from 0.54 to 1.45 for hang gliders. These are curve extrema, not bootstrap-band limits. The directional imbalance supports treating east, north and radial displacement separately. It describes the heterogeneous retained archive; it does not identify whether route choice, terrain or wind produces the imbalance. The centred regional analysis below asks how much directional structure remains after subtracting each subgroup’s mean. Launch-referenced positions can retain reacquisition offsets across missing intervals; within-segment increments avoid bridging those intervals but do not independently validate their endpoints.

3.2.2 Regional principal axes

For a region g , define the centred displacement covariance

$$\boldsymbol{\mu}_g(\tau) = \mathbb{E}[\mathbf{X} \mid g], \quad \boldsymbol{\Sigma}_g(\tau) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_g)(\mathbf{X} - \boldsymbol{\mu}_g)^\top \mid g] = U_g \text{diag}(\lambda_1, \lambda_2) U_g^\top, \quad (3.10)$$

with $\lambda_1 \geq \lambda_2$. The ratio and whitening below require $\lambda_2 > 0$. The leading eigenvector describes the direction of greatest spread, modulo 180°; it is an axis, not a signed wind direction. The ratio λ_1/λ_2 measures anisotropy of the covariance. An axis becomes poorly determined when the eigenvalues are close. Covariances pool all supported increment origins within each take-off box, so flights with more origins receive more weight. The four lags are exactly 10, 100, 1000 and 10000 s, evaluated directly on the 10-s coordinate grid. Increment origins are spaced by the lag within each segment; no increment crosses a segment boundary.

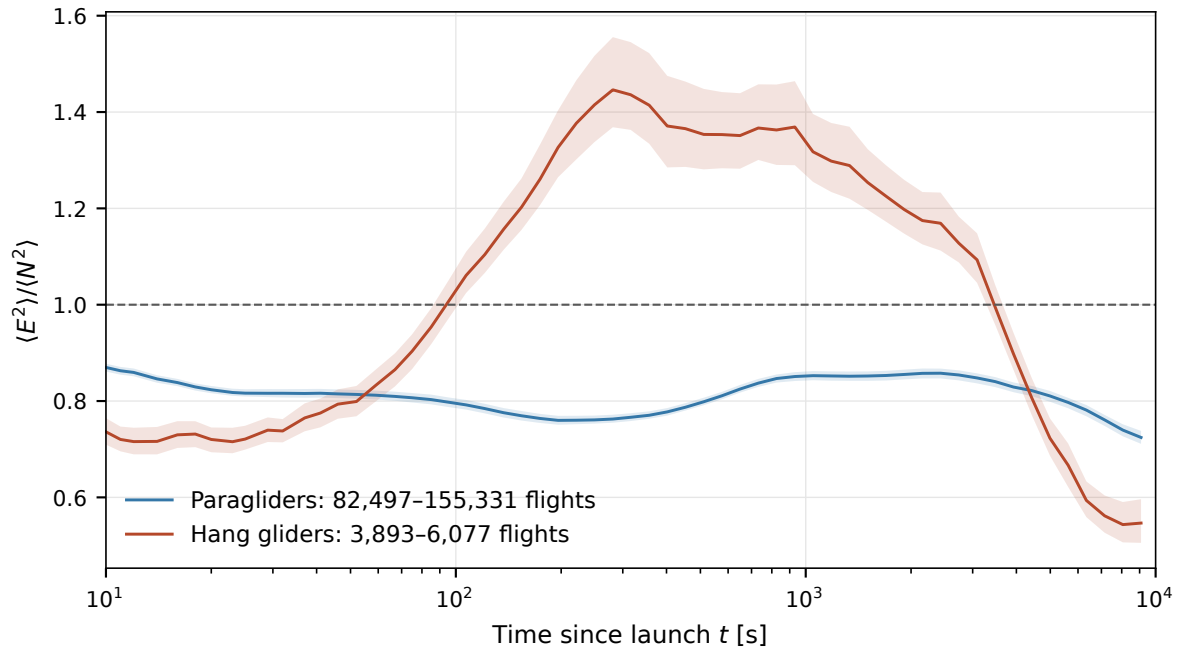


Figure 3.3. Archive east–north second-moment ratio against elapsed time since launch. Curves show the observed ratio; shading gives the pointwise 10–90% range from 200 bootstrap resamples of whole take-off-site-and-day groups. Only times with at least 30 paired flights and 10 contributing clusters are shown; legend counts give the minimum and maximum flight support along each curve. These are display minima, not a precision criterion. Unity is a necessary condition for isotropy, not sufficient evidence of it; these bands are not a simultaneous confidence statement across all times.

The panels compare lags, so they use one fixed population: the flights and segments of the C_{10000} cohort, which are supported at every lag. A change in shape across lags then cannot come from a change in who contributes. Measurements at a single lag, such as the terrain and wind comparisons that follow, need no such constraint and use every flight with a supported increment at that lag. At least eight flights are required for display; this condition supplies neither an uncertainty interval nor a precision guarantee.

At 1,000s, the regional comparison is

Region	Flights	λ_1/λ_2	Axis from east
Alps	33,175	2.14	66.4°
Pyrenees	2,360	2.49	168.2°
Channel coast	1,085	2.30	29.3°

Angles run counterclockwise from east. The Pyrenean axis lies closer to east–west; its eigenvalue ratio increases towards long lags, where few increments remain (Fig. 3.4), leaving both route geometry and sampling as possible contributors. The coastal comparison also has unequal principal variances, with thousands of flights at the reference lag. Its geographical label does not provide an isotropic control.

Environmental constraints across displacement scales. In all three regions, the observed ratio $A_g(\tau) = \lambda_1(\tau)/\lambda_2(\tau)$ increases between each pair of successive measured lags. The centred displacement cloud therefore becomes more elongated as the observation interval lengthens. This suggests that terrain-guided routes and persistent wind effects become more visible in displacement geometry when an increment spans enough time to explore their spatial scale or accumulate their directional contribution. Here τ is the duration of an increment

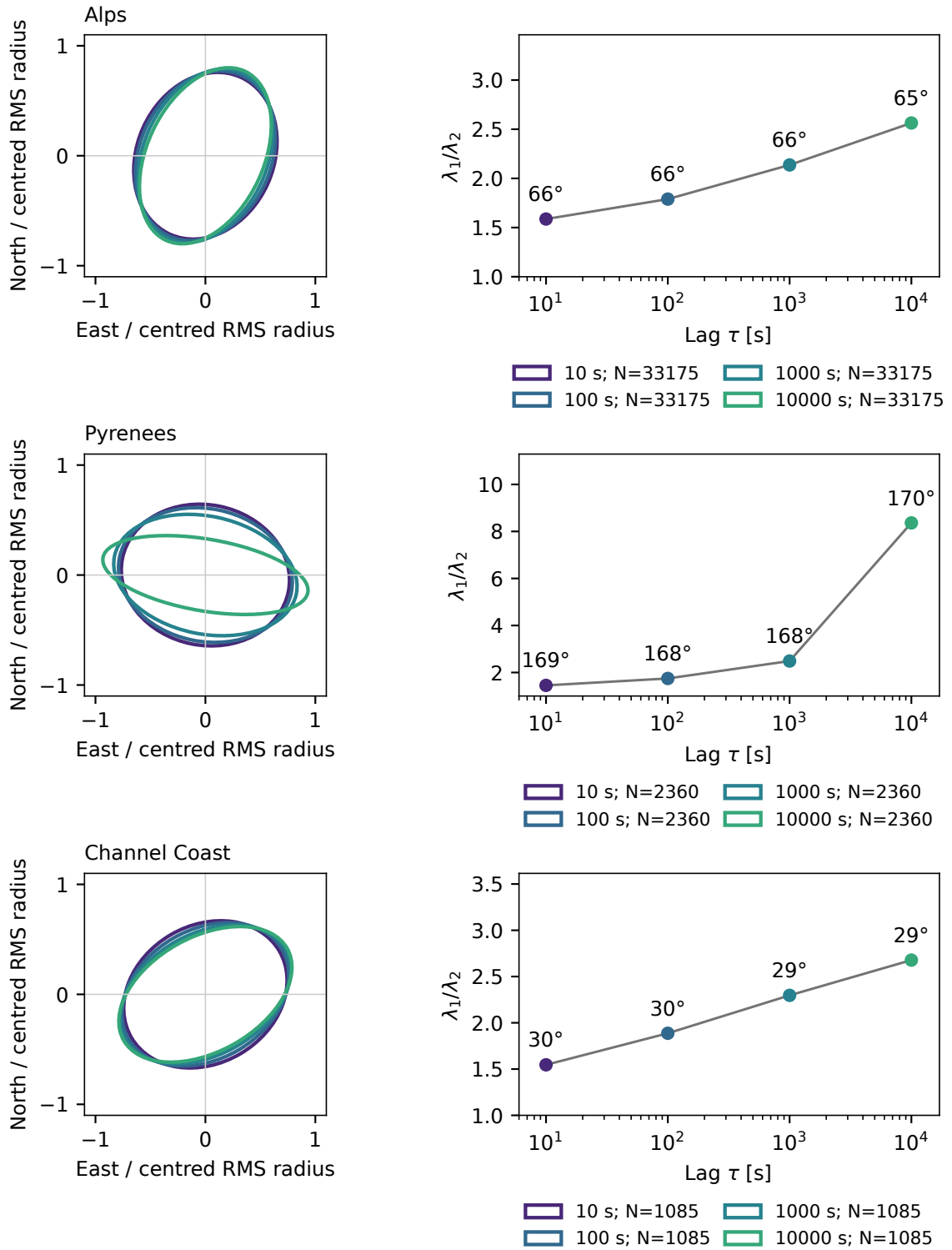


Figure 3.4. Regional principal-axis diagnostics for the fixed C_{10000} paraglider cohort. Covariances are centred before diagonalisation. The principal direction is an axis modulo 180° and its interpretation depends on the separation of the two eigenvalues. Region counts and lag dependence delimit the evidence. These measurements describe displacement geometry and do not identify wind. Ellipses and matching coloured markers show all four lags: 10, 100, 1000 and 10000 s. Ellipses are normalised by the square root of the covariance trace, removing amplitude differences. Legends give the number of cohort flights, which is the same at every lag; angle labels accompany the eigenvalue ratios.

starting anywhere inside a supported segment; it is not the time elapsed since take-off. The four estimates establish an ordering on the sampled grid, rather than a positive derivative at every intermediate lag. Where the eigenvalues are differentiable and positive,

$$\frac{d \log A_g}{d\tau} = \frac{\lambda'_1}{\lambda_1} - \frac{\lambda'_2}{\lambda_2}. \quad (3.11)$$

An increasing ratio measures faster proportional growth along the principal axis; both variances may still increase. It does not measure the strength of a force.

Time needed to explore terrain constraints. A short increment may sample only a small part of a valley, ridge or accessible flight corridor. Over a longer interval, the pilot may approach terrain or leave usable lift and have to turn, follow the valley or remain near the ridge. The transverse displacement can then grow more slowly than displacement along the route. This gives a concrete interpretation to the idea of taking time to “feel the boundary”: the increment must span enough of the route for its geometry to restrict the observed displacement. In an idealised corridor of width L , transverse displacement is bounded by the corridor width while longitudinal displacement can continue to grow. A crossing-time estimate is $L/v_\perp$ for approximately persistent transverse motion, or of order $L^2/D_\perp$ for diffusive transverse exploration. These are dimensional examples, not fitted time scales for the flights. Actual mountains are neither reflecting walls nor impermeable boundaries at every altitude. Pilots may anticipate a turn, and terrain-linked lift can influence headings immediately. The ratios already exceed one at 10 s, so the observation does not imply that orography is irrelevant at short lags.

Wind coherence and accumulated displacement. The horizontal wind contribution along a flight is $\Delta \mathbf{r}_w(t, \tau) = \int_0^\tau \mathbf{w}_h(t+s) ds$. A persistent component accumulates with a consistent sign; rapidly changing components can partly cancel. The covariance of this contribution is

$$\Sigma_w(t, \tau) = \int_0^\tau \int_0^\tau \text{Cov}[\mathbf{w}_h(t+s), \mathbf{w}_h(t+u)] ds du. \quad (3.12)$$

For a stationary scalar component with autocovariance $C_w(s)$, this reduces to $2 \int_0^\tau (\tau-s) C_w(s) ds$, the velocity-correlation relation underlying Taylor’s dispersion formula [2]. Over an interval shorter than its coherence time, nearly constant random wind gives a contribution proportional to τ^2 ; well beyond an integrable correlation time, the leading growth is generally linear in τ when the integrated autocovariance is nonzero. Waiting longer does not itself make wind more coherent. A weak persistent contribution can nevertheless become visible after accumulating enough displacement relative to other sources of spread.

For the centred PCA, a constant wind vector shared by every observation only adds $\tau \mathbf{w}_h$ to the mean and disappears on centring. The proposed wind mechanism must therefore involve variability between flights or increment origins, temporal or spatial changes of wind, or a response of the chosen route. As an illustrative calculation, let the background displacement have covariance $2D\tau I_2$ and let an independent wind component W act along a fixed unit axis $\mathbf{e}$. Suppose W varies across the ensemble with variance σ_w^2 but remains constant within an increment. Then

$$\Sigma(\tau) = 2D\tau I_2 + \sigma_w^2 \tau^2 \mathbf{e} \mathbf{e}^\top, \quad \frac{\lambda_1}{\lambda_2} = 1 + \frac{\sigma_w^2 \tau}{2D}. \quad (3.13)$$

In this example a small σ_w delays the scale at which the directional contribution competes with the background, of order $2D/\sigma_w^2$. The wind must remain coherent over that scale for this argument to apply. Neither the diffusive background nor the independent wind model has

been fitted here. In the flights, air-relative motion and wind can be correlated, adding cross covariances to Eq. (3.12); a broadly isotropic distribution of persistent wind directions would also fail to select a common axis. Thus the growing ratio supports an interpretation in terms of accumulated environmental guidance, while its attribution requires fixed flight/origin support, terrain scales along the routes and wind matched to flight time and altitude. Changing route mixtures and the declining long-lag population can also contribute to the observed growth.

Rotation, whitening and second-order isotropy. Rotation into the principal frame gives uncorrelated components at the lag at which the covariance was fitted. It does not establish independence or remove temporal correlations. It cannot improve radial collapse because

$$|U_g^\top \mathbf{X}| = |\mathbf{X}|. \quad (3.14)$$

Whitening with the reference covariance, $\mathbf{Z}(\tau) = \Sigma_g(\tau_0)^{-1/2}[\mathbf{X}(\tau) - \boldsymbol{\mu}_g(\tau)]$, changes lengths. It sets the covariance to the identity at the reference lag τ_0 ; it need not do so at other lags, and it does not imply a rotationally invariant distribution. If instead each lag has its own positive-definite covariance, then

$$\mathbf{Z}_\tau = \Sigma_g(\tau)^{-1/2}[\mathbf{X}_\tau - \boldsymbol{\mu}_g(\tau)], \quad \text{Cov}(\mathbf{Z}_\tau) = I_2 \quad (3.15)$$

by the defining identity of whitening [3]. This does enforce second-order isotropy of each centred increment distribution separately: all unit-vector projections have variance one and the covariance ellipse is a circle. With estimated covariances, the equality holds on the same weighted sample used to fit the transformation; it is not guaranteed on new flights. Higher-order angular structure can remain, and the resulting normalisation does not test a common scaling law.

Second-order isotropy of an entire centred vector process would also require its two-time covariance $K(s, t) = \text{Cov}[\mathbf{Y}(s), \mathbf{Y}(t)]$ to satisfy $QK(s, t)Q^\top = K(s, t)$ for every spatial rotation Q and every pair of times. Equal one-time covariances do not ensure this. For example, two independent stationary components with unit variance and autocorrelations $\exp(-a|h|)$ and $\exp(-b|h|)$, with $a, b > 0$ and $a \neq b$, have covariance I_2 at every time but different directional memory. Their two-time covariance is diagonal with unequal entries at nonzero separation, so a quarter-turn changes it. Moreover, applying a different matrix to each increment lag generally breaks the additivity needed to interpret those vectors as increments of a single transformed trajectory. A control that tests whether one spatial transformation and one scalar scale factor suffice instead fits the transformation at a fixed reference lag and evaluates covariance shape, angular distributions and two-time dependence on other scales and flights.

Independent terrain orientation

An independent geographical reference helps distinguish an observed displacement axis from an assumed mountain direction. NOAA's ETOPO 2022 elevation model [4] supplies this reference within the same Alpine and Pyrenean take-off boxes. The construction can be pictured as making a binary map of the high terrain and fitting an ellipse to the selected map cells: the ellipse's long axis is the indicative direction of the mountain area.

Construction. The numerical recipe is:

1. Crop the ETOPO elevation map to the same Alpine or Pyrenean box used to classify flight take-offs.

2. Retain every fourth cell in longitude and latitude. Starting from the 15-arc-second ETOPO grid gives one-arc-minute spacing and approximately one sixteenth of the original number of cells.
3. Select the retained cells with elevation $h_i \geq 1,000$ m. This produces a binary highland footprint: elevation above the threshold gives no additional weight.
4. Convert the centre of every selected cell from longitude–latitude to local WGS84 coordinates $\mathbf{q}_i = (E_i, N_i)$, in metres from the centre of the regional box. Use zero projection height because only the plan-view orientation is required.
5. Assign the area weight $w_i = \cos \varphi_i$, where φ_i is the cell latitude, to compensate approximately for the decreasing physical width of a longitude–latitude cell away from the equator.
6. Compute the weighted centre and the 2×2 spatial covariance of the selected cell centres.
7. Diagonalise that covariance. The eigenvector with the larger eigenvalue is the long axis of the footprint; report its angle counterclockwise from east, modulo 180° .

In step 6 the quantities are

$$\bar{\mathbf{q}} = \frac{\sum_i w_i \mathbf{q}_i}{\sum_i w_i}, \quad C_{\text{terrain}} = \frac{\sum_i w_i (\mathbf{q}_i - \bar{\mathbf{q}})(\mathbf{q}_i - \bar{\mathbf{q}})^\top}{\sum_i w_i}. \quad (3.16)$$

The eigenvector of C_{terrain} with the larger eigenvalue is the long axis of the area-weighted cell cloud.

Interpretation. No flight coordinate enters this calculation, and neither the threshold nor the axis is fitted to the flight PCA. The method estimates the mean elongation of all terrain above the chosen threshold inside the box. It does not trace a crest, identify individual valleys or weight the terrain actually crossed by a flight. The box boundary, plateaux and the curvature of a mountain chain can therefore affect the result. Repeating the construction at 500 and 1,500 m measures sensitivity to the operational footprint definition.

The angles, counterclockwise from east and modulo 180° , are

Region	Terrain axis	Flight axis	Separation
Alps	47.6°	65.2°	17.6°
Pyrenees	170.5°	170.5°	<1°

Using thresholds of 500, 1000 and 1,500 m places the terrain axis in 42.1° to 47.9° in the Alps and 169.7° to 175.4° in the Pyrenees. These spans describe sensitivity to the footprint definition; they are not confidence intervals. Decimal values identify the calculation, not the precision of a physical mountain direction.

The Pyrenean long-lag spread is closely aligned with the independent highland axis. This supports a geographical association between route orientation and the range. In the Alps, both axes occupy the northeast–southwest sector, but the flight axis is about 18° more northward than the mean highland axis; simple parallel alignment is a poorer description there. The curved Alpine chain and its truncation by the box can contribute to this difference. Neither comparison identifies a causal terrain effect: wind, route choice and the long-flight population can also contribute. The PCA uses 46,336 Alpine and 4,331 Pyrenean flights at this lag. Their increments can extend beyond the take-off box, and their tangent frames differ from the single regional terrain frame. This is an indicative regional comparison, not a measure of terrain exposure along individual tracks or a test of pilot skill.

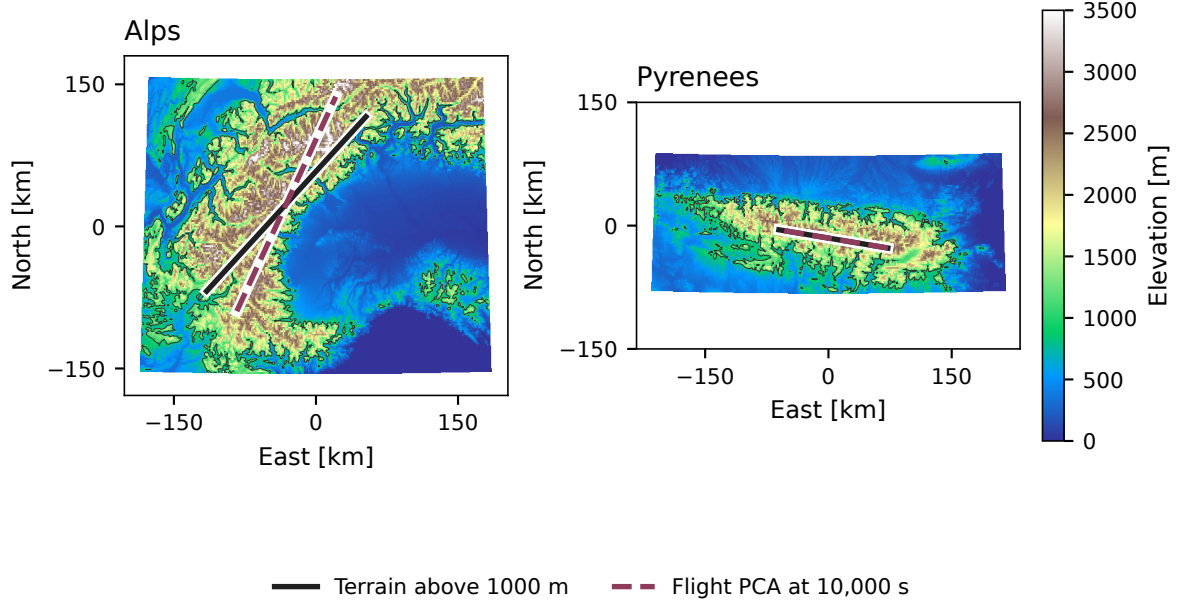


Figure 3.5. Independent terrain orientation compared with the paraglider displacement covariance at 10,000s. Elevation and the 1,000 m contour come from ETOPO 2022 [4]. The solid dark line is the highland-footprint axis; the dashed coloured line is the flight PCA axis, translated to the same centre for comparison. Line lengths carry no displacement or ridge-length information. Map coordinates are kilometres east and north of each box centre.

Alpine boxes chosen for a reliable terrain direction. The Alpine box above contains a chain that does not run along a line. From the Maritime Alps to the Bernese Alps the ridge line turns through roughly a quarter of a turn, from a north–south trend in the south to an east–north–east trend in Switzerland. One terrain axis and one PCA axis cannot describe such a shape, and the Alpine row of the table above compares two averages taken over different bends. Asking whether the direction of motion follows the direction of the relief needs boxes in which the relief has one direction.

The boxes are selected from the elevation map alone. A rectangle with edges on a grid of 0.2° , inside the Alpine take-off box, sides of at least 0.8° and an area of at least 1.5 square degrees, qualifies when its crest footprint, the cells at or above 2,000 m, meets all of the following.

1. It has at least 400 cells and an elongation $\lambda_1/\lambda_2 \geq 2.5$.
2. Its axis moves by at most 10° when the threshold is changed to 1500 and to 2,500 m. The spread is the largest deviation from the circular mean of the three axes.
3. Its axis moves by at most 10° , in the same sense, when any one box edge is shifted by $\pm 0.2^\circ$. This measures the dependence on where the rectangle happens to cut the chain.
4. Its bend is at most 10° . The footprint cells are cut into three groups of equal size along the long axis and the bend is the angle between the segment joining the centroids of the first two groups and the segment joining the last two. It is zero for a straight band and large for a chain that turns inside the box.

A box must also contain at least 300 flight take-offs. This is the only condition that involves flights, and it involves no direction. Qualifying boxes are ranked by the largest of the three instabilities (threshold spread, edge spread, bend), then by area, and taken in that order when they share no area with a box already chosen. The whole Alpine box fails the last rule, with a

bend of 37° , and the selection was completed and stored before any flight axis was computed.

The procedure returns 3 boxes (Fig. 3.6). For each, the flight axis is the long axis of the centred 10,000 s displacement covariance of the flights that took off inside it, computed with the regional-PCA estimator of Fig. 3.4: nonoverlapping origins inside a segment, each with equal weight. A single lag needs no fixed population, so every flight with a supported increment contributes.

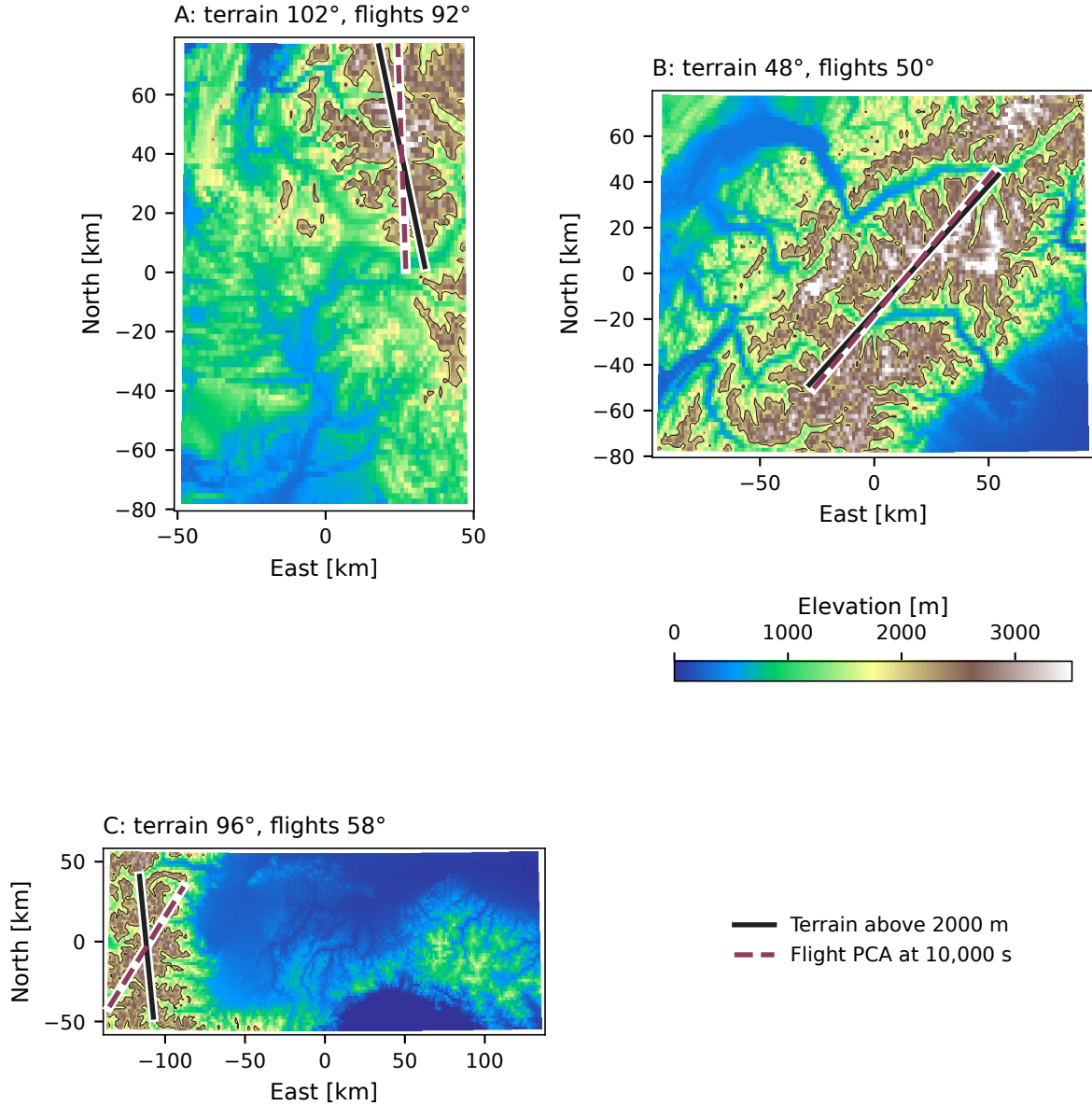


Figure 3.6. The three Alpine boxes selected from terrain alone, one panel each. Solid dark line: axis of the crest footprint above 2,000 m. Dashed line: flight PCA axis at 10,000 s, translated to the same centre. Panel titles give both angles counterclockwise from east. Thin contours mark 2,000 m; line lengths carry no information. Panels have different scales. Elevation comes from ETOPO 2022 [4].

Angles run counterclockwise from east, modulo 180° .

	Box (E; N)	Terrain		Flights		λ_1/λ_2	Sep.
		Axis	λ_1/λ_2	Number	Axis		
A	5.4° to 6.6°; 43.8° to 45.2°	101.8°	4.1	12,587	92.0°	2.60	9.8°
B	6.0° to 8.4°; 45.2° to 46.6°	48.0°	6.1	19,220	50.3°	4.78	2.3°
C	6.6° to 10.0°; 44.2° to 45.2°	95.6°	5.5	670	58.2°	1.38	37.4°

In box B, which covers the Mont Blanc massif and the Bernese Alps and holds most flights, the two axes lie 2.3° apart. Terrain thresholds of 1500 and 2,500 m move its terrain axis to 45.4° and 52.0° , so the agreement does not depend on one threshold. In box A, the Écrins, the Vercors and Haute-Provence, the axes lie 9.8° apart. The terrain axis of this box moves by 9.0° under edge shifts, which is of the same size as the separation. Box C, the Cottian and Maritime Alps east of A, holds few flights and their covariance is almost round ($\lambda_1/\lambda_2 = 1.38$), so its flight axis is weakly determined and the 37.4° separation is not evidence of misalignment. The reliability criteria concern the terrain axis only. They cannot make a flight axis well determined, and a box can have a clear terrain direction and a nearly isotropic set of flights.

Two of three boxes therefore show flight axes within about 10° of the terrain axis, against 17.6° for the whole Alpine box. The comparison stays descriptive: it rests on three boxes, gives no uncertainty interval for either angle, and does not control for wind or route choice.

An independent coastal wind reference

The relevant vertical reference for the Channel Coast is the altitude of the flights contributing to the 10,000 s PCA. The cleaned GNSS altitude z is interpolated onto the same 10 s segment grids as the horizontal coordinates. Trapezoidal integration gives a mean altitude along each of the 2,013 nonoverlapping displacement windows; averaging those means with equal window weights gives 1,006 m. The 1,680 flights and their segment boundaries are exactly those of the PCA. Giving each flight equal weight instead gives 980 m; the middle half of window-mean altitudes lies between 854 m and 1,160 m. These are absolute recorded altitude proxies, not take-off altitude or height above the terrain. The receiver datums have not been harmonised (Section 2.7.5); the comparison interprets them approximately as altitude above mean sea level, with limited physical precision.

ERA5 hourly pressure-level wind and geopotential from ECMWF/Copernicus [5] supply an independent reference at that altitude. The values are accessed through a pinned snapshot of Earthmover’s public, cloud-optimised copy of ERA5 on AWS [6]. Earthmover is the distributor of this losslessly stored copy, not the producer of a separate meteorological reanalysis. The snapshot provides 2016–2025 values at the same nine 0.25° cells used for the regional surface-wind reference. These cells are nearest to the centres of a regular 3×3 partition of the Channel Coast take-off box. The extracted levels are 1000, 925, 850 and 700 hPa. For each cell and hour, geopotential Φ gives geopotential height $H = \Phi/g_0$ and approximate geometric height [7]

$$h = \frac{R_\oplus H}{R_\oplus - H}, \quad g_0 = 9.80665 \text{ m s}^{-2}, \quad R_\oplus = 6,371,000 \text{ m.} \quad (3.17)$$

The east and north wind components are linearly interpolated between the two hourly level heights bracketing 1,006 m. Both bounds must be above the model terrain. All selected profiles satisfy these conditions; no below-ground bound or vertical extrapolation enters the averages. A fixed pressure level is therefore not assigned a fixed altitude.

The main comparison separates *all hours throughout the year* from *09–17 UTC inclusive throughout the year*. April–September, with the same two hourly selections, is a separate

seasonal sensitivity. The ERA5 winds at 10 and 100 m above ground, retrieved through Open-Meteo [8, 9], remain lower-level controls. Their meteorological *from* directions ϕ , clockwise from north, are converted using $u = -s \sin \phi$, $v = -s \cos \phi$. All comparisons average the velocity components over the selected hours and weight cells by $\cos(\text{latitude})$ to approximate represented area. The resulting *towards* angle,

$$\theta_w = \text{atan2}(\langle v \rangle, \langle u \rangle), \quad (3.18)$$

is counterclockwise from east and reduced modulo 180° to compare with the undirected PCA axis. Neither height interpolation nor temporal averaging acts directly on degree angles.

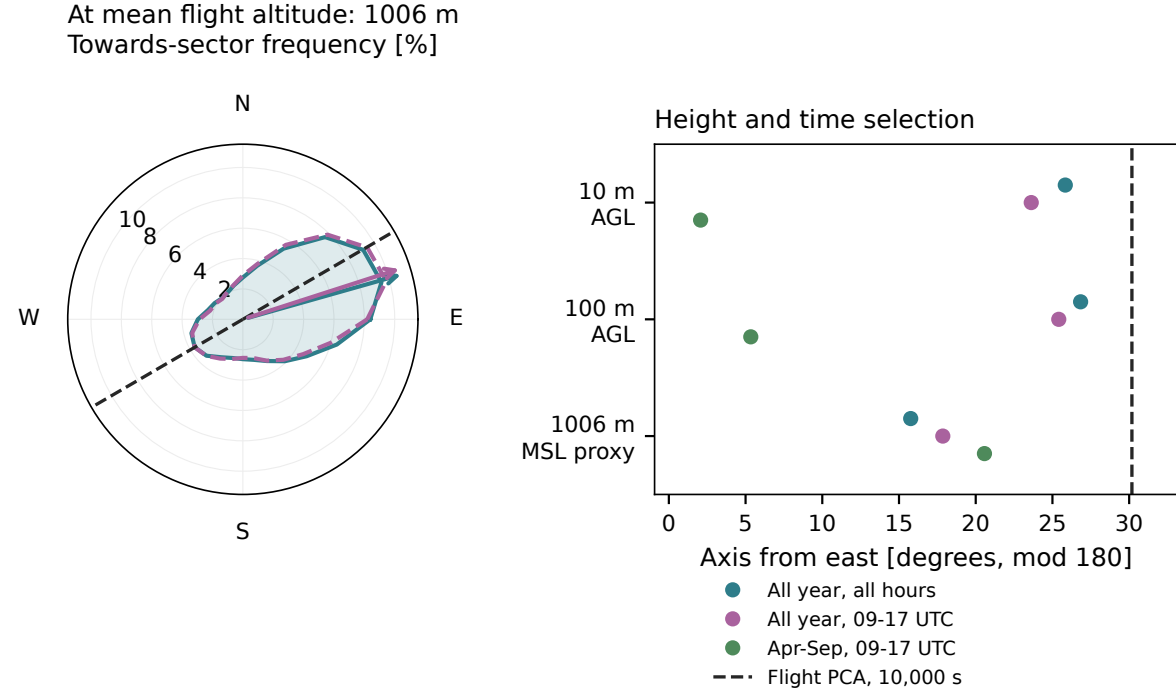


Figure 3.7. ERA5 wind at mean coastal flight altitude and the paraglider PCA at 10,000s. Left: frequencies of air motion *towards* 15-degree sectors at 1,006 m, for all-year all hours and all-year 09–17 UTC. Speeds below 0.5 ms^{-1} are excluded only from the frequency curves; arrows show the vector means using all selected hours. Right: wind axes at flight altitude, approximately above mean sea level, and at 10 and 100 m above ground (AGL). The warm daytime selection is shown as a separate check. The dashed black line is the same flight covariance axis in every comparison.

At mean flight altitude	Wind axis	PCA separation	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
April–September, all hours	16.0°	14.2°	0.35
April–September, 09–17 UTC	20.6°	9.6°	0.37

Here $\rho_w = \|\langle \mathbf{w} \rangle\| / \|\mathbf{w}\|$ measures directional cancellation, not confidence in an angle or the fraction of winds near it. The reference PCA angle is 30.2° .

At flight altitude the annual all-hour wind axis is 15.8° , separated from the flight axis by 14.4° . Restricting the same year-round population to 09–17 UTC gives 17.9° and a 12.3° separation. Thus the wind reference and displacement spread share a broadly eastward orientation, with an estimated offset of more than ten degrees. The warm-season selections

retain separations of about 10–14 degrees. At this altitude the temporal checks therefore change the comparison less than at 10 and 100 m AGL, where the annual near alignment gives way to a larger warm-daytime offset (Figure 3.7). Surface alignment cannot simply be transferred to the altitude at which these flights take place.

A height sensitivity evaluates the two quartiles of window-mean altitude and the equal-flight mean. At the lower quartile, some hours have no above-ground pressure level below the target. The comparison therefore excludes 868 of the 87,672 hours, about 1%, from *every* tested height and cell, and recomputes the mean-height reference on those same hours. The main estimates above retain all their selected hours. On common support, the quartile-height annual axes span 14.3° – 17.2° for all hours and 16.4° – 19.3° for 09–17 UTC. Equal-flight altitude weighting changes the corresponding mean-height axes by less than 0.3° ; restricting the main altitude to common support changes its annual axes by less than 0.1° . These checks preserve the annual offset. They are sensitivity ranges, not confidence intervals or a calibration of the GNSS datum.

The comparison remains descriptive. The wind climatology is not matched to flight dates, positions or instantaneous altitudes, and a take-off box need not contain whole flight routes. Moreover, the wind reference is a mean velocity while the PCA describes *centred* displacement covariance. A common constant additive wind shifts every displacement at fixed lag by $\tau \mathbf{w}$ and disappears under centring; variable winds, route choices and flight selection can still affect the covariance. The angular resemblance therefore does not establish a wind-driven mechanism or resistance to wind by one equipment group. The useful next question is whether winds matched to the observed flight conditions explain the remaining offset and covariance orientation.

The complete selected hourly pressure-level observations are preserved losslessly, with coordinates, timestamps, surface geopotential, source snapshot and hashes, in `revisions/channel-wind-flight-altitude-2026-09-13/`. The archive occupies about 32.22 MB compressed; the earlier 10/100-m inputs occupy another 6.33 MB. Both have SSD copies and offline reuse instructions. Only the regional extract used here is stored. Keeping its hourly resolution supports vertical interpolation before averaging, the directional distributions and separate time selections at modest storage cost.

3.2.3 Terrain, equipment and wind

Figure 3.8 compares east/north second-moment ratios over 10 s to 10,000 s. Their medians are

Region	Position ratio	Velocity ratio
Alps	0.67	0.81
Pyrenees	1.67	1.19

These uncentred moments retain mean motion, as in Eq. (3.9). The stronger Pyrenean east–west preference is consistent with the massif constraining headings, without identifying geography as its sole cause.

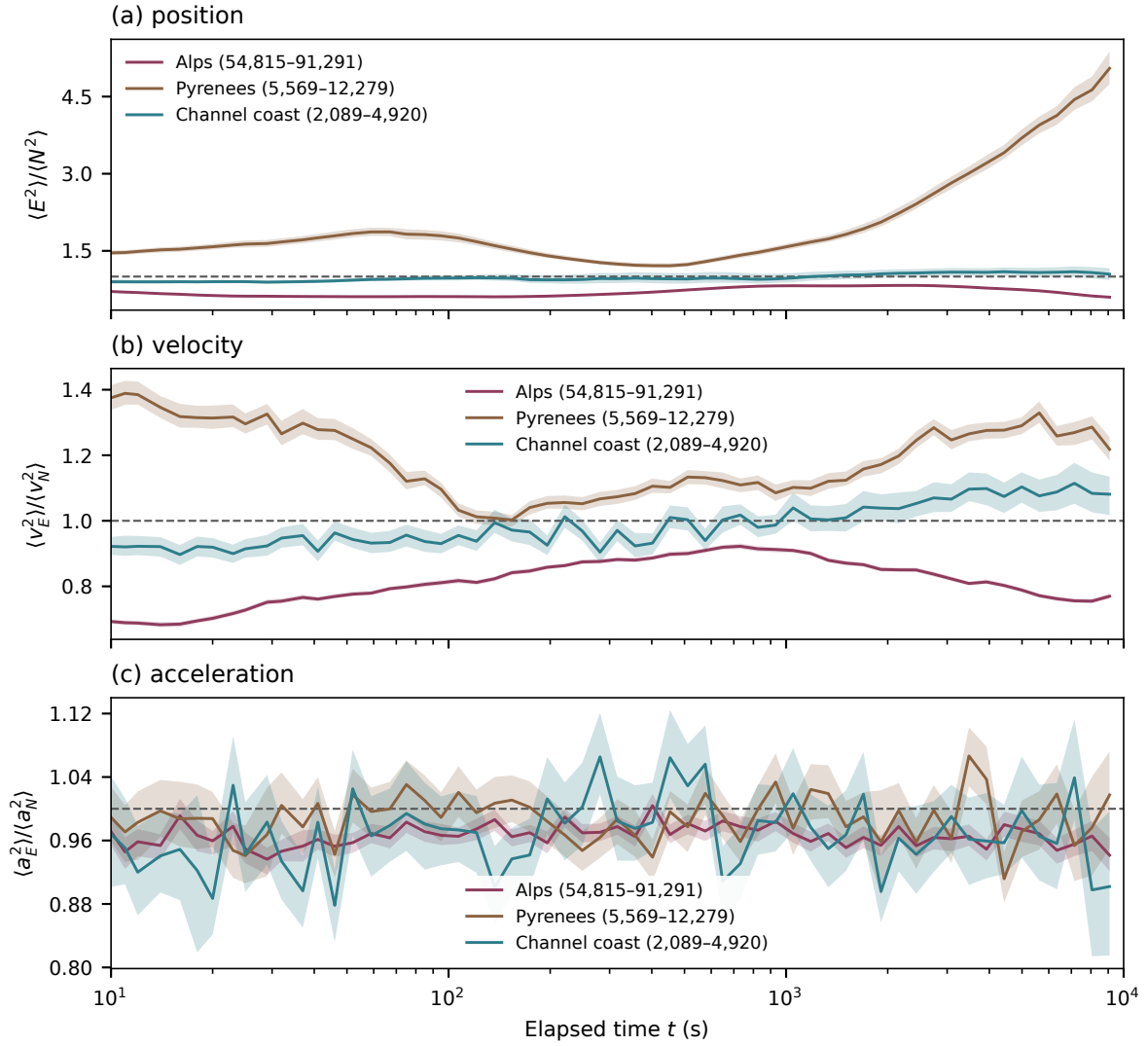


Figure 3.8. Paraglider east/north second-moment ratios by take-off region, against time since launch. Position, velocity and acceleration all retain mean motion. Shading gives pointwise 10–90% site-day bootstrap ranges; times require 30 flights and 10 clusters, with flight-count ranges in the legends. Missing site/date records form singleton groups. Unity marks equal component moments, leaving cross moments and angular structure untested. Panel limits follow each quantity; no scaling exponent is fitted.

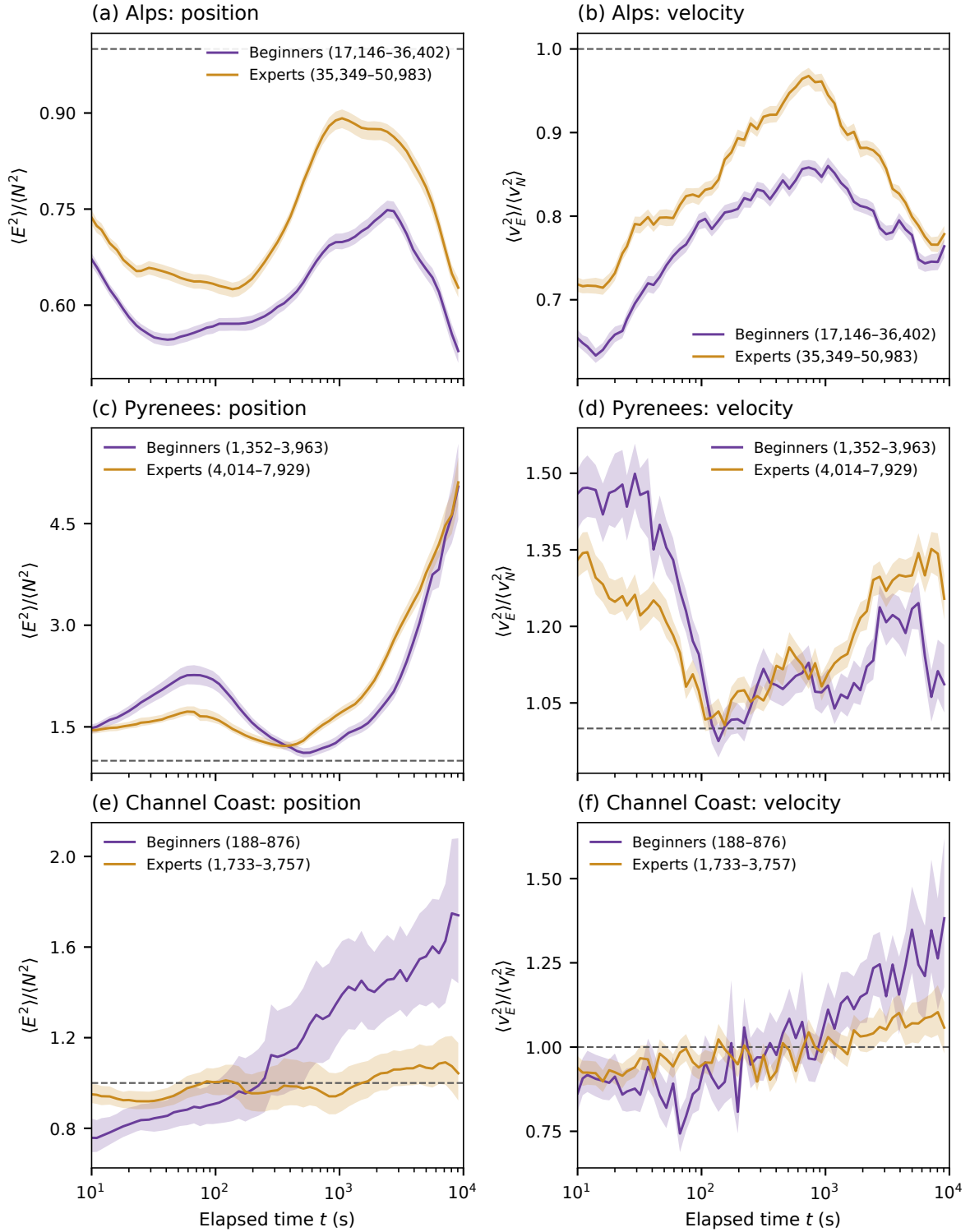


Figure 3.9. Paraglider second-moment ratios within each take-off region: beginners (EN A/B, violet) and experts (EN C/D/CCC, ochre), using equipment as an experience proxy. Tandem, non-certified and unclassified records are excluded. Pointwise bootstrap bands and support minima follow Fig. 3.8.

Equipment within each region. EN A/B and EN C/D/CCC provide the beginners and experts proxies of Sec. 2.4. In the Alps, the C/D/CCC position and velocity ratios remain closer to unity at all displayed supported times. The median position ratio is 0.70, versus 0.61 for A/B. The Pyrenees and coastal contrasts do not have the same ordering across all times and quantities.

An interpretation to test. The smaller east–north imbalance of the C/D/CCC group in the Alps and over much of the coastal comparison motivates a hypothesis: equipment and pilot decisions may moderate the directional constraints associated with wind and terrain. A wing with different flight characteristics may permit different routes, while a pilot’s choices may alter how a route follows the local lift and wind. Similar mountain profiles could reflect a shared geographical constraint, with a difference in its magnitude between groups. This is a possible explanation of the observed contrasts, rather than a measured decomposition of their causes.

The Pyrenean curves limit that explanation. Their ordering reverses with elapsed time, and the C/D/CCC group does not remain uniformly closer to one in either position or velocity. Near 9,086 s, the position ratios are about 5.05 for A/B and 5.11 for C/D/CCC. The coastal position contrast is larger at the same time, about 1.74 versus 1.04, but its support is also unequal: 188 versus 1,733 paired flights. The bands are pointwise 10–90% site–day bootstrap ranges, not simultaneous confidence bands or a direct confidence interval for a between-group difference.

What proximity to unity establishes. The ratio compares two ensemble raw second moments. It does not measure the isotropy of each flight or the ability of an individual pilot to make progress against wind. For example, a centred Gaussian with covariance

$$\begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$$

has equal east and north second moments but a principal-variance ratio of 19. For a general centred law, a covariance proportional to the identity establishes only a second-order condition, rather than rotational invariance of the full distribution. A mixture of strongly east-directed and strongly north-directed routes can also balance ensemble moments while each route remains directional. Deliberately following a downwind cross-country route may be successful flight despite that directionality.

Equipment, terrain and selection. EN A/B and EN C/D/CCC are equipment groups used as experience proxies. Actual pilot skill is unmeasured, and an experienced pilot can fly an A/B wing. Certification classes describe tested flight behaviour and associated pilot requirements; the class is not itself a measured aerodynamic performance ranking [10]. Equipment, pilot selection and intended task therefore remain entangled in these comparisons.

The Channel Coast box is also not a wind-only experiment. It includes coastal cliffs and valleys, including those documented in the Pays de Caux [11]. Wind and terrain may jointly orient lift and routes. Moreover, the boxes classify take-off locations rather than terrain exposure along the entire flight. The additional Poitou-Charente and Champagne-Lorraine comparisons in Fig. 3.10 extend the geographical description, but these broad boxes are not verified terrain polygons or meteorologically matched controls. The Poitou-Charente curves cross with broad overlapping bands; the later Champagne-Lorraine position contrast also leaves the C/D/CCC ratio above one. Thus the coastal pattern does not establish a general flatland mechanism.

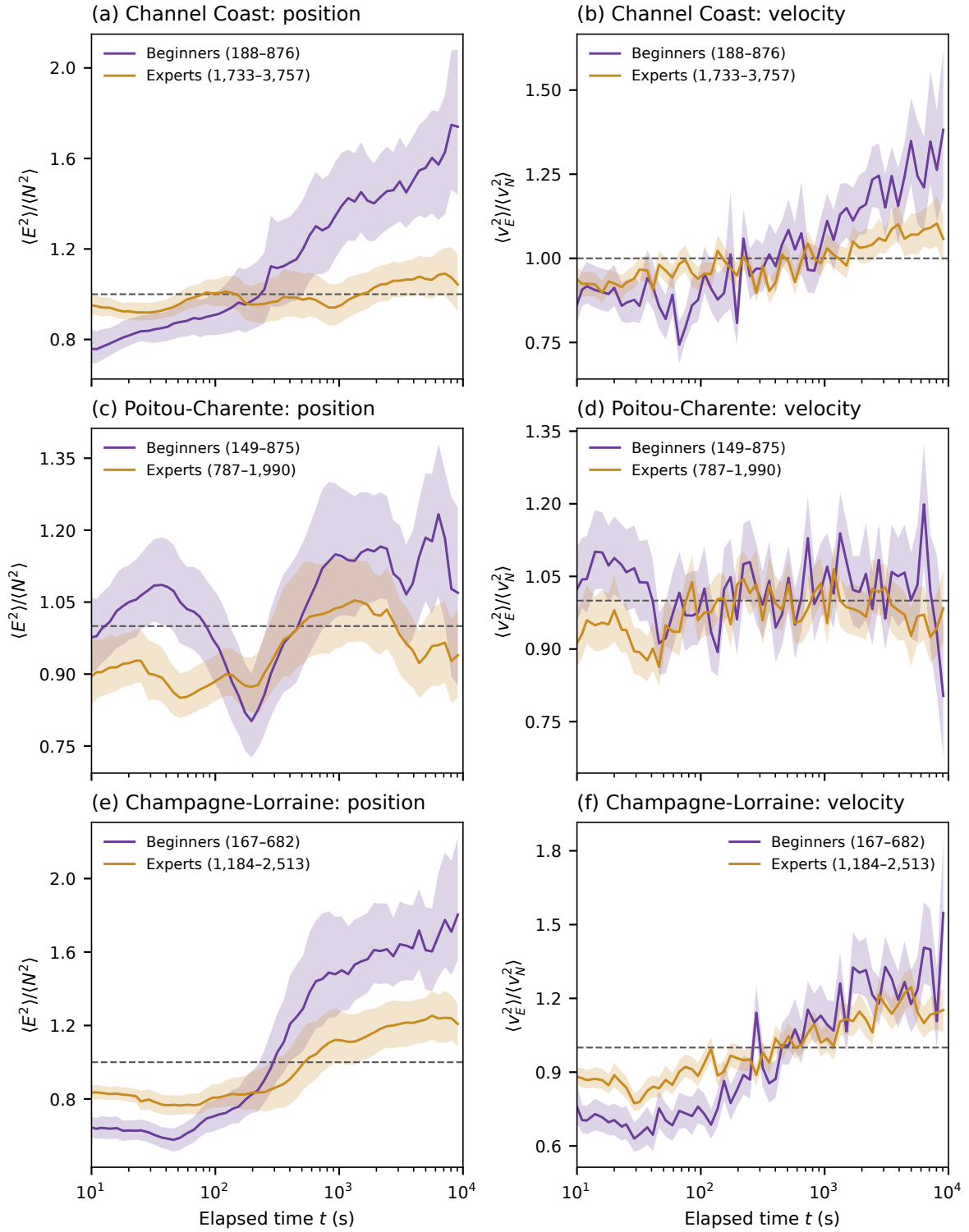


Figure 3.10. Equipment contrasts in three take-off boxes outside the mountain comparison: Channel Coast, Poitou-Charente and Champagne-Lorraine. Position and velocity ratios use the paired support, pointwise 10–90% site-day bootstrap bands and display minima of Fig. 3.9. The coastal row is repeated to compare it directly with the two additional regions. The boxes label take-off locations; they do not measure terrain exposure along each route or match wind conditions between equipment groups.

Discriminating comparisons. A first comparison would restrict both equipment groups to overlapping take-off sites and days, with comparable duration and task. This would test whether the contrast survives closer matching of observed conditions; the current cluster bootstrap adjusts the uncertainty for one grouping structure without performing that matching. Mean ground drift, centred directional variability and the mixture of individual route directions should then be distinguished. The independent regional references in Sections 3.2.2 and 3.2.2 provide an initial alignment comparison; wind matched to flight times and local terrain along tracks would test the proposed mechanisms more directly. Where records permit it, comparisons of the same pilot using different wings could help assess equipment and pilot selection separately. Before describing any of these contrasts as resistance to wind, the target needs to be defined: upwind progress, air-relative performance and the choice of route are different outcomes.

Wind and ground velocity. The kinematic relation

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h \quad (3.19)$$

prevents ground velocity alone from separating still-air motion and wind. Route choice and net progress also enter the mean. Differentiation removes a constant wind contribution but retains changing wind and manoeuvres.

If the wind field is $\mathbf{w}_h(\mathbf{r}, z, t)$, its derivative along the trajectory is

$$\mathbf{a}_{\text{ground}} = \frac{d\mathbf{v}_{\text{air}}}{dt} + \partial_t \mathbf{w}_h + (\mathbf{v}_{\text{ground}} \cdot \nabla_h) \mathbf{w}_h + \dot{z} \partial_z \mathbf{w}_h. \quad (3.20)$$

Thus even a time-independent wind field contributes to acceleration when the glider crosses spatial gradients.

Separating these effects requires terrain alignment and independent wind estimates within matched region and site-day groups. Thermal-centre drift may help after segmentation, subject to assumptions about thermal motion. Randomly rotating flights offers an alignment control: it balances pooled directions while preserving every radial displacement, without reconstructing still-air motion.

3.2.4 Regional changes of velocity and direction

Second differences permit a regional comparison that is insensitive to an added constant velocity. The operator A_2 of Eq. (4.2) and the second moment V_2 are defined with the preliminary transport measurements of Chapter 4. Their physical scale can be expressed without fitting an exponent. For the two adjacent interval-mean ground velocities,

$$\bar{\mathbf{v}}_1 = \frac{\mathbf{r}(t + \tau) - \mathbf{r}(t)}{\tau}, \quad \bar{\mathbf{v}}_2 = \frac{\mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)}{\tau}, \quad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\mathbb{E}|\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2}. \quad (3.21)$$

Thus D_2 , in m s^{-1} , measures changes of the mean velocity vector between adjacent intervals. Both changes of direction and changes of its magnitude contribute. This identity requires neither stationary increments nor self-similarity. It does not measure wind speed.

The focused measurement uses all 108,189 paraglider and 4,747 hang-glider flights in the three regional take-off boxes. It reads the verified common-grid coordinates used by the regional PCA, without repeating cleaning or limiting the number of flights. Eighteen physical lags span 10 s to 10,000 s; the four display lags agree with the updated PCA at 10, 100, 1000 and 10,000 s. Segment boundaries remain mandatory.

Three support conventions answer different questions. The available-origin curves use every supported origin for each order separately, spaced by τ within each segment. The matched-order comparison uses the same origins and flights for orders one, two and three, and therefore requires 3τ of continuous support. These stencils overlap even though consecutive origins are separated by τ . A further common-origin control uses every 10-s origin supporting a continuous 3000-s window, and retains that exact set for every order and every lag through 1,000 s. The 1000-s limit is a declared support convention; it avoids selecting only flights with 30,000-s continuous records merely to compare the smaller scales.

Within each flight, eligible origins are pooled before flights receive equal weight. The report also retains the pooled-origin result used as a weighting sensitivity. This distinguishes the estimator here from the pooled-origin PCA in Sec. 3.2. Uncertainty uses 500 bootstrap draws of complete site–day groups, jointly across lags and difference orders. Missing group keys occur in three Alpine paraglider flights and are treated as singleton groups. Pointwise 95% intervals require at least 20 contributing groups and 90% finite replicates; this display threshold is not a guarantee of coverage. Regional RMS-change ratios additionally receive approximate simultaneous 95% bands over their supported lag grid, obtained from the bootstrap maximum standardised log-ratio deviation. The interpretation assumes independently resampleable groups; dependence across sites, dates or repeated pilots can remain.

For directionality, retain the vector $\mathbf{d}_2 = A_2 \mathbf{r}$ before taking its squared norm. Its mean and centred covariance satisfy

$$\boldsymbol{\mu}_2 = \mathbb{E} \mathbf{d}_2, \quad C_2 = \text{Cov}(\mathbf{d}_2), \quad V_2 = \text{tr } C_2 + |\boldsymbol{\mu}_2|^2. \quad (3.22)$$

The covariance includes variation between flight means as well as within flights. The ratio of its eigenvalues measures directional imbalance of the dispersion; the scalar V_2 alone cannot do so. As before, the leading axis has no signed direction and becomes poorly determined near equal eigenvalues.

Table 3.2. Paraglider diagnostics at 1,000 s, with equal flight weights and origins shared by all three difference orders. The eigenvalue ratios refer to centred covariances at the indicated order.

Region	Flights	D_2 [m/s]	Ratio, order 1	Ratio, order 2	V_3/V_2
Alps	89,181	5.19	2.28	1.83	2.60
Pyrenees	12,020	4.78	2.17	1.42	2.79
Channel Coast	4,679	3.77	1.92	1.37	2.42

At this lag, second differencing reduces the covariance anisotropy relative to ordinary increments in all three regions, but leaves a directional contrast (Table 3.2). The order-two axes are about 65.45° , 171.09° and 30.61° counterclockwise from east. At 10 s the second-difference covariance is much closer to equal principal variances, especially on the coast; an angle there should not be interpreted as a resolved environmental orientation.

Holding origins fixed through 1000 s gives D_2 values of 5.31 m s^{-1} in the Alps, 4.89 m s^{-1} in the Pyrenees and 3.72 m s^{-1} on the coast at 1000 s. The Alpine-to-coastal ratio is 1.426, with simultaneous band 1.403–1.449; the Alpine-to-Pyrenean ratio is 1.086, with band 1.074–1.097. The ordering changes with lag: at 10 s the coastal and Pyrenean values exceed the Alpine value. Similar-looking mountain curves therefore do not establish regional equality, and a single regional amplitude ratio does not describe the whole interval.

A descriptive standardisation further restricts the common-origin population to shared strata of declared open/closed task, retained duration (up to 2 h, 2–4 h, over 4 h), season period (through 2009, 2010–2019, 2020 onward) and equipment (EN A/B or EN C/D/CCC). Each retained stratum needs at least ten flights per region; its target mass is proportional

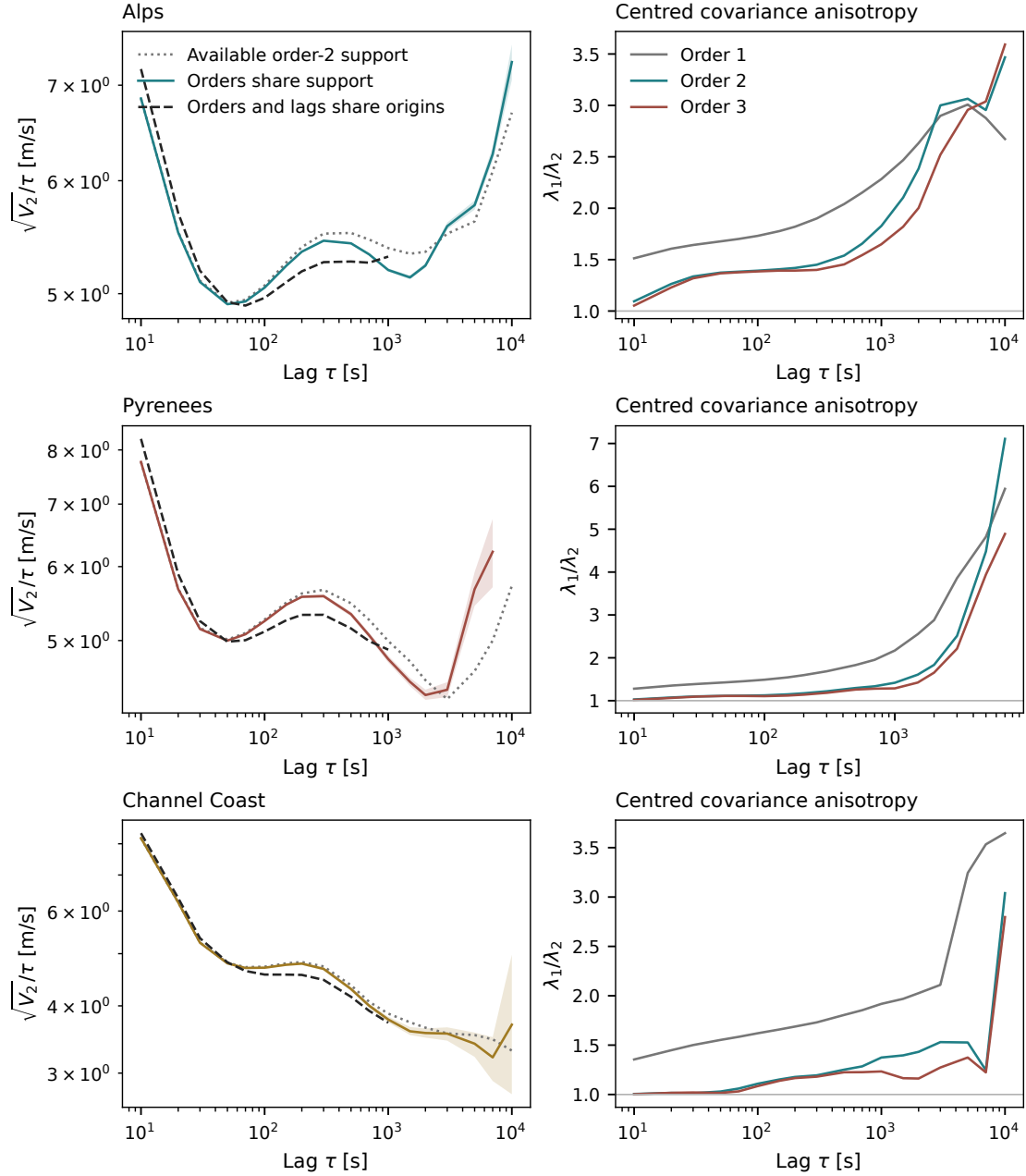


Figure 3.11. Regional changes of ground velocity in paragliders. Left: D_2 on available order-two support, on support shared by orders one to three, and on origins common to all orders and lags through 1000 s. Shading is the pointwise site-day interval for the matched-order curve. Right: covariance eigenvalue ratios on the same matched origins. Curves are omitted where fewer than eight flights contribute; all numerical cells remain in the report. Support at the long end is shown in Fig. 3.16.

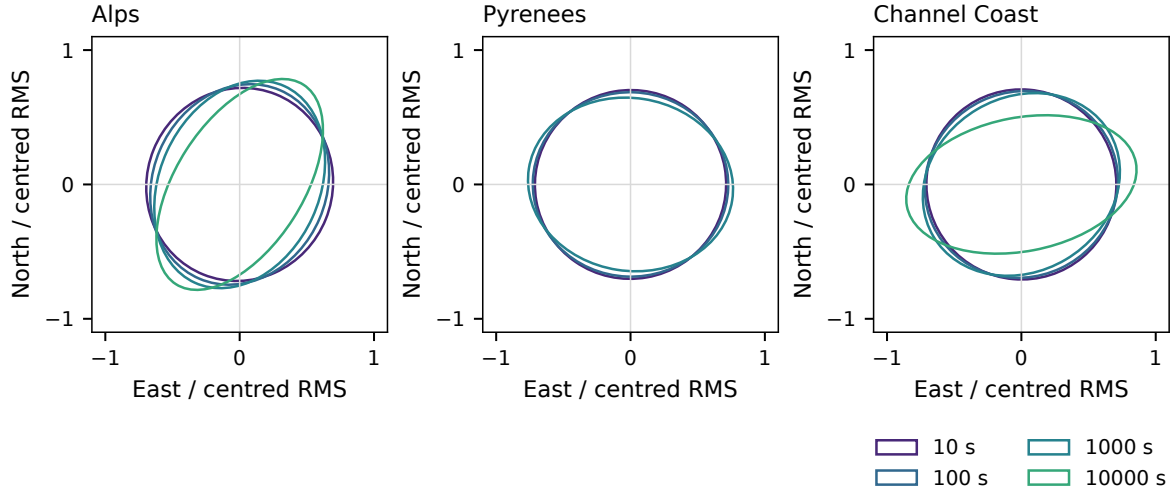


Figure 3.12. Centred second-difference covariance ellipses, normalised by their own trace, on matched-order origins. Colours use the same four physical lags as the regional PCA. Normalisation removes amplitude and the ellipses omit the mean. The Pyrenean 10,000-s ellipse is not shown because only three flights contribute.

to its smallest regional count. Unknown context labels are excluded from this control. The resulting 23 strata retain 65,537, 9,933 and 4,366 flights. Their standardised 1000-s values are respectively 5.13 m s^{-1} , 4.62 m s^{-1} and 3.72 m s^{-1} . The coastal contrast remains in these point estimates; calibrated intervals for the standardised contrasts are not supplied. No three-region common equipment/context strata survive for hang gliders.

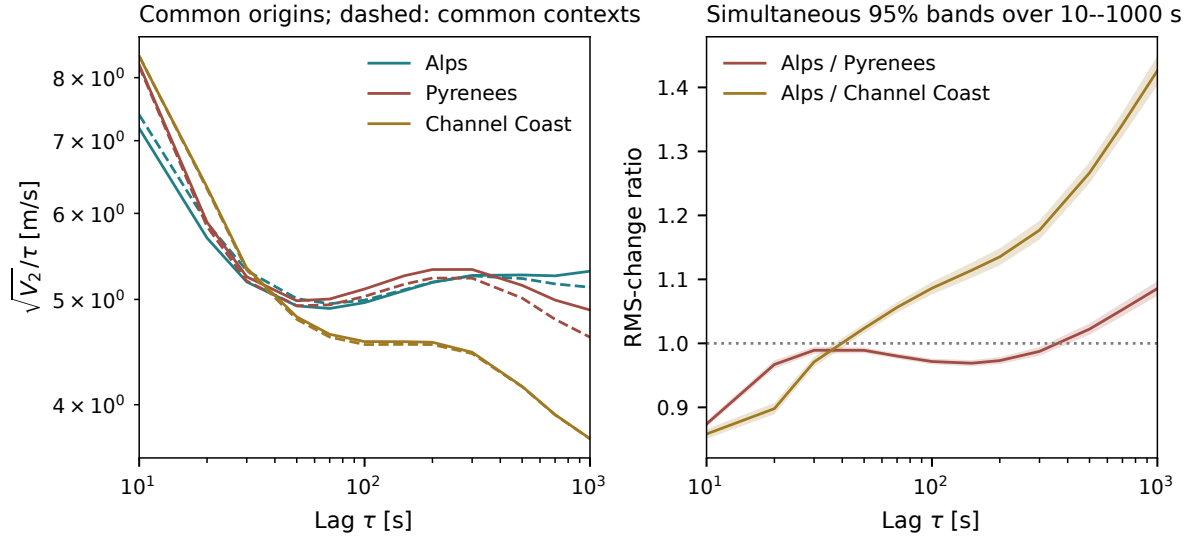


Figure 3.13. Paraglider population controls. Left: common-origin curves, with the common-context standardisation dashed. The dashed curves are descriptive point estimates. Right: regional ratios of the unstandardised common-origin curves; shading is simultaneous over the 18-lag grid's supported subset through 1000 s. These controls do not match meteorology, local terrain exposure or pilot identity.

These findings constrain a wind explanation without identifying wind. If populations differed only by an added constant velocity and otherwise shared the same residual dynamics, their second differences would agree. Their measured differences require more than that particular explanation. Spatial or temporal wind variation, route geometry, manoeuvres and population selection remain possible contributors. The coast has smaller changes of interval-

mean velocity at 1000 s, rather than a larger signal that could simply be labelled stronger wind. An environmental attribution requires independent wind information and terrain alignment, ideally within sites under contrasting conditions.

3.2.5 Third differences and the distribution of changes

The third difference extends the cancellation by one polynomial degree:

$$A_3\mathbf{r} = \mathbf{r}(t + 3\tau) - 3\mathbf{r}(t + 2\tau) + 3\mathbf{r}(t + \tau) - \mathbf{r}(t), \quad V_3 = \mathbb{E}|A_3\mathbf{r}|^2. \quad (3.23)$$

For $\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}t^2$ with constant vector acceleration, $V_2 = |\mathbf{a}|^2\tau^4$ while $V_3 = 0$. A smooth curved trajectory generally retains both signals: locally $A_3\mathbf{r}$ is proportional to $\mathbf{r}^{(3)}\tau^3$. Constant-speed circular motion is therefore not eliminated by taking order three.

Under the same centred stationary-increment second-moment law used in Eq. (4.4), algebra gives

$$V_3 = K(15 - 6 \cdot 2^{2H} + 3^{2H})\tau^{2H}, \quad \frac{V_3}{V_2} = \frac{15 - 6 \cdot 2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1. \quad (3.24)$$

The power law must hold through 3τ . Thus the order ratio, in addition to the two slopes, is a consistency condition on that model. It equals three for Brownian motion plus an arbitrary constant velocity. Agreement between orders would not constitute independent data or certify removal of every environmental trend.

The observed order ratios change with lag (Fig. 3.14). At 100 s they are about 2.29 in both mountain regions and 2.38 on the coast; at 1000 s they are 2.60, 2.79 and 2.42. These curves supply a sensitivity and model-consistency diagnostic, without establishing a scale-independent Hurst parameter. Order three also increases the effect of independent position errors: variance σ^2 per coordinate adds $6\sigma^2$ to the order-two scalar-component moment and $20\sigma^2$ to order three. Correlated errors and smoothing alter those expressions. The 10-s end therefore remains sensitive to acquisition and preprocessing resolution; a calibrated sensor-noise subtraction is not performed here.

The distribution of $|A_2\mathbf{r}|/\tau$ supplies information lost by its second moment. We retain equal-flight histogram probabilities and the squared-change contribution of every bin, including zeros and both tails. Reported quantiles are bin bounds, not interpolated exact sample quantiles. At 1000 s, the largest tenth of changes accounts for approximately 42.7–46.5% of the squared-change signal in the Alps, 39.8–43.7% in the Pyrenees and 38.4–42.8% on the coast. Each range reflects the unresolved boundary bin, not sampling uncertainty. This measures concentration of the observable; it does not identify a flight phase, turbulence or a tail-generating mechanism.

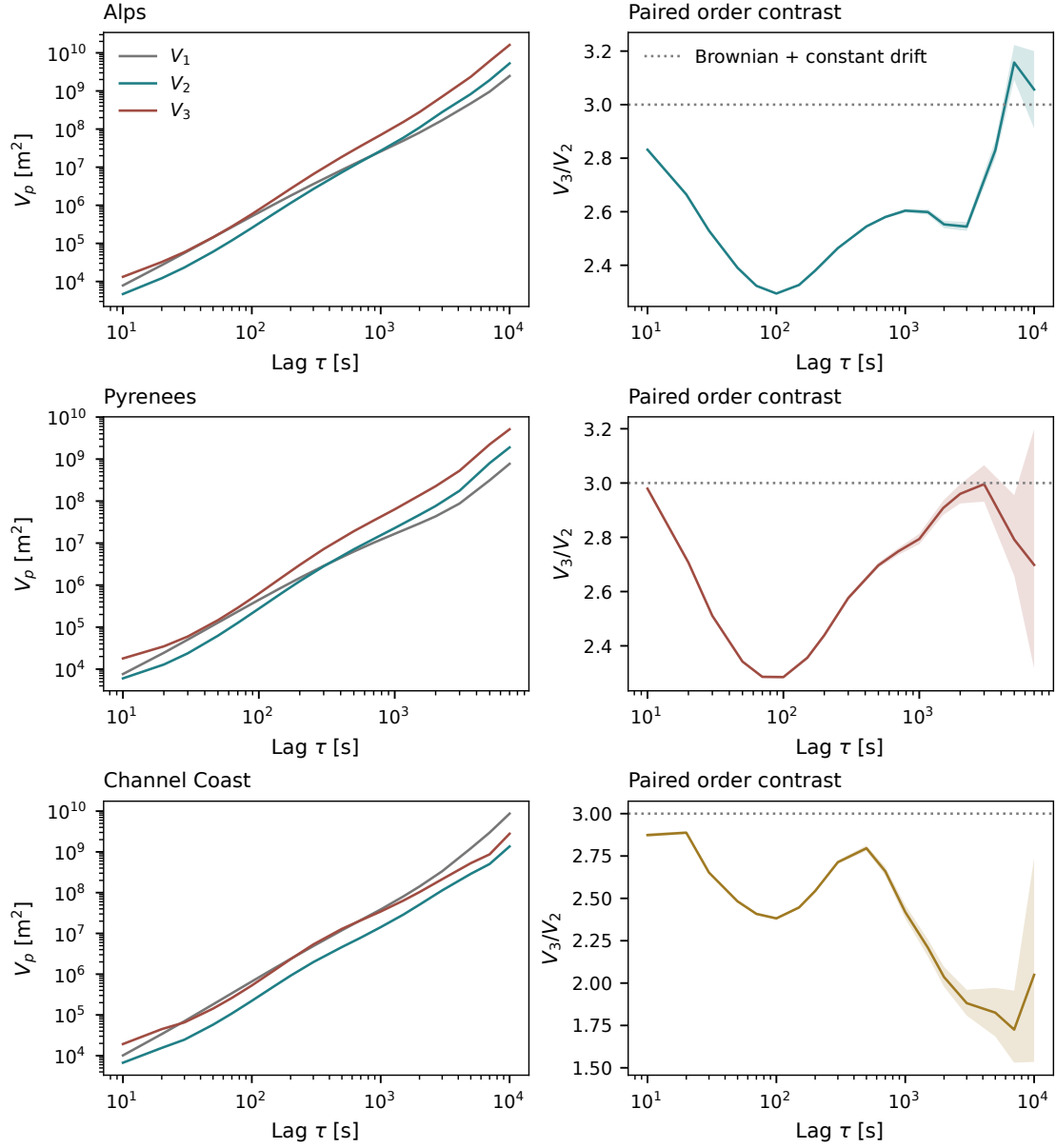


Figure 3.14. Paraglider variations and their paired order ratio on identical origins. Different prefactors make the amplitudes of V_2 and V_3 directly unequal even under one scaling model. Right: V_3/V_2 with pointwise site-day intervals; the dotted line is the Brownian-plus-constant-velocity reference, not a fit. Fewer than eight contributing flights suppress a displayed curve cell.

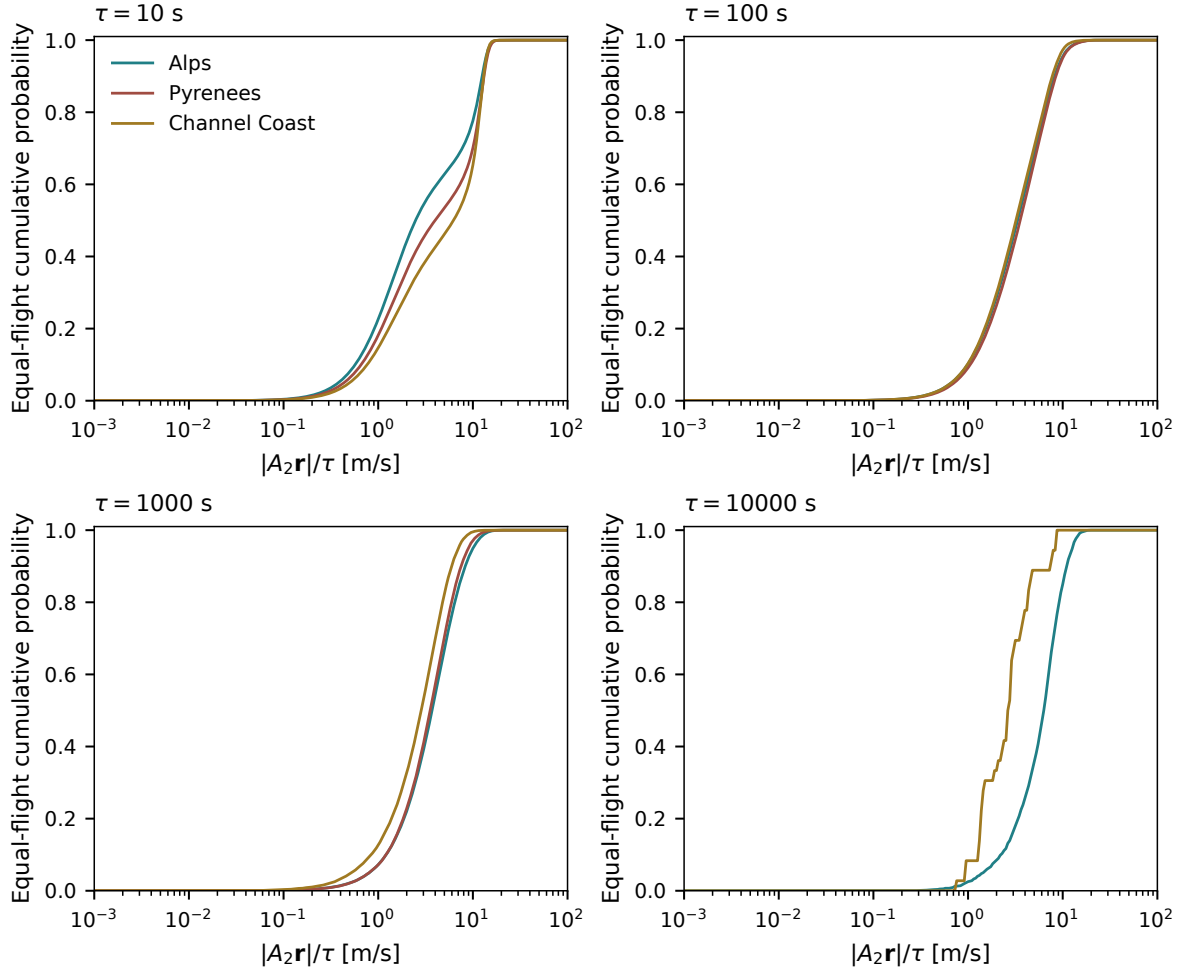


Figure 3.15. Equal-flight distributions of changes of interval-mean velocity at the four PCA display lags, using matched-order support. Curves are accumulated from fixed logarithmic bins; the complete report retains zero and overflow mass. No distribution is drawn for fewer than eight flights. The changing long-lag population must accompany comparisons between panels.

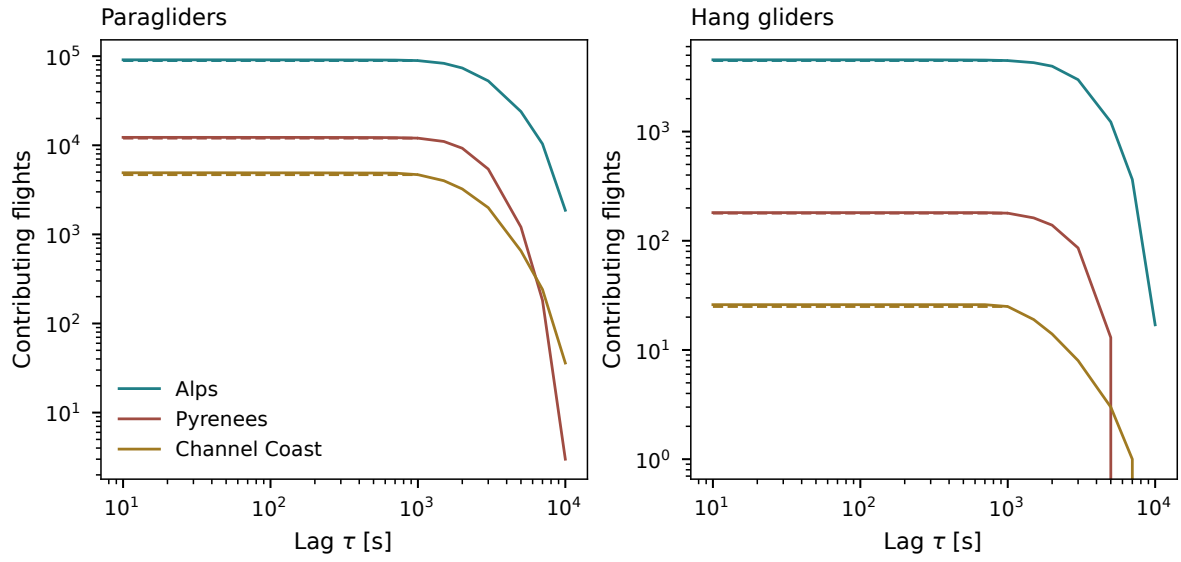


Figure 3.16. Contributing flights for matched orders (solid) and common origins through 1000 s (dashed). A matched observation requires 3τ of continuous support. At 10,000 s the paraglider counts are 1,865, 3 and 36; hang-glider counts are 17, 0 and 0. All regional eligible flights enter where supported.

Chapter 4

Preliminary global transport

Chapter 3 defined displacement growth for the complete archive. This chapter introduces the drift-removing second difference and turns to the distributional structure of displacement: whether displacement quantiles rescale together, how components and joint laws depend on direction, how flight duration and equipment composition shape the result, and what these measurements constrain before flight phases are assigned in Chapter 5. Table 3.1 in Chapter 3 lists the flight populations used throughout both chapters.

A first section removes constant velocity with a second difference and sets route closure against it. Quantiles then test scalar rescaling, with a fixed-population control, before the component and joint laws address vector scaling and dependence, and time-averaged curves are grouped by region and initial altitude. Duration and equipment composition then check how the observed growth depends on the selected flights. A final section states the resulting constraints for the phase analysis in Chapter 5.

4.1 Drift removal and route closure

4.1.1 Removing a constant velocity

A constant velocity contributes a quadratic term to an uncentred MSD. If $\mathbf{r}(t) = \mathbf{v}_d t + \mathbf{Y}(t)$ and the residual increment has zero mean, then

$$\mathbb{E}|\mathbf{X}(t, \tau)|^2 = |\mathbf{v}_d|^2 \tau^2 + \mathbb{E}|\mathbf{Y}(t + \tau) - \mathbf{Y}(t)|^2. \quad (4.1)$$

Without the zero-mean assumption there is also a cross term. The two contributions add as functions of lag; their fitted exponents do not add.

A useful control is the second difference,

$$A_2 \mathbf{r}(t; \tau) = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad (4.2)$$

$$V_2(\tau) = \langle |A_2 \mathbf{r}(t; \tau)|^2 \rangle. \quad (4.3)$$

Since $A_2(\mathbf{r}_0 + \mathbf{v}_d t) = 0$, this observable is exactly insensitive to a constant velocity. It avoids estimating drift from the endpoints, which would impose a return constraint on the residual trajectory. Its cost is a window of length 2τ . We retain it for this specific purpose, alongside $V_1(\tau) = \langle R^2 \rangle$.

To relate the second-difference slope to H , suppose the residual is a centred finite-variance process with stationary increments and $\mathbb{E}|\mathbf{Y}(t + \tau) - \mathbf{Y}(t)|^2 = K\tau^{2H}$. Writing adjacent increments as $\mathbf{I}_1$ and $\mathbf{I}_2$ gives $2\mathbb{E}[\mathbf{I}_1 \cdot \mathbf{I}_2] = K[(2\tau)^{2H} - 2\tau^{2H}]$, hence

$$V_2(\tau) = K(4 - 2^{2H})\tau^{2H}. \quad (4.4)$$

For $K > 0$ and $0 < H < 1$, half the slope of V_2 estimates H under these assumptions; fractional Brownian motion is one example [12]. At $H = 1$, the prefactor vanishes and the logarithmic slope is undefined.

On a smooth curved trajectory, $A_2\mathbf{r} = \ddot{\mathbf{r}}\tau^2 + O(\tau^3)$, so $V_2 \propto \tau^4$ at sufficiently small lag. Half that slope is not the Hurst exponent of a stationary-increment position process. Agreement between differences of order two and three is a consistency check; it does not prove that all environmental trends have disappeared. In particular, $b_1 - b_2$ is not a measured fraction of transport due to wind or course geometry.

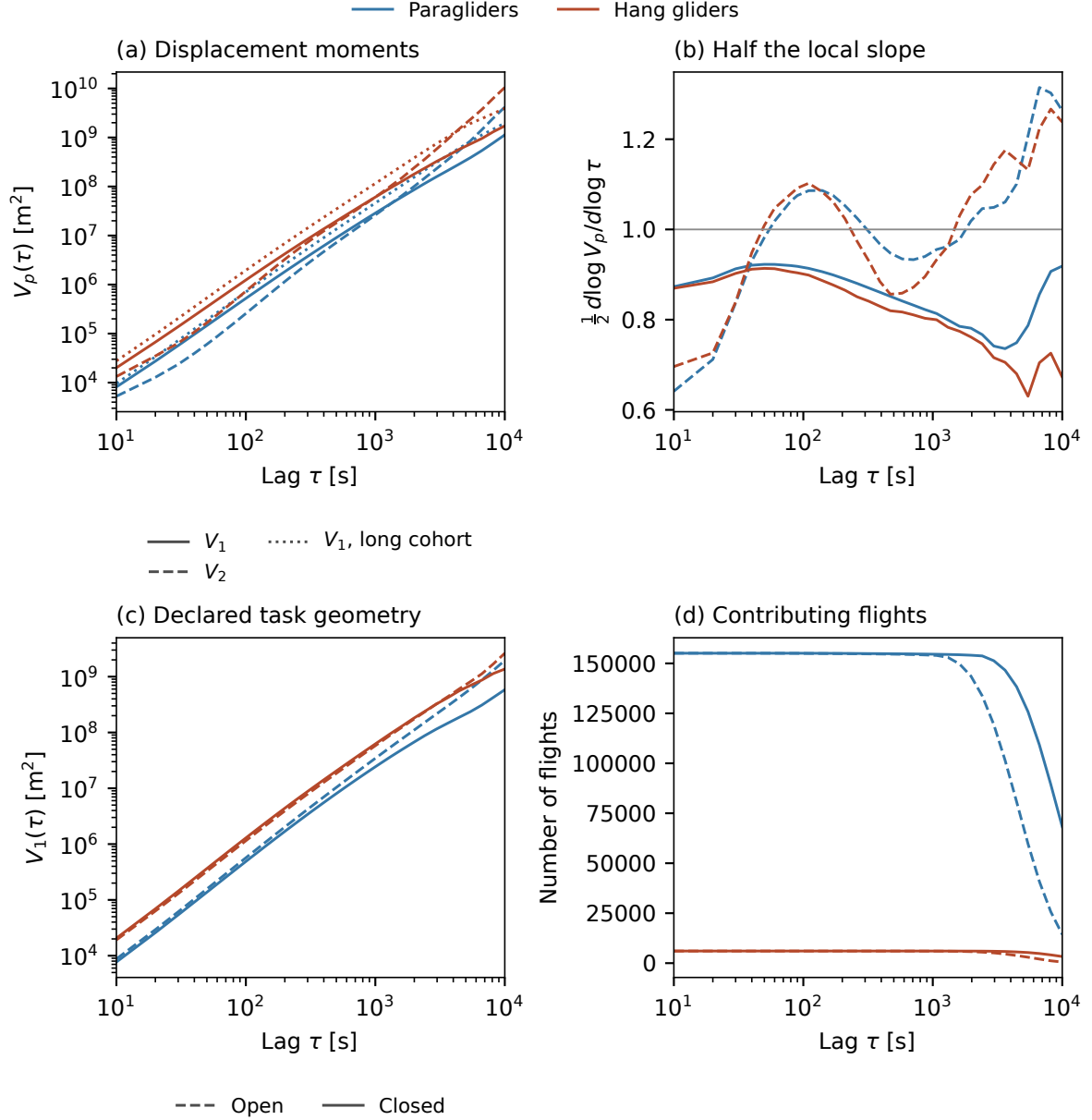


Figure 4.1. Displacement and constant-velocity control over 10s to 10,000s in the eligible archive. The figure separates the second moments from their local slopes and the declared open and closed tasks. Coverage is shown explicitly; the support of a second difference is smaller because it spans 2τ . Slopes refer to this interval and this eligible population.

Both disciplines have $b_1 > 1$ (Table 4.1). Restricting the analysis to flights with a segment of at least 20,000s shifts the slopes: the result depends on the selected population. At 10,000s,

plain increments retain 68,207 paraglider and 3,313 hang-glider flights, with 84,997 and 3,920 non-overlapping windows. Second differences retain only 14,360 and 563 flights because they span 2τ .

Table 4.1. Effective equal-flight slopes on 10 s to 10,000 s. The fixed cohort contains flights with at least one segment of 20,000 s. These are descriptive slopes without calibrated confidence intervals.

Discipline	Population	Flights	b_1	b_2
paraglider	all selected	155085	1.703	2.003
	fixed long cohort	14360	1.790	2.059
hang glider	all selected	6060	1.649	1.985
	fixed long cohort	563	1.744	2.025

The measured b_2 values near or above two, together with the curvature in Fig. 4.1, prevent a direct interpretation as $2H$ for a scale-free stationary-increment residual. The control removes constant velocity but does not yield an intrinsic Hurst exponent here.

A plausible origin of the apparent exponent. The cancellation itself does not impose stationary increments or a power law on the remaining motion. To see what V_2 measures, write the velocities averaged over two consecutive intervals as $\bar{\mathbf{v}}_1 = \mathbf{I}_1/\tau$ and $\bar{\mathbf{v}}_2 = \mathbf{I}_2/\tau$. Then, exactly,

$$D_2(\tau) = \sqrt{\langle |\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2 \rangle}, \quad V_2(\tau) = \tau^2 D_2^2(\tau), \quad b_{2,\text{loc}} = 2 + 2 \frac{d \log D_2}{d \log \tau}.$$

The averages use the same flights, origins and weights as V_2 at each lag. In the saved curves over 10 s to 10,000 s, D_2 varies nonmonotonically between approximately 4.9 m s^{-1} to 7.3 m s^{-1} for paragliders and 7.6 m s^{-1} to 11.5 m s^{-1} for hang gliders: its magnitude changes much less than the three-decade lag range. Thus b_2 near two can arise when the change between interval-mean velocity vectors remains of comparable magnitude across the fitted lags. It does not establish $H = 1$: in Eq. (4.4), that exact boundary case would give $V_2 = 0$. The small excesses above two are descriptive estimates, not calibrated evidence for $H > 1$.

Finite turning and velocity-persistence times provide a physical reason for curvature. Constant-velocity cancellation retains circling, acceleration, changes between glides and climbs, and changes in wind along a trajectory. These motions introduce characteristic times rather than guaranteeing a scale-free residual. At short lags, smoothing and cadence may also affect the observed change; the smooth-trajectory τ^4 limit above is a possible local regime, not an assertion that the measured interval lies in that limit.

Nonstationarity is not required for this interpretation. In a single stationary-increment population, the general identity is $V_2(\tau) = 4M_2(\tau) - M_2(2\tau)$; the power-law expression follows only if the same law holds at both τ and 2τ . For the stationary exponentially correlated velocity example of Eq. (7.11), with persistence time t_p and velocity variance σ_v^2 , this gives

$$V_2(\tau) = 2\sigma_v^2 t_p^2 \left[2 \frac{\tau}{t_p} - 3 + 4e^{-\tau/t_p} - e^{-2\tau/t_p} \right].$$

Its short-lag growth is $4\sigma_v^2 \tau^3/(3t_p)$ and its long-lag growth is $4\sigma_v^2 t_p \tau$. A slope near two can therefore occur in a crossover even though the position increments are stationary and there is no globally self-similar position process. This example establishes a mechanism, not a fit to the flight curves; their bends need not follow one exponential persistence time.

Population selection can compound these physical scale effects. Fixing the long-flight cohort does not bring b_2 below two in either discipline (Table 4.1), so disappearing short flights alone do not explain the result. That control still changes the admitted origins and their within-flight weights with lag. Nor can the empirical V_1 and V_2 curves be inserted into the stationary identity without matching their observations and weights. The plausible working explanation is therefore that the fitted interval combines velocity and route time scales with changing observation support, rather than isolating one self-similar residual. Anisotropy alone would not invalidate Eq. (4.4). To distinguish these explanations, the next checks should hold flights and origins fixed through 2τ , examine the increment law across flight time and phases, and seek a stable scaling interval outside cadence and smoothing effects. The current evidence does not identify one physical cause or prove that no useful restricted-regime H exists.

Reworked in thesis Sections 3.2 and E.2. Original analysis retained below.

4.1.2 Open and closed routes

A declared closed task rewards return towards the starting region. An open task allows continued progress away from it. This difference can change long-lag displacements even if speed and short-time behaviour are similar. Task classes are compared separately in Figure 4.1; a declaration is not a measurement that a particular retained segment closes exactly.

A circular trajectory makes the distinction precise. For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ with period $T_c = 2\pi/\omega$,

$$V_1(\tau) = 4a^2 \sin^2\left(\frac{\pi\tau}{T_c}\right), \quad V_2(\tau) = 16a^2 \sin^4\left(\frac{\pi\tau}{T_c}\right). \quad (4.5)$$

The displacement initially grows, then decreases as the lag approaches a complete return. The second difference also retains this geometry. Removing constant drift cannot remove closure, a turn, or a spatially varying forcing.

The selected paraglider tasks include 67,003 open and 87,807 closed flights; hang gliders contribute 2,183 and 3,877. Their effective b_1 values are respectively 1.770 versus 1.623, and 1.688 versus 1.628. At order two the ordering reverses: 1.937 versus 2.036 and 1.892 versus 2.007. This is consistent with different route geometry affecting the two observables differently; it is not a quantitative decomposition into drift and fluctuations.

A complementary descriptor compares net displacement with the path actually observed on the common grid. Let t_f^- and t_f^+ be the first and last retained grid times, and let L_f^{obs} sum only the within-segment chords:

$$L_f^{\text{obs}} = \sum_{s \in S_f} \sum_{j=0}^{N_{fs}-2} |\mathbf{r}_{fs,j+1} - \mathbf{r}_{fs,j}|, \quad \chi_f^{\text{obs}} = \frac{|\mathbf{r}_f(t_f^+) - \mathbf{r}_f(t_f^-)|}{L_f^{\text{obs}}}, \quad L_f^{\text{obs}} > 0. \quad (4.6)$$

For one uninterrupted observed path, the triangle inequality gives $0 \leq \chi_f^{\text{obs}} \leq 1$. Across separated segments, the denominator omits unobserved travel while the numerator spans it, so this bound need not hold. Reacquisition offsets can also affect the endpoint displacement. The full collector contains 181 paraglider and two hang-glider values above one, all among flights with multiple eligible segments. This is an observed-path descriptor, not a reconstruction of the complete distance flown.

In the eligible archive, median χ_f^{obs} is 0.365 for open paraglider tasks and 0.027 for closed ones; hang-glider medians are 0.177 and 0.026. These medians describe the endpoint geometry of retained observations, with the gap limitation above. Restricting to flights with one eligible observed segment gives closely similar open/closed medians: 0.364/0.027 in paragliders and 0.177/0.026 in hang gliders. This sensitivity check preserves the descriptive ordering; one

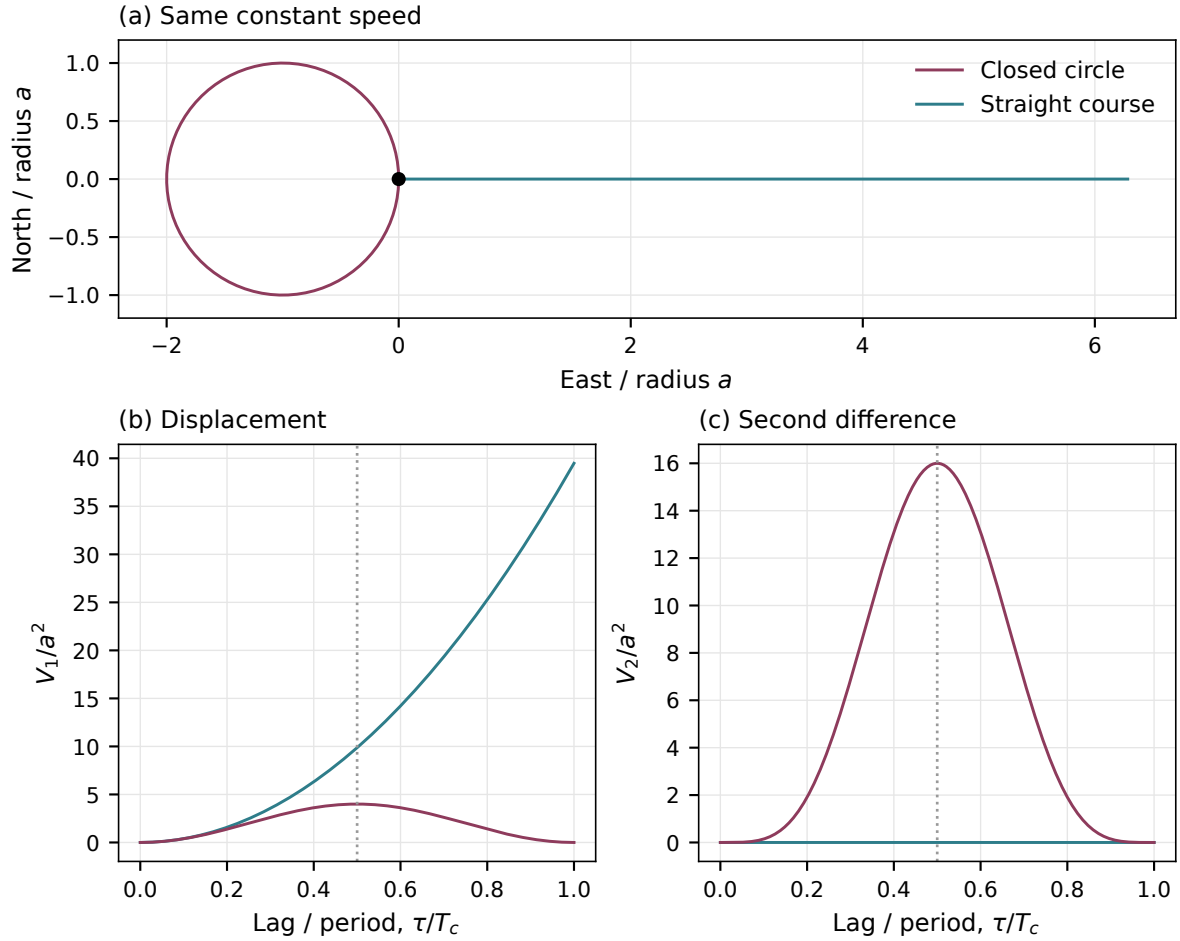


Figure 4.2. Analytical straight and circular paths at the same speed. The circle is continued periodically to evaluate second differences. Both V_1 and V_2 vanish after a complete period: constant-velocity removal preserves the effect of route closure.

eligible segment need not cover the complete original flight. For an uninterrupted observed path, small χ_f^{obs} means little net progress relative to its length. None of the increment statistics crosses a segment boundary.

Matching task groups by duration and region, and comparing both physical lag and τ/T_f , would help separate route geometry from population composition.

4.2 Quantiles and self-similarity

4.2.1 The scaling prediction

For an H -self-similar process starting at the origin, $(\mathbf{r}(ct_1), \dots, \mathbf{r}(ct_n)) \stackrel{d}{=} c^H(\mathbf{r}(t_1), \dots, \mathbf{r}(t_n))$ for every $c > 0$ and every finite choice of times. Self-similarity therefore constrains joint distributions, beyond the MSD alone. With stationary increments, it implies the one-lag prediction

$$\mathbf{X}(t, \tau) \stackrel{d}{=} (\tau/\tau_0)^H \mathbf{X}(t, \tau_0). \quad (4.7)$$

Writing $a_\tau = (\tau/\tau_0)^H$, and assuming the corresponding densities exist, the two-dimensional and radial laws obey different Jacobians:

$$p_{\mathbf{X}}(\mathbf{x}; \tau) = a_\tau^{-2} F(\mathbf{x}/a_\tau), \quad p_R(r; \tau) = a_\tau^{-1} f(r/a_\tau). \quad (4.8)$$

The radial law follows by integrating the joint density over angle, including the polar Jacobian r . It does not require isotropy. In particular, a two-dimensional density at the origin would scale as a_τ^{-2} , not as a_τ^{-1} . These are consequences of the proposed vector scale family, not statements that the measured laws already satisfy it. A shared scale factor does not require the radial and component densities to have the same shape.

East and north scaling. Projecting Eq. (4.7) onto either geographic axis gives the signed component laws

$$X_E(t, \tau) \stackrel{d}{=} a_\tau X_E(t, \tau_0), \quad X_N(t, \tau) \stackrel{d}{=} a_\tau X_N(t, \tau_0). \quad (4.9)$$

Their one-dimensional densities therefore satisfy

$$p_E(x; \tau) = a_\tau^{-1} f_E(x/a_\tau), \quad p_N(y; \tau) = a_\tau^{-1} f_N(y/a_\tau), \quad (4.10)$$

where $f_E(u) = \int F(u, v) dv$ and $f_N(v) = \int F(u, v) du$ are marginals of the reference joint density in Eq. (4.8). Neither independence nor isotropy is required: the same exponent H does not require $f_E = f_N$.

The component quantile analysis below uses the magnitudes $Y_j = |X_j|$, for $j \in \{E, N\}$. These obey

$$|X_j(t, \tau)| \stackrel{d}{=} a_\tau |X_j(t, \tau_0)|, \quad Q_{|X_j|, p}(\tau) = a_\tau Q_{|X_j|, p}(\tau_0). \quad (4.11)$$

For $y > 0$, their densities are $p_{|X_j|}(y; \tau) = p_j(y; \tau) + p_j(-y; \tau)$ and have the same one-dimensional factor a_τ^{-1} . A log-log quantile slope is defined only for positive finite quantiles; this qualification matters for absolute components and prevents applying that fit directly to negative or zero signed quantiles.

If scaling is assumed separately for the two marginals, one may instead write $X_j(t, \tau) \stackrel{d}{=} (\tau/\tau_0)^{H_j} X_j(t, \tau_0)$ with distinct H_E and H_N . This is a weaker marginal hypothesis: even equal exponents would not establish the vector law without its component dependence (Sec. 4.3).

Define the radial quantile at probability p by

$$Q_p(\tau) = \inf\{r : \Pr[R(t, \tau) \leq r] \geq p\}, \quad 0 < p < 1. \quad (4.12)$$

Equation (4.7) implies

$$Q_p(\tau) = a_\tau Q_p(\tau_0), \quad H_p = \frac{d \log Q_p}{d \log \tau} = H \quad \text{when } 0 < Q_p(\tau_0) < \infty. \quad (4.13)$$

The four probabilities $p = .25, .50, .75, .90$ describe small, typical and larger displacements. Quantiles use ranks, complementing moments without requiring a finite fourth moment. Common quantile slopes are necessary for the proposed one-lag scaling law; they do not establish self-similarity across several times.

4.2.2 A fixed population across lags

At each lag, all percentiles belong to the same displacement distribution. They do not label fixed groups of slow or fast flights. A separate sampling effect arises when that distribution draws on different flights or weights at different lags.

If flight f supplies $n_f(\tau)$ admissible increments, pooling them defines

$$\hat{F}_\tau^{\text{pool}}(x) = \sum_{f \in \mathcal{F}_\tau} \frac{n_f(\tau)}{\sum_g n_g(\tau)} \hat{F}_{f, \tau}(x), \quad \hat{F}_{f, \tau}(x) = \frac{1}{n_f(\tau)} \sum_k 1\{R_{fk}(\tau) \leq x\}. \quad (4.14)$$

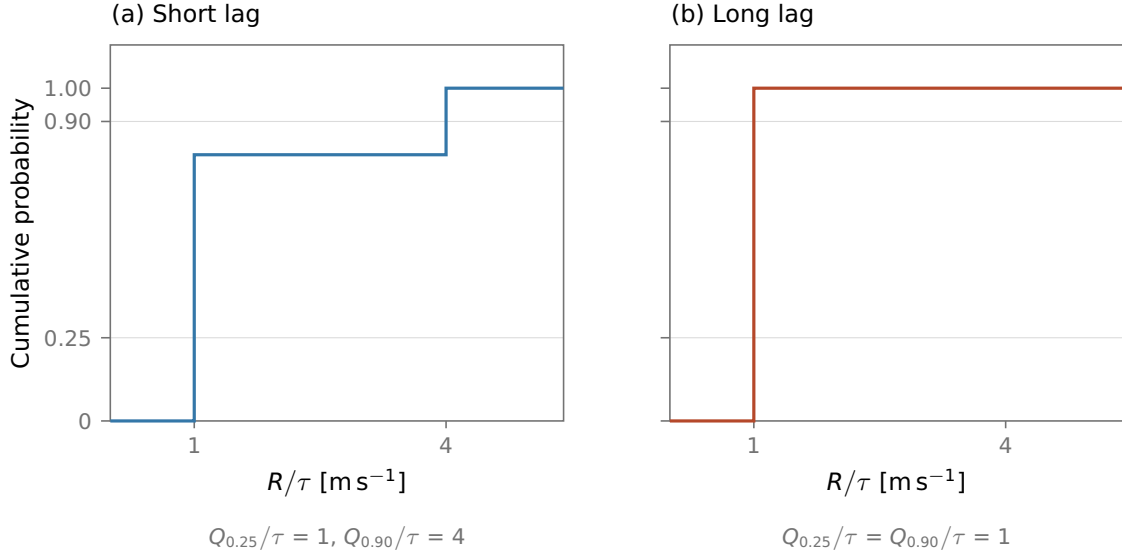


Figure 4.3. A sampling effect can imitate different quantile growth rates. Consider two straight flights at 1 m s^{-1} lasting 20,000 s each and two at 4 m s^{-1} lasting 5,000 s each. At a 10-s lag, the fast flights supply 1/5 of the non-overlapping windows, so the 90th percentile of R/τ is 4. At 10,000 s they supply none, and that percentile is 1. Every flight is exactly ballistic, yet the pooled upper quantile grows more slowly than τ between these lags. A fixed cohort of the two long flights has $Q_p(\tau) = (1 \text{ m s}^{-1})\tau$ at every displayed probability level. The example is constructed, not measured from the archive.

As lag grows, short flights disappear and window counts change. Both membership $\mathcal{F}_\tau$ and mixture weights therefore vary. Quantiles depend nonlinearly on these weights: averaging flight-specific quantiles would give a different quantity. Figure 4.3 illustrates the resulting bias.

The analysis therefore fixes the sampled population once and keeps the same flights, weights and time origins at every lag. A flight joins the cohort $\mathcal{F}_*$ when it has at least one retained segment of 20,000 s. That threshold is an operational choice; the minimum duration mathematically required to observe a 10,000 s increment is shorter. It supplies at least 1,001 common 10-s origins in one qualifying segment. Other eligible segments of these same flights also contribute when they support the required interval, so the cohort fixes flights rather than only their longest segments. Its conclusions remain conditional on long flights.

Let $\tau_{\max} = 10,000 \text{ s}$ and keep every starting time t_{fk} on the 10-s grid whose complete interval $[t_{fk}, t_{fk} + \tau_{\max}]$ lies within one retained segment. Then

$$\hat{F}_\tau^{\text{fixed}}(x) = \frac{1}{|\mathcal{F}_*|} \sum_{f \in \mathcal{F}_*} \frac{1}{n_f^*} \sum_{k=1}^{n_f^*} 1\{R_f(t_{fk}, \tau) \leq x\}, \quad \hat{Q}_p(\tau) = \inf\{x : \hat{F}_\tau(x) \geq p\}. \quad (4.15)$$

Flights, origins and weights are now independent of τ , so quantiles measured at different lags describe one sampled law. The convention inverts the empirical CDF without interpolating between order statistics. The common origins overlap at most lags; they preserve the sampled population without making its increments independent. The cohort contains 14360 paraglider and 563 hang-glider flights, retaining respectively 21,739,115 and 791,712 origins at every lag. These overlapping origins are not independent replicates.

Table 4.2. Radial quantile slopes in the fixed population of Eq. (4.15). Within each discipline the lag set is the one used also for the component quantiles: 10 s to 10,000 s for paragliders and 10 s to 10,000 s for hang gliders. These are descriptive slopes, without calibrated confidence intervals; coverage of the target interval does not establish that a single power describes it.

Percentile	Paragliders	Hang gliders
25	0.976	1.000
50	0.899	0.895
75	0.890	0.863
90	0.898	0.868

The four slopes differ, and they are not ordered monotonically in p (Table 4.2). The 25th-percentile slope is the largest in both disciplines, while $H_{.90} > H_{.75}$.

This contrast cannot come from membership, weights or origins changing with lag, since all three are held fixed. It can reflect phase changes, route geometry, environment and sampling uncertainty within the selected long flights. The roughly five-and-a-half-hour segment requirement limits this conclusion to that population.

4.2.3 Relative spread

What a self-similar process requires of these quantiles. Throughout this chapter τ is a *lag*: the separation between two instants of the same flight in the increment $\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t)$ of Eq. (3.1), with the origin t taken inside a retained segment. It is never the time elapsed since take-off, and every flight supplies many origins t at each τ . Only Fig. 3.1a uses elapsed time, and it is labelled as such.

Let the process be H -self-similar with stationary increments. The law of $R(t, \tau) = |\mathbf{X}_f(t, \tau)|$ is then independent of the origin t and obeys

$$R(t, \tau) \stackrel{d}{=} (\tau/\tau_0)^H R(t, \tau_0), \quad (4.16)$$

where $\stackrel{d}{=}$ denotes equality in distribution. Stationarity of the increments is what allows the pooling over origins; self-similarity alone constrains displacement from the start of the process rather than from an arbitrary t . Quantiles are equivariant under multiplication by a positive constant, so Eq. (4.16) gives $Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0)$ at every probability p , with the same exponent H . The common factor then cancels in any ratio of two quantiles:

$$\frac{Q_{0.90}(\tau)}{Q_{0.25}(\tau)} = \frac{Q_{0.90}(\tau_0)}{Q_{0.25}(\tau_0)} \quad \text{for every } \tau. \quad (4.17)$$

A ratio that varies with lag therefore contradicts Eq. (4.16) for every choice of H , since the ratio is unchanged by any rescaling of R . This is a necessary condition, and a collapse of rescaled distributions cannot restore it.

The test is blind to the exponent. For isotropic Gaussian increments the radius is Rayleigh distributed and Eq. (4.17) evaluates to $\sqrt{\ln 0.10 / \ln 0.75} \simeq 2.83$ at every lag, for fractional Brownian motion of any H as much as for Brownian motion. A constant ratio constrains the shape of the law; the growth exponent has to be measured separately.

Reading a changing ratio. The interpretation of an ordering such as $H_{0.25} > H_{0.90}$ follows directly from

$$\frac{d}{d \log \tau} \log \left(\frac{Q_{0.90}(\tau)}{Q_{0.25}(\tau)} \right) = H_{0.90}(\tau) - H_{0.25}(\tau), \quad (4.18)$$

which vanishes identically under Eq. (4.17). A negative right-hand side means that the ratio of large to small displacements decreases. Both quantiles may increase in absolute distance. The lower one increases faster in *proportion*, so the distribution becomes less spread out by this relative measure. For example, a slope difference of -0.10 maintained over a tenfold increase of lag multiplies the ratio by $10^{-0.10} \simeq 0.79$.

On a common lag grid, the fitted slope of $\log(Q_{.90}/Q_{.25})$ also equals the difference of the fitted quantile slopes, by linearity of least squares. Its sign need not imply monotonic change. In the controlled population, the paraglider ratio rises from 1.97 at 10 s to a sampled maximum of 4.20 at 90 s; the hang-glider ratio rises from 2.27 to 5.52 at 80 s. After these early maxima, the ratios fall to local minima of 2.23 at 2,410 s and 2.32 at 1,970 s, respectively. They then rise to 3.01 and 2.60 at 10,000 s. The negative fitted contrast summarises this curved evolution; it does not describe a monotone narrowing over the full interval.

This ratio measures relative quantile spread, leaving the outer 10% and local density shape unspecified. Observations can also change rank with lag.

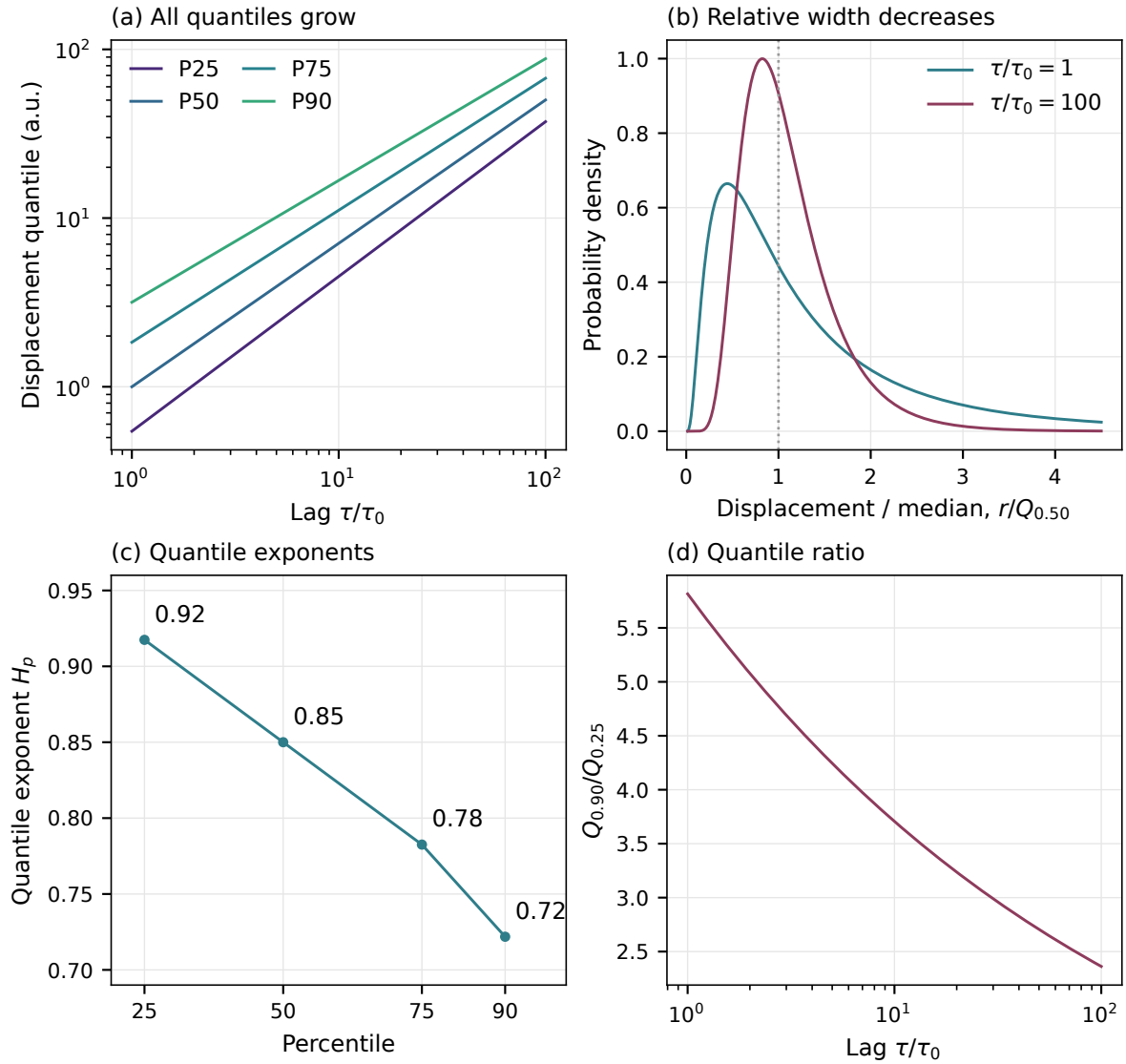


Figure 4.4. Analytical illustration of decreasing quantile exponents, not a fit to flight data. Let $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$, where Z is standard Gaussian and $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$ on the illustrated interval. Then $H_p = 0.85 - 0.1\Phi^{-1}(p)$. All four quantiles grow, but the distribution narrows after division by its median. The declining ratio makes that change explicit.

Rank-dependent growth in the controlled population challenges a common rescaling exponent. Section 4.3.2 measures the moment spectrum on the same fixed population and finds it curved in the direction these rank exponents predict. A model must reproduce the quantile curves and their shape changes as well as their average growth.

4.3 Vector scaling and dependence

The radial quantiles describe displacement magnitude. We now ask whether the same rescaling describes the signed east–north vector, including its dependence structure. The population and origin controls of Sec. 4.2.2 remain fixed throughout this comparison. An anisotropic distribution can obey a common scalar rescaling; isotropy is not an assumption of the test.

4.3.1 Testing east, north and radial scaling

The absolute east and north components and the radius change shape over 10 s to 10,000 s, even in the fixed population. To interpret their comparison, first consider two ways marginal scaling can fail to determine vector scaling.

Component exponents can differ. For centred independent Gaussian components with variances $a\tau^{2H_E}$ and $b\tau^{2H_N}$,

$$\mathbb{E}R^2 = a\tau^{2H_E} + b\tau^{2H_N}, \quad \frac{1}{2} \frac{d \log \mathbb{E}R^2}{d \log \tau} = \frac{H_E a \tau^{2H_E} + H_N b \tau^{2H_N}}{a \tau^{2H_E} + b \tau^{2H_N}}. \quad (4.19)$$

Each Gaussian marginal scales exactly, yet the radial effective exponent varies with lag. No cross-correlation is needed.

Equal marginal exponents also leave component dependence undetermined. For example, the equiprobable pairs

$$\{(1, 2), (-1, -2)\} \quad \text{and} \quad \{(1, -2), (-1, 2)\} \quad (4.20)$$

have identical signed marginals and radius $\sqrt{5}$, but opposite cross-covariance. Thus even matching both marginals and the radial law does not establish a common joint law. The Cramér–Wold characterization uses all signed linear projections [13]. Conversely, scalar scaling of the joint vector always implies radial scaling, including for anisotropic laws. Operator self-similarity, with a matrix exponent, is a broader hypothesis [14].

Paired component and radial quantiles. On the fixed observations of Sec. 4.2.2, compare

$$Y_E = |X_E|, \quad Y_N = |X_N|, \quad Y_R = R = \sqrt{X_E^2 + X_N^2}. \quad (4.21)$$

These are magnitudes of displacement over a lag. Absolute values discard direction; R measures the distance between the endpoints, rather than the path flown between them.

Every curve uses fixed flights and 10-s origins, equal total weight per flight, and no drift subtraction. The four probabilities give twelve slopes $H_{j,p}$ per discipline, for $j \in \{E, N, R\}$. The population contains 14360 paraglider and 563 hang-glider flights from the retained snapshot (cleaning version 2.3.0).

We bootstrap whole flights with replacement 400 times. Each resample keeps all origins of a flight together, using its multiplicity at every lag and for all three quantities. Recomputed mixture quantiles preserve the pairing in contrasts such as $H_{N,p} - H_{E,p}$ and $H_{j,.90} - H_{j,.25}$. The pointwise 95% percentile intervals assume independent flights within this fixed population. Shared site–day dependence and simultaneous coverage are not accounted for.

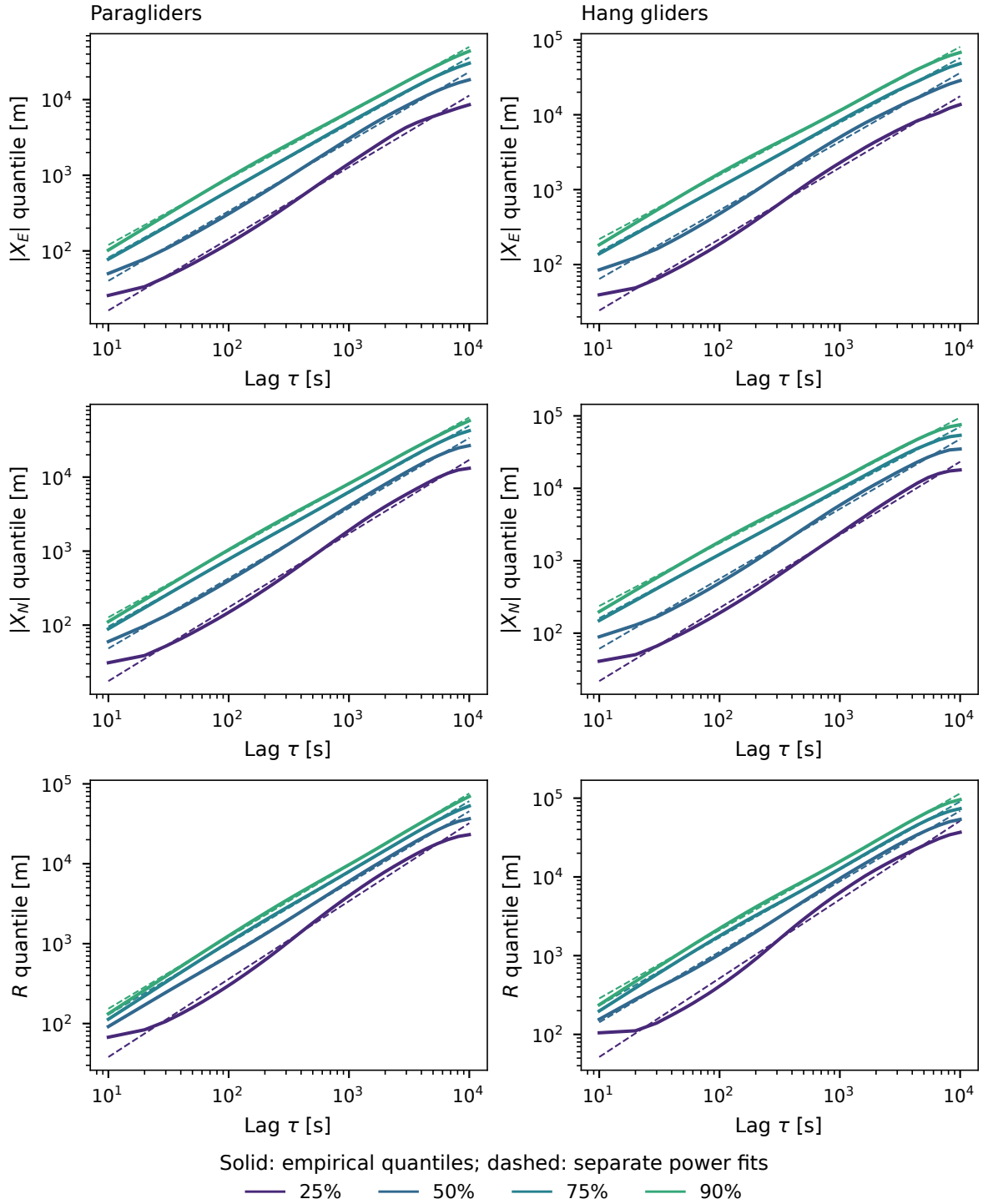


Figure 4.5. Four quantiles for each absolute component and the radius in the fully controlled population. Solid curves are empirical weighted quantiles; dashed lines are their separate log-log fits over 10 s to 10,000 s. Columns distinguish disciplines and rows distinguish quantities. A visibly curved quantile can have a well-defined regression slope without following a single power law.

Table 4.3. Separate effective quantile exponents over 10 s to 10,000 s, using fixed flights, origins and weights. The exponents refer to magnitudes in metres; the corresponding squared-displacement slopes are exactly twice these values.

Discipline	Quantity	$H_{.25}$	$H_{.50}$	$H_{.75}$	$H_{.90}$
Paragliders	$ X_E $	0.947	0.921	0.882	0.873
	$ X_N $	0.996	0.948	0.904	0.900
	R	0.976	0.899	0.890	0.898
Hang gliders	$ X_E $	0.954	0.917	0.862	0.855
	$ X_N $	1.010	0.965	0.881	0.867
	R	1.000	0.895	0.863	0.868

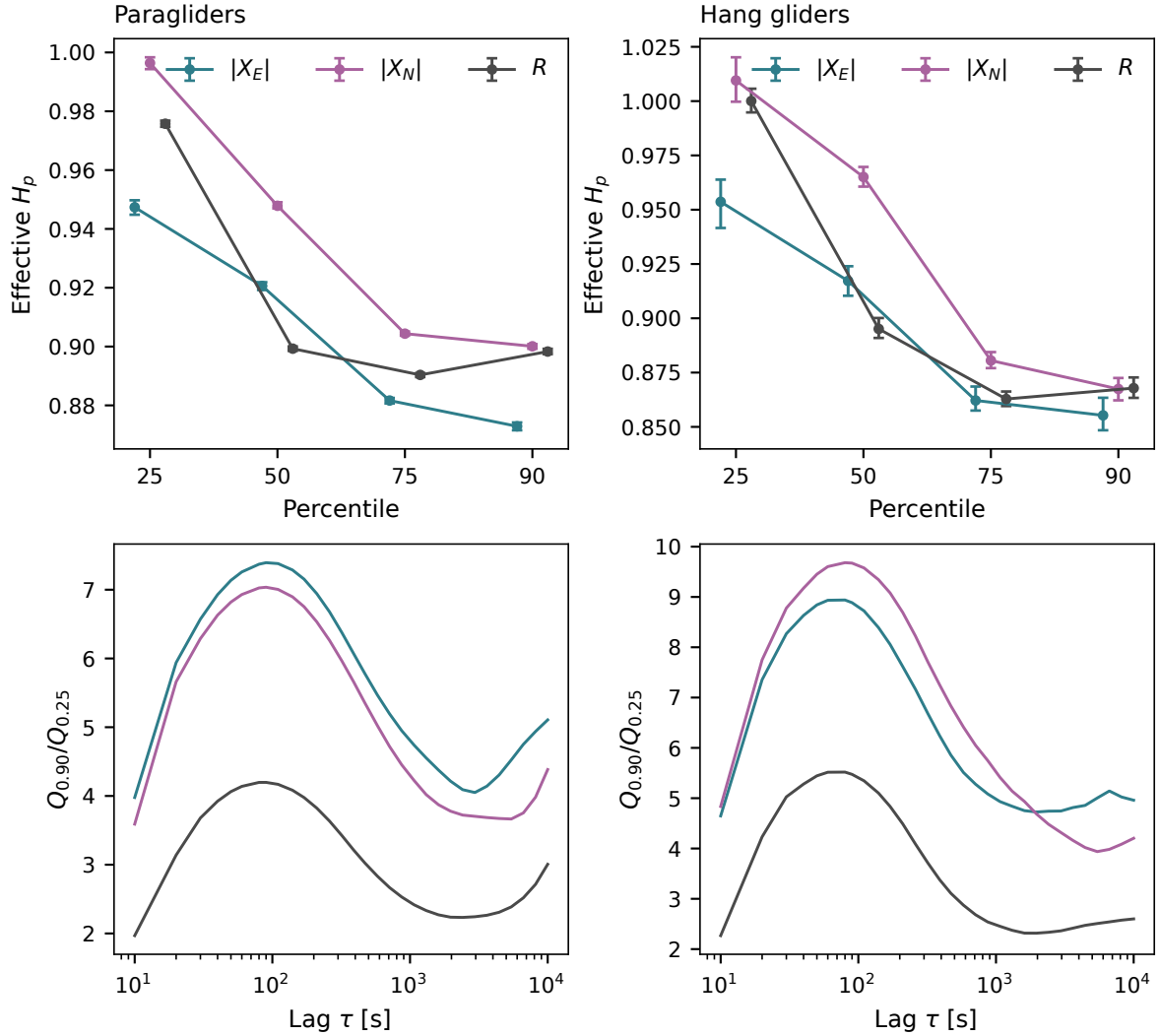


Figure 4.6. Top: the twelve separate magnitude exponents per discipline, with pointwise 95% whole-flight bootstrap intervals. Small horizontal offsets distinguish quantities at the same percentile. Middle: the corresponding ratios $Q_{.90}/Q_{.25}$ for east, north and radius. Ratio changes test distribution shape without choosing a rescaling exponent. Bottom: excess kurtosis of $|X_E|$, $|X_N|$ and R at every lag, with pointwise 95% whole-flight bootstrap intervals; the horizontal line marks the Gaussian value zero. All curves use the fixed population; interval assumptions are stated in the text.

The lower-quartile slope exceeds the 90th-percentile slope for east, north and radius in both disciplines. Two paired contrasts summarize the differences:

Contrast	Paragliders	Hang gliders
$H_{R,.90} - H_{R,.25}$	-0.077	-0.132
95% interval	$[-0.079, -0.076]$	$[-0.139, -0.124]$
$H_{N,.50} - H_{E,.50}$	0.027	0.048
95% interval	$[0.026, 0.029]$	$[0.040, 0.054]$

Intervals are conditional on the fixed population and the independence assumption above. Figure 4.6 shows the non-monotone spread ratios. Both contrasts are fitted over 10 s to 10,000 s, where the lower-quartile curve is poorly described by a power law. The comparison with a restricted reference window is awaiting regenerated fixed-population estimates and their paired bootstrap intervals.

In both disciplines, north–east contrasts are positive at all four ranks, with their paired pointwise intervals excluding zero. The radial rank contrasts are negative, also with intervals excluding zero. These comparisons support rank- and direction-dependent effective growth within the fixed long-flight population under the stated resampling assumptions. They do not establish a single power law at each rank. The radial median slope lies below both component median slopes: different observations can occupy the component and radial ranks.

4.3.2 Moment spectrum

The quantile exponents above describe fixed ranks of the displacement law. Moments weigh the whole law at once, with a weight set by the order, so the two families read the same increments through different lenses and constrain a model jointly.

On the fixed population of Sec. 4.2.2, with F fixed flights, flight f contributing the origin set $\mathcal{O}_f$ of size n_f , and every flight carrying equal total weight, define

$$M_q(\tau) = \frac{1}{F} \sum_{f=1}^F \frac{1}{n_f} \sum_{t \in \mathcal{O}_f} R_f(t, \tau)^q, \quad q > 0, \quad (4.22)$$

and the exponents

$$\zeta(q) = \frac{d \log M_q(\tau)}{d \log \tau}, \quad \nu(q) = \frac{\zeta(q)}{q}, \quad (4.23)$$

both fitted by least squares in $\log \tau$ over the window fixed below. Under Eq. (4.16) the whole law rescales by $(\tau/\tau_0)^H$, so $M_q(\tau) \propto \tau^{qH}$ and $\nu(q) = H$ at every order. Dividing by q is what puts the spectrum on the same footing as the quantile exponents H_p : both measure $d \log(\text{length})/d \log \tau$.

An exact identity links the two families. Writing the moment as an integral over ranks and substituting the fitted quantile laws $Q_p(\tau) = A_p \tau^{H_p}$,

$$\begin{aligned} M_q(\tau) &= \int_0^1 Q_p(\tau)^q dp = \int_0^1 A_p^q \tau^{qH_p} dp, \\ \nu(q) &= \int_0^1 H_p w_p(q, \tau) dp, \quad w_p(q, \tau) = \frac{Q_p(\tau)^q}{\int_0^1 Q_{p'}(\tau)^q dp'}. \end{aligned} \quad (4.24)$$

So $\nu(q)$ is a weighted mean of the rank exponents whose weight grows with the size of the displacement. As $q \rightarrow 0$ the weight becomes uniform and ν tends to the plain average of H_p ; as q grows the weight concentrates on the largest displacements and ν tends to the exponent of the upper ranks. A spectrum flat in q is what one rescaling exponent predicts. A spectrum

falling with q says the upper ranks grow more slowly than the lower ones, which is the same statement Eq. (4.18) makes without any fit.

Equation (4.24) also sets the sensitivity of the test. A moment of positive order gives less weight to small displacements, so a flat spectrum alone is weak evidence for a common exponent across all ranks.

Population. Equation (4.22) inherits the requirements of Sec. 4.2.2 unchanged. A moment pooled over all eligible flights weights every lag by the population that still supplies windows there, so its log-log slope mixes displacement growth with a change of membership. That pooled spectrum gives ν between 0.843 and 0.883 in paragliders and between 0.832 and 0.854 in hang gliders, rising with q in both. It is reported here only as the uncontrolled comparison, for the same reason Table 4.2 gives for the quantiles.

Reference window and measured spectrum: pending results. The fixed-population moment spectrum, the comparison of fitting windows and the paired bootstrap intervals require regeneration of the numerical report. The two tables and spectrum figure are therefore not included in this version. Quantitative claims about the spectrum's curvature or an exclusion of the Lévy-walk benchmark are deferred until those results are available. The definitions and available quantile measurements remain applicable.

4.3.3 Distribution collapse

To test a scale family, fit one exponent $H_{j,*}$ per quantity, shared by all four ranks, with separate intercepts $b_{j,p}$, by minimising

$$\sum_{p \in \{.25, .50, .75, .90\}} \sum_{\tau \in \mathcal{L}} \left[\log \frac{Q_{j,p}(\tau)}{1 \text{ m}} - b_{j,p} - H_{j,*} \log \frac{\tau}{\tau_0} \right]^2, \quad \tau_0 = 1,000 \text{ s.} \quad (4.25)$$

Ranks and lags have equal weight. On the common grid, $H_{j,*}$ is the mean of the four separate slopes. The reference time fixes the intercept without changing the exponent.

With $a_j(\tau) = (\tau/\tau_0)^{H_{j,*}}$, a density collapse plots

$$u = y/a_j(\tau), \quad a_j(\tau) p_{Y_j}(a_j(\tau)u; \tau). \quad (4.26)$$

The fitted exponent rescales both axes at every lag, without tuning the histogram overlay. Densities use probability mass divided by bin width over the full positive support; any exact zero is stored separately as an atom. Quantiles and empirical-CDF distances use the observations directly, independently of binning.

The full positive support can approach numerical zero. Summing all bins that intersect $[0, 1 \text{ mm})$, together with the zero atom, bounds the probability in that interval below 3×10^{-5} in every displayed law. Very narrow bins can therefore have a high density while carrying little probability. These values do not establish sub-millimetre physical resolution; the median-normalised CDF panels below make the central probability structure easier to compare than the full-range logarithmic densities.

To distinguish an imperfect power fit from a failure of any single scale factor, we also plot $Y_j/Q_{j,.50}(\tau)$. A shape-preserving scale family must collapse under this normalisation even if its scale is not a power of lag. We measure separation by

$$D = \max_{\tau, \tau' \in \mathcal{L}_{\text{shown}}} \sup_u |\hat{F}_\tau^{\text{scaled}}(u) - \hat{F}_{\tau'}^{\text{scaled}}(u)|. \quad (4.27)$$

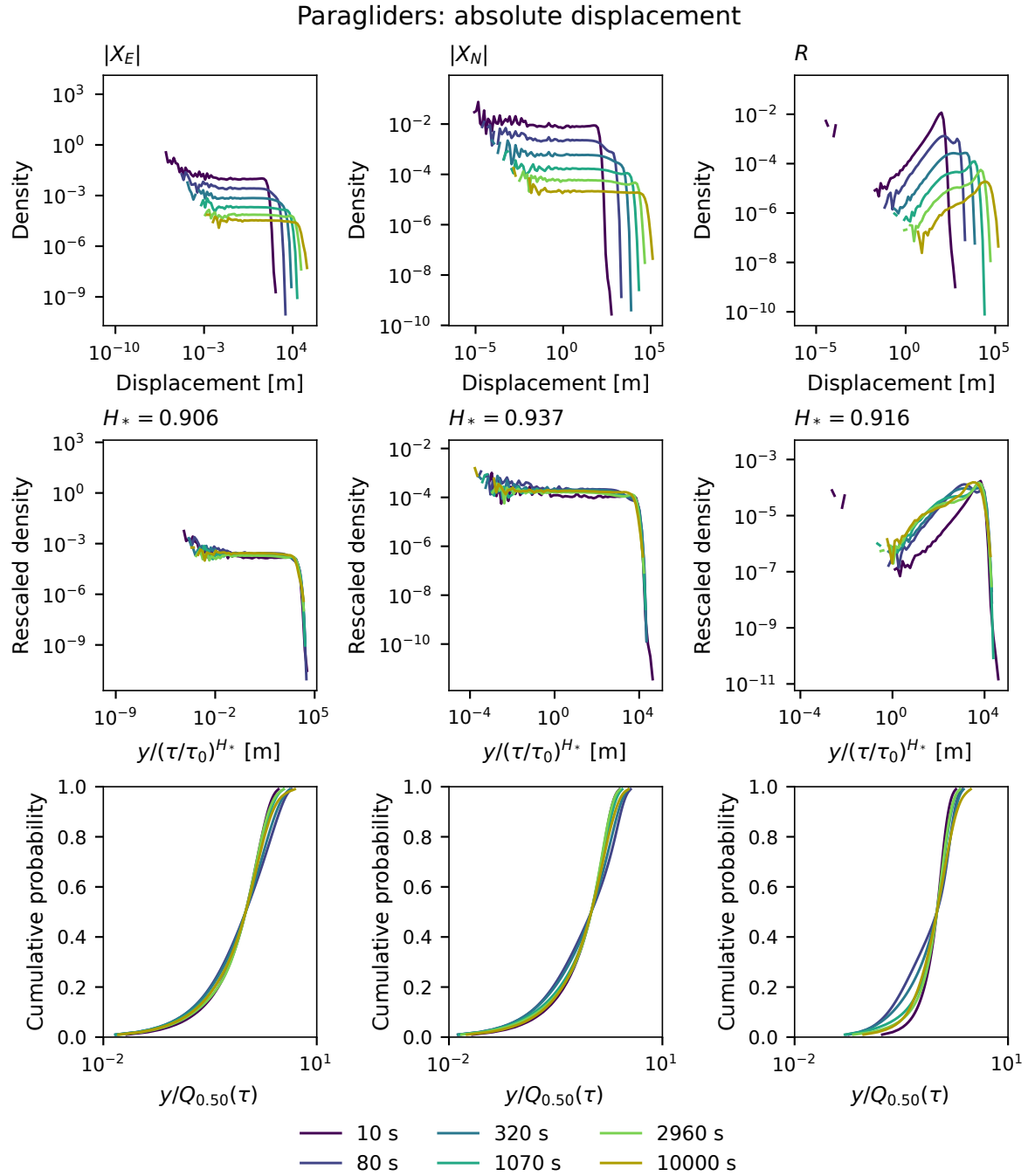


Figure 4.7. Paraglider displacement distributions with fixed flights, weights and origins. Top: raw densities of $|X_E|$, $|X_N|$ and R . Middle: their rescaling by the common exponent fitted separately for each quantity over 10 s to 10,000 s. Bottom: cumulative distributions after division by each lag’s own median, drawn from the 1st–99th empirical percentiles. Colours identify six lags selected across the measured interval. Median normalisation removes an arbitrary scale change, including one that is not a power law; remaining separation therefore reveals changes of shape. The quantitative distances use the full ECDFs.

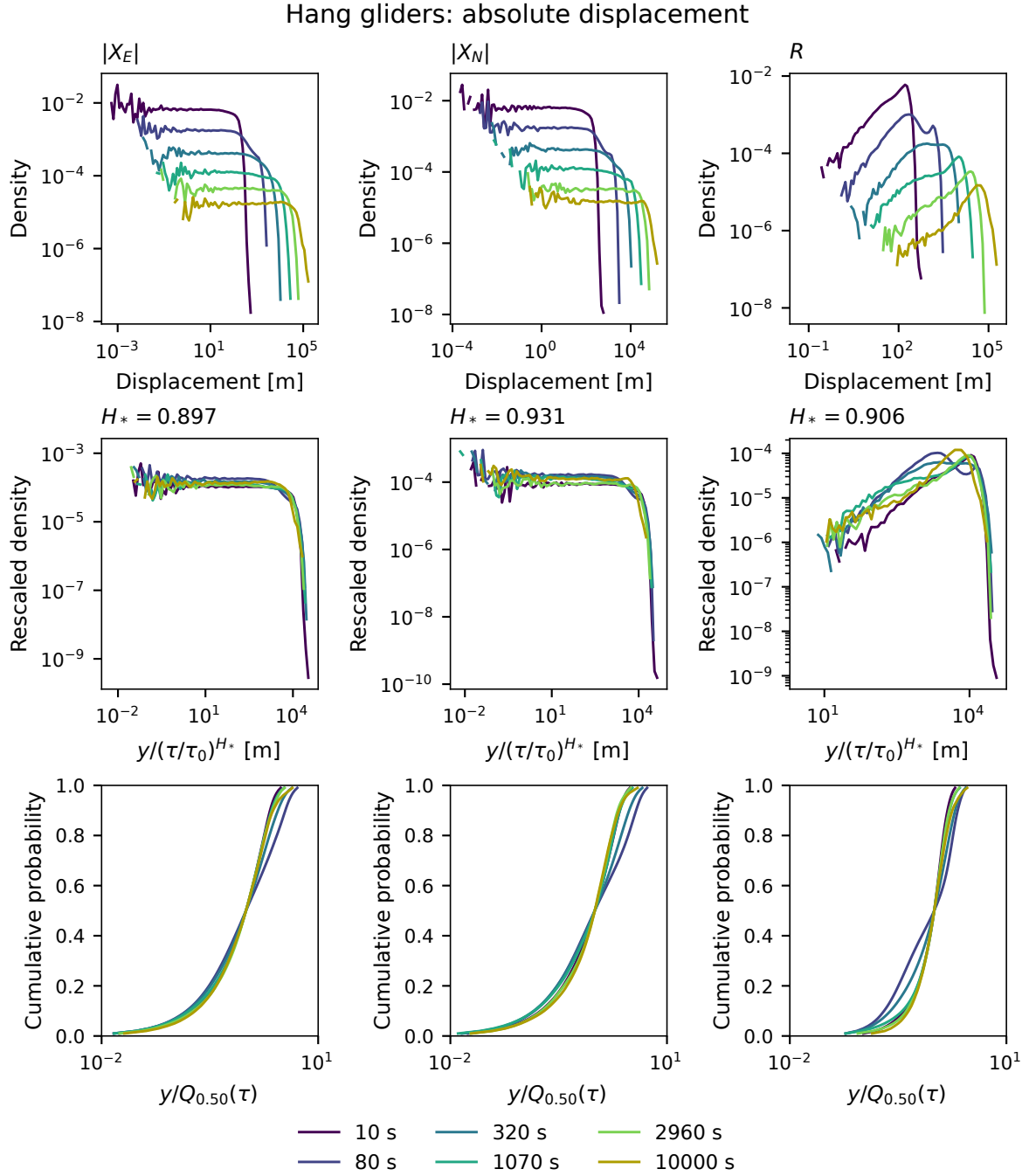


Figure 4.8. The same distribution and collapse diagnostics for hang gliders. The calculation uses the same origins at every lag and the same definition of $H_{j,*}$ as Figure 4.7; it does not borrow the paraglider exponent.

Table 4.4. Candidate common magnitude exponent and largest pairwise ECDF separation among the six displayed lags. D_{power} uses the fitted power scale; D_{median} uses each lag’s own median. A distance of 0.10 is a ten-percentage-point cumulative probability difference somewhere in the rescaled distribution.

Discipline	Quantity	H_*	D_{power}	D_{median}
Paragliders	$ X_E $	0.906	0.177	0.147
	$ X_N $	0.937	0.203	0.178
	R	0.916	0.273	0.194
Hang gliders	$ X_E $	0.897	0.185	0.179
	$ X_N $	0.931	0.269	0.213
	R	0.906	0.331	0.229

We evaluate the supremum at the empirical jumps for the six displayed lags. It is a descriptive probability difference. Weighted, paired and overlapping observations do not support an independent-sample Kolmogorov–Smirnov p -value.

Approximate collapse and its limits. Figures 4.7 and 4.8 show a clear visual improvement after power rescaling: the absolute east and north densities approximately superpose over a broad part of their displayed range. This is evidence of approximate marginal scaling on the measured interval, with fitted exponents near 0.9. The radial curves also move into a broadly common range, especially around their main peak and upper falloff, but retain more visible differences on the rising side and around the peak.

The visual agreement is not exact distributional collapse. The maximum power-rescaled ECDF separations are 0.177, 0.203 and 0.273 for east, north and radius in paragliders, and 0.185, 0.269 and 0.331 in hang gliders (Table 4.4). For example, 0.177 is a difference of 17.7 percentage points in cumulative probability for some pair of displayed lags, not a relative density error. Logarithmic density axes can make such differences appear small. Median normalisation reduces the radial distances to 0.194 and 0.229, but leaves separation for all three quantities. The rank-dependent slopes and changing quantile ratios provide further evidence that one power scale is an incomplete description.

We therefore describe the east and north results as approximate collapse of the absolute-component one-lag laws, rather than established self-similarity of either signed component process. Taking absolute values loses sign information, and these plots do not test joint scaling across multiple times. Nor do different fitted $H_{E,*}$ and $H_{N,*}$ establish one common vector exponent. The measured discrepancies have no calibrated sampling-error threshold: they quantify the residual differences without constituting a formal rejection test or excluding useful scaling on a more restricted range.

Why the radial density need not have a plateau. Scaling compares a given quantity across lags; it does not require the component and radial scaling functions to coincide. At a fixed lag, polar coordinates give the exact relation

$$p_R(r; \tau) = r \int_0^{2\pi} p_{\mathbf{X}}(r \cos \theta, r \sin \theta; \tau) d\theta, \quad r > 0. \quad (4.28)$$

If the joint density is continuous and finite at the origin, with a nonzero value there, then $p_R(r; \tau) \sim 2\pi p_{\mathbf{X}}(0, 0; \tau)r$ as $r \downarrow 0$. In contrast, a continuous finite signed marginal density gives $p_{|X_E|}(x; \tau) \rightarrow 2p_E(0; \tau)$, and similarly for north. A nonzero marginal value therefore gives a plateau, while the regular radial density starts linearly from zero. This distinction does not require isotropy. Geometrically, a small absolute east displacement selects a strip; a small

Table 4.5. Common magnitude exponent $H_{j,*}$ fitted on three lag windows with identical observations and weights. The differences expose the cost of summarising curved quantile growth by a single power. All twelve separate slopes and their paired bootstrap contrasts for each window are retained in the numerical report.

Discipline	Quantity	10–10000 s	60–1970 s	210–1970 s
Paragliders	$ X_E $	0.906	0.952	0.956
	$ X_N $	0.937	0.977	0.985
	R	0.916	0.953	0.950
Hang gliders	$ X_E $	0.897	0.951	0.943
	$ X_N $	0.931	0.981	0.992
	R	0.906	0.957	0.948

radius requires both components to lie in a small disk. Their probabilities are respectively of order ϵ and ϵ^2 under these regularity conditions.

An exact scale-family example makes the distinction explicit. For independent centred Gaussian components with common standard deviation $\sigma_\tau = \sigma_0 a_\tau$, the absolute-component and radial densities are

$$p_{|X_E|}(x; \tau) = \frac{\sqrt{2/\pi}}{\sigma_\tau} e^{-x^2/(2\sigma_\tau^2)}, \quad p_R(r; \tau) = \frac{r}{\sigma_\tau^2} e^{-r^2/(2\sigma_\tau^2)}, \quad x, r \geq 0. \quad (4.29)$$

Both collapse exactly after division of their argument by a_τ and multiplication of their density by a_τ , although only the component density has a plateau. The extra r becomes $a_\tau u$, leaving the same overall factor a_τ^{-1} for the radial density in Eq. (4.8). This example explains why a rising radial curve is compatible with scaling; it is not a Gaussian fit to the flights. The observed small-radius histograms need not have reached the regular asymptotic regime, particularly near numerical zeros. Their residual separation across lags, rather than the absence of a plateau, is the relevant limitation on collapse.

Squared displacements. For any of the nonnegative magnitudes Y_j and $S_j = Y_j^2$, the inverse-CDF quantiles satisfy exactly

$$Q_{S_j,p}(\tau) = Q_{Y_j,p}(\tau)^2, \quad \beta_{j,p} = 2H_{j,p}, \quad p_{S_j}(s; \tau) = \frac{p_{Y_j}(\sqrt{s}; \tau)}{2\sqrt{s}}, \quad s > 0. \quad (4.30)$$

The squared-density collapse therefore uses scale a_j^2 and exponent $2H_{j,*}$; bootstrap slope intervals double too. Squared-law plots appear in Appendix 4.A. Squaring is a monotone bijection for nonnegative values, so corresponding ECDF distances are unchanged. It cannot repair a failed collapse. Radial quantiles must still be computed from paired components: a quantile of $X_E^2 + X_N^2$ is generally not the sum of their quantiles.

Dependence on the fitting interval. We repeat the common-exponent fit on 60s to 1,970s and 210s to 1,970s, retaining the same flights, origins and probabilities. These windows measure sensitivity; their endpoints are not estimated breakpoints.

The intermediate-window exponents move closer to one (Table 4.5). Growth therefore depends on rank, direction and lag even after fixing the observations. Circling, partial returns, extended glides and weather could contribute, but this analysis does not separate their effects. Since the component laws themselves change shape, the data also do not isolate the special case of scaling marginals with changing dependence alone.

4.3.4 Signed joint distributions

Spatial and temporal joint laws. Here *joint* refers to the east and north components of the same displacement $\mathbf{X}(t, \tau) = \mathbf{r}(t + \tau) - \mathbf{r}(t)$. At each lag, the two-dimensional law $p_\tau(x_E, x_N)$ describes their simultaneous dependence, including the shape and orientation of the displacement cloud. Comparing these laws at several lags still compares separate two-dimensional distributions.

A temporal joint law also retains the association between displacements in different intervals. For example, two consecutive intervals of duration τ give the four-dimensional random vector

$$(X_E(t, \tau), X_N(t, \tau), X_E(t + \tau, \tau), X_N(t + \tau, \tau)).$$

Its law describes how the first displacement is related to the next, such as whether an eastward displacement tends to be followed by another eastward displacement or a return. The separate one-lag laws do not generally determine this joint law. They can nevertheless constrain temporal dependence through the distribution of accumulated displacements; they are not devoid of temporal information.

Process self-similarity requires the joint scaling of every finite collection of observation times, as defined in Sec. 4.2. With stationary increments, the collapse of the one-lag vector laws is a necessary consequence, but even exact collapse would not establish that full property. The maps and distances below assess this one-lag consequence; they do not estimate a four-dimensional temporal joint law. The subsequent held-out comparison in Sec. 4.3.5 adds projections of two consecutive intervals and quantifies approximate agreement on selected scales.

The measured spatial comparison. Figures 4.9 and 4.10 retain the signs of (X_E, X_N) on the same flights and origins. Both coordinates are divided by $s_0(\tau/1,000\text{ s})^{H_*}$, where H_* is the mean of all twelve slopes and $s_0 = Q_{R,.50}(\tau_r)/(\tau_r/1,000\text{ s})^{H_*}$, with $\tau_r = 1,070\text{ s}$, the available lag nearest the reference time. Thus the recorded median is adjusted to the common reference by the same power. Geographic axes remain fixed, without lag-specific centring, rotation or whitening.

The six lags share 64 bins per axis within $[-6, 6]$, plus overflow cells. With equal total weight per flight, let $p_{\tau,ij}$ be the joint cell mass. The distance between two binned laws is

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|, \quad (4.31)$$

including overflow. This comparison resolves differences at the chosen bin size; it is not a calibrated rejection test. Energy distance is an alternative for full multivariate laws with finite first moments [15], but inference would still require calibration for paired flights and dependent origins.

The candidate vector exponents are $H_* = 0.920$ for paragliders and 0.911 for hang gliders. After this common rescaling, the central concentration and orientation still evolve with lag. In particular, the long-lag law occupies a smaller spread in the rescaled coordinates; the geographical mean also changes. These changes concern the signed vector distribution, beyond the magnitude comparisons. At the chosen resolution, pairwise distances range from 0.099 to 0.370 for paragliders and from 0.136 to 0.441 for hang gliders. The 10-s versus 10000-s distances are 0.370 and 0.403, respectively. Overflow mass is zero to numerical precision, so the measured separation is not caused by omitting a tail outside the displayed square.

The fixed flights and origins rule out changing membership as the source of these particular differences. They do not distinguish changes of marginal shape, mean motion and component

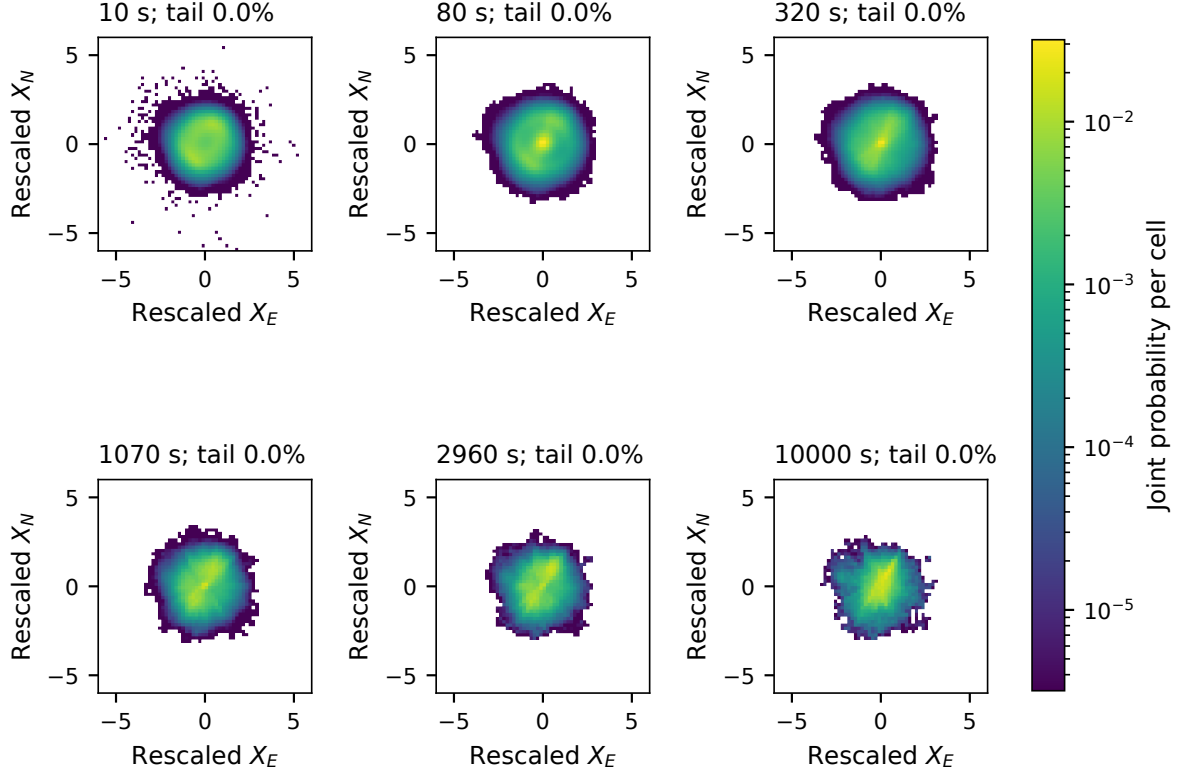


Figure 4.9. Signed joint displacement laws for the fixed paraglider population, after one scalar power rescaling of both axes. Colours show probability per common cell. Each panel reports the mass outside the displayed square; overflow cells are retained in numerical comparisons. These are descriptive distributions, not iid hypothesis tests.

dependence. A larger binned distance also does not rank the disciplines by strength of a physical mechanism: the finite-flight sampling error and sensitivity to bin size have not been calibrated.

4.3.5 Testing approximate scaling beyond one-lag magnitudes

The visible collapse motivates a stronger, independently evaluated claim: does one scalar exponent approximately preserve signed displacement laws and their dependence across two consecutive intervals? The answer is substantially more favourable on 100 s to 1,600 s than across the full range. This is evidence for approximate scaling at a stated temporal and distributional resolution, rather than exact self-similarity of the whole process.

A temporal observable and common observations. Write $\mathbf{X}_1 = \mathbf{r}(t + \tau) - \mathbf{r}(t)$ and $\mathbf{X}_2 = \mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)$. The measured vector is

$$\mathbf{Z}_\tau = (X_{1,E}, X_{1,N}, X_{2,E}, X_{2,N}), \quad \tilde{\mathbf{Z}}_\tau = \frac{\mathbf{Z}_\tau}{s_0(\tau/\tau_0)^{H_*}}. \quad (4.32)$$

The invertible transformation $(\mathbf{X}_1, \mathbf{X}_2) \mapsto (\mathbf{X}_1, \mathbf{X}_1 + \mathbf{X}_2)$ connects this observable to positions relative to the same origin at times τ and 2τ . For an H -self-similar process with stationary increments, its law must be invariant under this common rescaling. Here the origins are pooled within observed flights; stationarity of those increments is not established by this comparison.

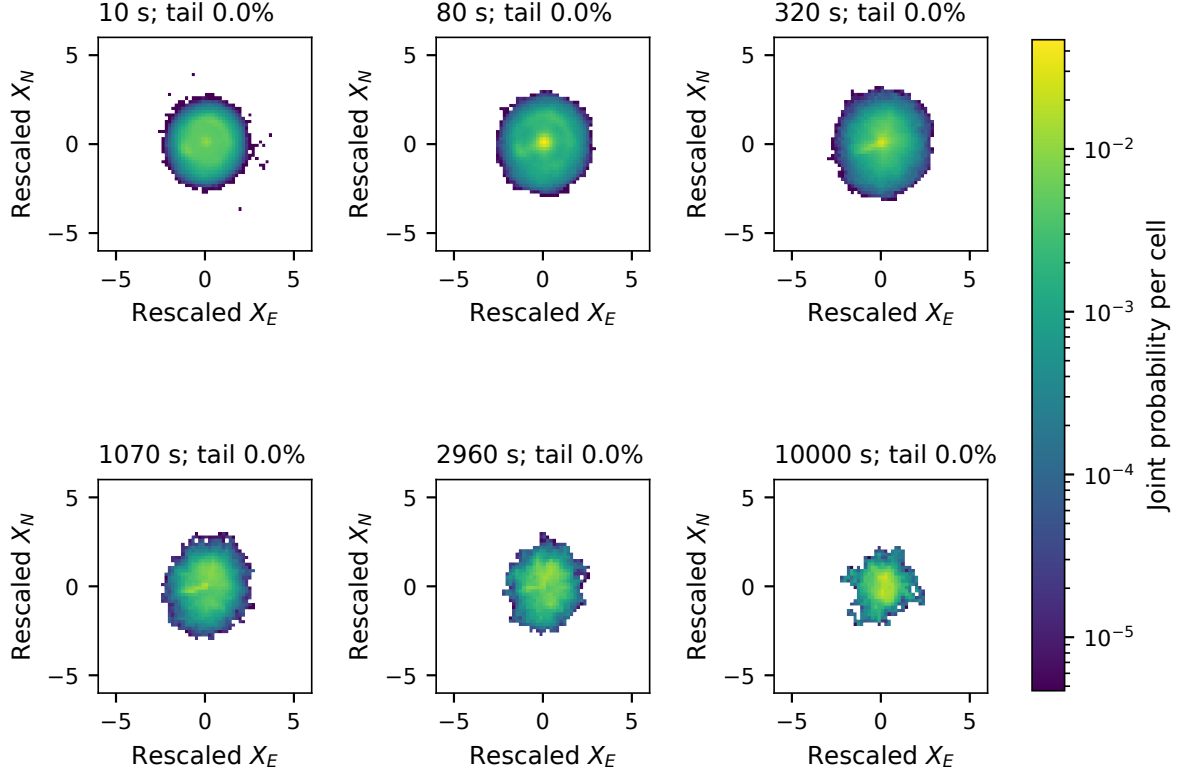


Figure 4.10. The same signed joint-law diagnostic for the fixed hang-glider population. Axes and colour scales remain common across lags within the discipline.

Three lag grids, in seconds, were fixed before running this extension:

$$\begin{aligned} \text{full:} & \quad (10, 100, 1000, 10000), \\ \text{intermediate:} & \quad (100, 200, 400, 800, 1600), \\ \text{long:} & \quad (1000, 2000, 4000, 8000). \end{aligned}$$

They probe the full, intermediate and long intervals suggested by the preceding descriptive analysis; their boundaries are not inferred crossover times. Within each grid, every 10-s origin with uninterrupted support through $2\tau_{\max}$ is retained in every eligible segment. The same flights and origins enter every lag. Cleaning boundaries are never crossed. Origins receive equal weight within a flight, and flights receive equal total weight. The required support is therefore 20000, 3200 and 16000 seconds, respectively. Different grids admit different populations; their comparison does not isolate a lag effect from flight selection.

Separate fitting and evaluation. A fixed hash of the calendar date assigns approximately half the dates to training and half to validation, with the same assignment in both disciplines. All eligible flights with valid dates enter one of these roles. Three intermediate-range paraglider flights have invalid catalogue dates and are excluded; no other eligible flight is excluded for this reason. On training flights only, the inverse-CDF quantiles of $|X_{1,E}|$, $|X_{1,N}|$ and $|\mathbf{X}_1|$ at probabilities .25, .50, .75, .90 are fitted against lag in logarithmic coordinates. H_* is the unweighted mean of the twelve slopes. The reference time τ_0 is the smallest lag of the grid, and s_0 is its training radial median. Both are then held fixed on validation flights. There is no lag-specific centring, rotation, whitening or marginal adjustment. For bounded memory, training quantiles use 8192 logarithmic bins between 10^{-6} and 10^7 metres, with separate underflow and overflow counts. Geometric bin centres bound the quantile discretisation error

Table 4.6. Training and validation populations for the two-interval comparison. Days and origins refer to validation; the training quantile fit supplies H_* .

	Lag range (s)	Train	Validate	Days	H	$n_{\text{orig, val}}$
Para	10–10000	7149	7211	902	0.899	3598452
Para	100–1600	74267	75561	2997	0.915	59782745
Para	1000–8000	13252	13725	1339	0.905	7680809
Hang	10–10000	281	282	198	0.884	119993
Hang	100–1600	2934	2980	938	0.899	2639959
Hang	1000–8000	609	631	348	0.914	294973

by 0.183%; the corresponding deterministic slope-error bound is recorded for each grid. This approximation concerns training only: validation probabilities are counted directly from the signed increments.

What agreement is tested. The four coordinate projections and all twelve unit-normalised pairwise sums and differences define a fixed set $\mathcal{U}$ of 16 directions in $\mathbb{R}^4$. In particular, $(X_{1,E} - X_{2,E})/\sqrt{2}$ compares consecutive eastward displacements and is sensitive to changes hidden by their separate marginals. CDFs are evaluated at the fixed dimensionless thresholds $\mathcal{A} = \{0, \pm 0.1, \pm 0.25, \pm 0.5, \pm 1, \pm 2, \pm 4, \pm 8\}$. For validation flights $f = 1, \dots, F$, with n_f common origins, define

$$\hat{F}_{\tau,u}(z) = \frac{1}{F} \sum_f \frac{1}{n_f} \sum_i 1\{u^\top \tilde{\mathbf{Z}}_{fi,\tau} \leq z\}, \quad \hat{D}_{\mathcal{U}'} = \max_{\tau < \tau', u \in \mathcal{U}', z \in \mathcal{A}} |\hat{F}_{\tau,u}(z) - \hat{F}_{\tau',u}(z)|. \quad (4.33)$$

The spatial set uses $E_1, N_1, (E_1 + N_1)/\sqrt{2}, (E_1 - N_1)/\sqrt{2}$; the temporal set contains the remaining twelve directions, involving the second interval. All projections would characterise a joint law, but this finite set does not [13]. It measures agreement at specified directions and thresholds, including signs and some temporal dependence, rather than a bound on the full four-dimensional distribution or its tails.

For uncertainty, 999 bootstrap replicates resample whole validation calendar dates, preserving all flights, origins and lag contrasts within a date. Let $c_{.95}$ be the 95th percentile of the maximum absolute bootstrap error of the signed CDF contrasts, over all lag pairs, directions and thresholds. The reported bounds are $[\max(0, \hat{D} - c_{.95}), \min(1, \hat{D} + c_{.95})]$. They are simultaneous within a discipline and lag grid and conditional on the training fit; they are not simultaneous across the six discipline–grid analyses. The bootstrap assumes independently resampleable dates. Persistent weather and pilots repeated across dates can violate that assumption. These are bootstrap confidence statements, not finite-sample guarantees. Tolerances of 0.05, 0.10 and 0.20 were fixed with the protocol. An upper bound below a tolerance supplies positive evidence of approximate agreement at that resolution; merely failing to reject equality would not do so.

A supported intermediate-range approximation. On 100 s to 1,600 s, the fitted exponents are 0.915 for paragliders and 0.899 for hang gliders. For paragliders, the signed east, north and spatial-projection discrepancies have upper bounds 0.0305, 0.0308 and 0.0313: all pass the declared 0.05 tolerance. Including temporal projections raises the upper bound to 0.0829, still below 0.10. Hang-glider spatial projections have upper bound 0.0640, below 0.10; their temporal discrepancy is 0.0893 with upper bound 0.1043. The latter narrowly misses the declared 0.10 criterion and passes only the coarser 0.20 criterion. Figures 4.12 and 4.13 show how the temporal difference exposes more separation than signed east and north alone. The

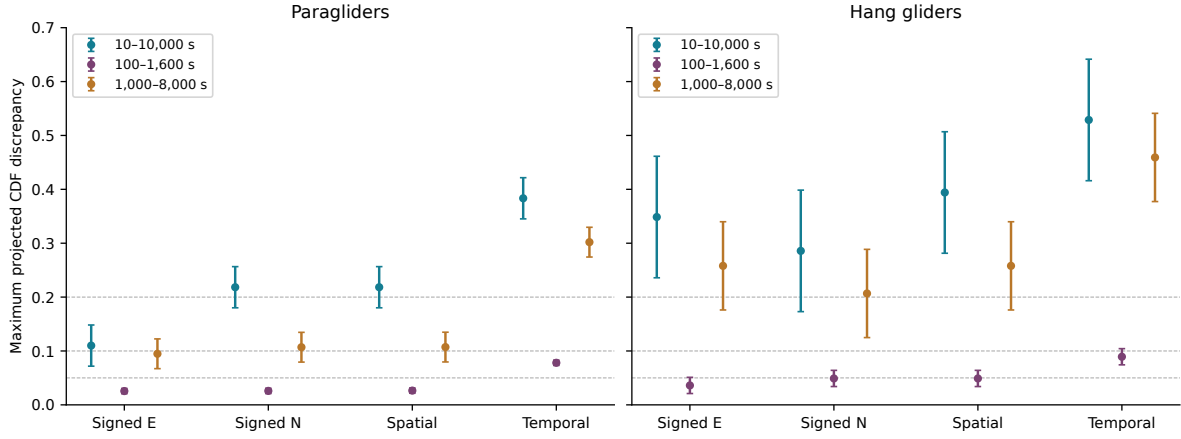


Figure 4.11. Held-out projected-CDF discrepancies and simultaneous 95% bootstrap bounds. Dashed lines mark the declared tolerances. One common training exponent rescales all four coordinates. Each grid retains its own fixed population.

Table 4.7. Maximum held-out projected-CDF discrepancy, with its simultaneous 95% upper bound in parentheses. A value of 0.05 means five percentage points of cumulative probability, not a 5% relative error in displacement.

	Lag range (s)	Signed E	Signed N	Spatial	Temporal
Para	10–10000	0.110 (0.148)	0.218 (0.257)	0.218 (0.257)	0.383 (0.422)
Para	100–1600	0.026 (0.030)	0.026 (0.031)	0.026 (0.031)	0.078 (0.083)
Para	1000–8000	0.095 (0.122)	0.107 (0.135)	0.107 (0.135)	0.302 (0.330)
Hang	10–10000	0.349 (0.461)	0.286 (0.399)	0.394 (0.507)	0.529 (0.642)
Hang	100–1600	0.036 (0.051)	0.049 (0.064)	0.049 (0.064)	0.089 (0.104)
Hang	1000–8000	0.258 (0.340)	0.207 (0.289)	0.258 (0.340)	0.459 (0.541)

plotted lines interpolate the tested CDF thresholds; they are not estimates of exact agreement between thresholds.

The full and long grids give much larger temporal discrepancies: their lower confidence bounds exceed 0.20 in both disciplines. Even on the intermediate grid the discrepancies are resolved above zero, so exact invariance is not supported. The positive conclusion is therefore *approximate scalar scaling of the signed spatial and tested two-interval laws over 100 s to 1,600 s*, strongest for paragliders. It is reasonable to call this evidence for an approximate self-similar

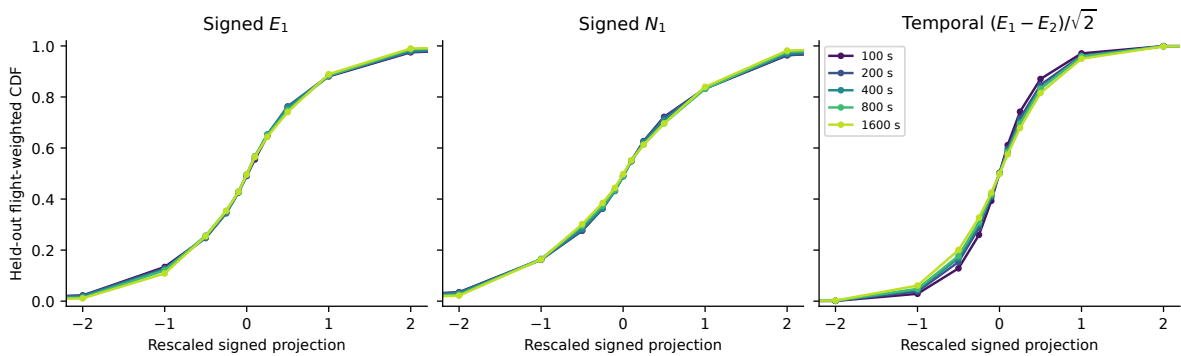


Figure 4.12. Paraglider validation CDFs on the intermediate grid. The first two panels retain component signs; the last measures a projection involving both consecutive intervals. Each flight contributes the same total probability.

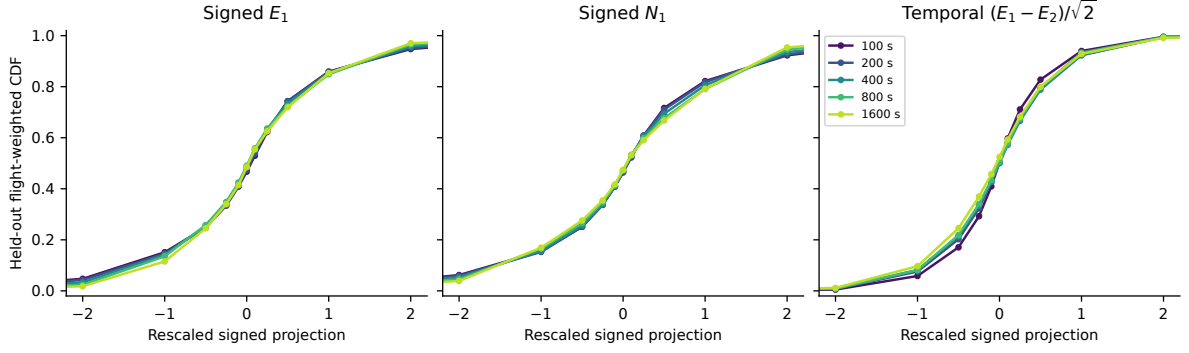


Figure 4.13. The same intermediate-grid signed and temporal comparison for hang gliders, using their independently fitted training exponent and length scale.

description on those scales, provided its projection resolution, tolerances and population are stated. It does not establish self-similarity for arbitrary observation times, increment stationarity, or an intrinsic Hurst exponent. Small-scale manoeuvres and longer-scale route constraints provide a plausible explanation for an intermediate interval with better agreement; the present analysis does not identify those mechanisms or disentangle them from cohort differences.

4.4 Time-averaged transport by region and initial altitude

Before examining directional structure, we ask whether the magnitude of horizontal transport differs across the three take-off regions and four initial-altitude bands. These are two separate groupings of the same eligible archive. The regional comparison uses the Alps, Pyrenees and Channel Coast boxes. The altitude comparison uses every eligible flight with a finite initial altitude, including flights outside those boxes. It is therefore not a comparison of four altitude groups within one common region.

The altitude bands retain the names introduced in the dataset chapter: Plains below 300 m, Hills from 300 to below 800 m, Low mountains from 800 to below 1,500 m, and High mountains from 1,500 m upward. Here the grouping variable is the cleaned trajectory’s GNSS origin altitude, `alt0`. The earlier exploratory map used the first raw altitude and allowed a pressure fallback; the present transport grouping uses neither that fallback nor the instantaneous altitude along the route. These historical labels do not measure local slope, relief or terrain exposure. A record beginning airborne can enter a high-altitude band over a low launch region.

4.4.1 Flight weighting and the fixed cohort

The measurement reuses the identified native-grid segment TA-MSDs for all 155,085 paraglider and 6,060 hang-glider flights with eligible segments. Segments contain at least eight fixes and have native cadence at most 10 s. At each requested lag, the within-flight average pools all supported displacement origins, and the group average gives each contributing flight equal weight:

$$m_f(\tau) = \frac{\sum_s n_{fs}(\tau) m_{fs}(\tau)}{\sum_s n_{fs}(\tau)}, \quad M_g(\tau) = \frac{1}{N_g(\tau)} \sum_{f \in g: n_f(\tau) > 0} m_f(\tau). \quad (4.34)$$

This differs from the equal-segment archive average introduced at the beginning of the chapter. No displacement crosses a segment boundary. For N_{fs} fixes at interval Δt_{fs} , the requested lag is rounded to $k = \text{round}(\tau / \Delta t_{fs})$ native steps and $n_{fs} = N_{fs} - k$. Stored support requires

$1 \leq k \leq \lfloor N_{fs}/2 \rfloor$ and $\tau \geq \Delta t_{fs}$. The 59 requested lags span 10–10,000 s; the nearest one to 1000 s is 1,049 s. This grid must not be relabeled as the exact decade grid used later for PCA and regional differences.

Every curve below keeps only the segments that support the maximum requested lag, 10,000 s, and reuses that same segment and flight set at all 59 lags. Membership and between-flight weights are therefore independent of τ , which is what allows curves measured at different lags to be read as one population. The available origins and the segment weights within a flight still change with lag. The comparison is conditional on long continuous records. The native rounding rule can admit slightly different boundary cases from the exact common-grid controls.

For Figs. 4.14 and 4.15, five hundred complete site-day bootstrap draws are paired across lags within each group. Shading gives pointwise 95% percentile intervals; at least 20 contributing groups and 90% finite replicates are required. Three paraglider flights have missing cluster keys and become singleton groups. No missing origin-altitude labels remain in this eligible population. The bands neither match weather or pilots nor calibrate simultaneous between-group tests. Displayed curves require at least eight flights, and their support is shown explicitly.

4.4.2 Regional contrasts in the fixed cohort

At 1,049 s, the paraglider RMS displacements $\sqrt{M_g}$ are 6.97 km in the Alps, 6.58 km in the Pyrenees and 9.65 km on the coast (Fig. 4.14), over 12,053, 251 and 290 flights. The coast leads the two mountain regions, whose mutual separation is small. These amplitudes belong to the long-flight cohort, which travels farther at intermediate lags than the archive as a whole.

The middle panels make a change of scaling easier to see than parallel-looking log-log curves. A constant $\sqrt{M_g}/\tau$ would correspond to quadratic growth of this moment. In the fixed cohort the coastal 1000–10,000-s log-slope is 1.98, against 1.62 in the Alps and 1.63 in the Pyrenees. The coastal curve stays close to quadratic growth over the last decade while the two mountain regions fall well below it, and no region exceeds two. These fitted slopes describe separate ranges of a bending curve; they are not Hurst estimates or calibrated hypothesis tests.

Hang gliders supply no regional comparison in this cohort. Only the Alps retain enough flights, 458, giving 11.20 km near the reference lag. The Pyrenean and coastal groups contain two flights each, below the eight required for display, so the regional ordering of Fig. 4.14 is a paraglider result alone.

4.4.3 Initial-altitude bands and their regional composition

The paraglider RMS displacements at the reference lag are 9.38 km, 9.07 km, 6.84 km and 7.06 km, in increasing altitude-band order. The two lower bands lie above both mountain-labelled groups, and the ranking is not monotonic in altitude: High mountains lies above Low mountains. The long-lag behaviour separates the same way. Over the last decade the Plains and Hills slopes are 1.97 and 1.93, against 1.62 and 1.64 in the two mountain bands.

For hang gliders the four bands hold 45, 68, 348 and 102 flights, giving 11.66 km, 12.94 km, 11.35 km and 9.81 km. Hills leads and High mountains trails, so the two disciplines do not share one band ranking. Neither the band label nor one pooled ranking identifies an effect of altitude on a pilot's dynamics.

Figure 4.16 shows why the two groupings are not independent environmental controls. The coastal cohort is concentrated in Plains, while Alpine and Pyrenean records mostly occupy the two higher bands. Within the Alpine paraglider cohort the four band amplitudes lie much

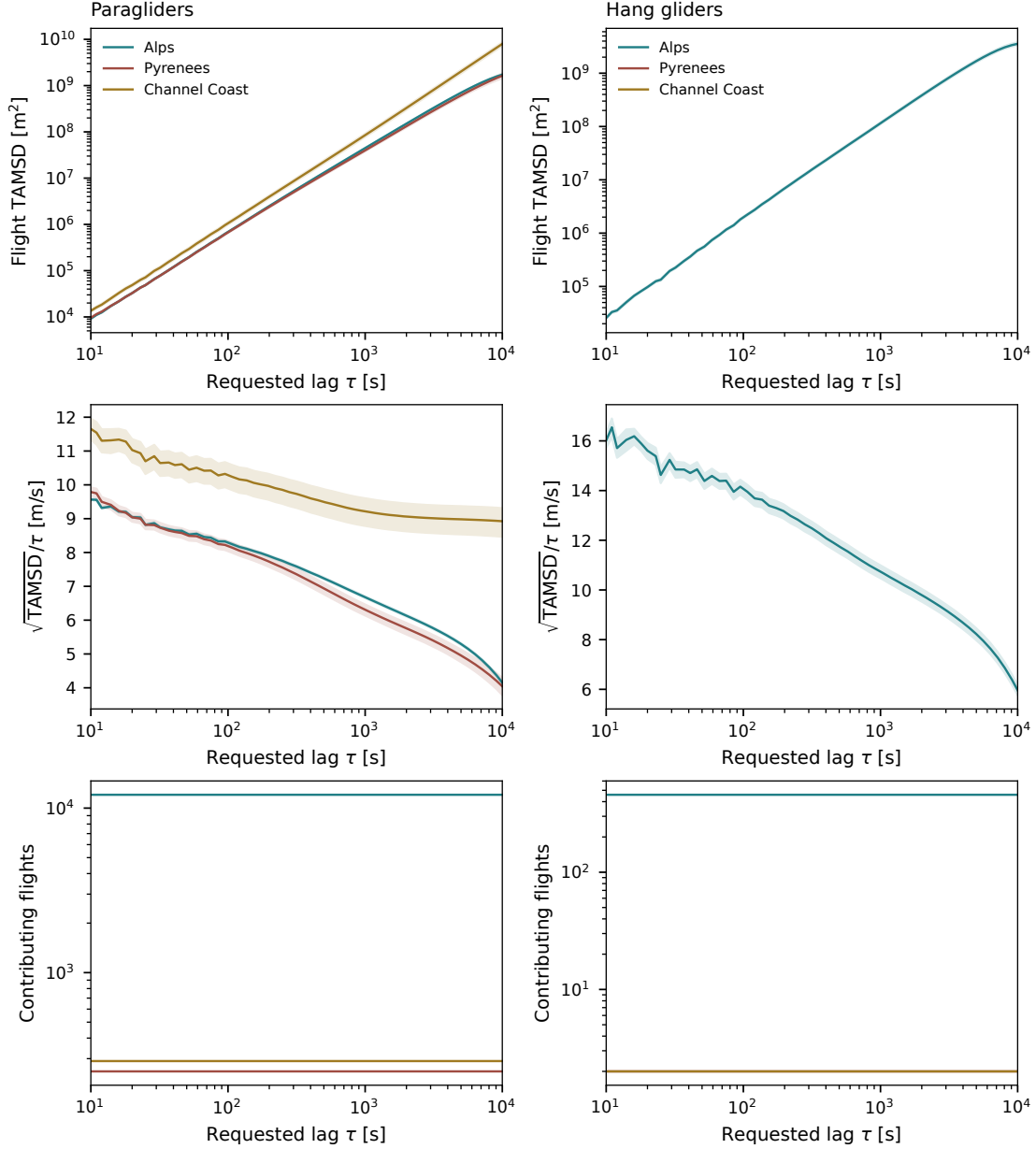


Figure 4.14. Horizontal flight TA-MSD by take-off region, separately for the two disciplines. Top: squared-displacement mean. Middle: its square root divided by the requested lag, the RMS magnitude of the interval-mean ground velocity up to native-grid rounding. Bottom: contributing flights, which are constant in τ because every curve retains the same long segments over the full lag range. Shading is the pointwise site-day interval. Hang-glider curves for the Pyrenees and coast are omitted because the cohort contains only two flights in each; their support remains shown.

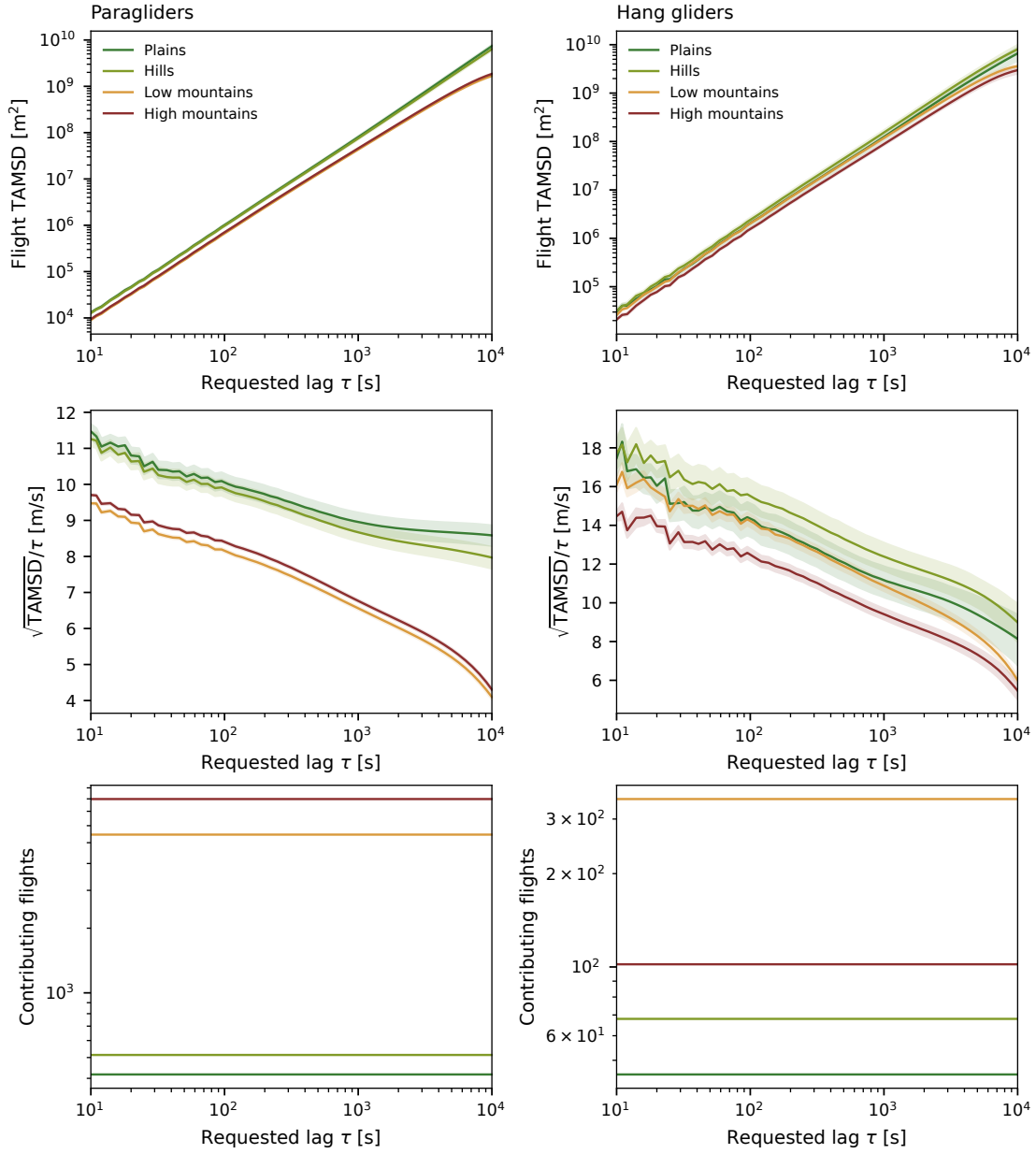


Figure 4.15. The same TA-MSD estimators grouped by cleaned GNSS origin altitude. The names Plains, Hills, Low mountains and High mountains denote the stated altitude bands, not measured terrain classes. All eligible regions enter this grouping. Lines, shading and support conventions match Fig. 4.14, including the fixed cohort.

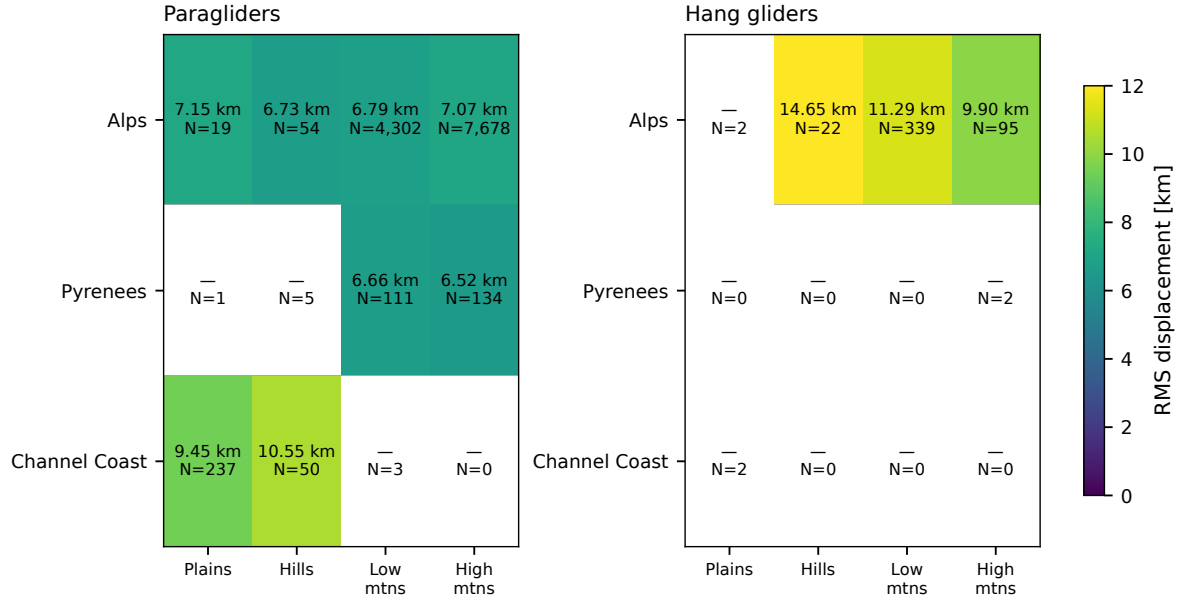


Figure 4.16. Descriptive cross-classification at the requested 1,049s lag, within the same fixed cohort. Each cell gives $\sqrt{M_g}$ and its contributing flight count. Values with fewer than eight flights are suppressed; counts remain visible. The same colour scale applies to both disciplines. These cells carry no calibrated intervals and do not standardise task, weather or pilot identity.

closer together than in the pooled altitude comparison above, and they do not decline with altitude. The Pyrenees supply only the two mountain bands, whose amplitudes are nearly equal. The coastal cells above Hills are empty in this cohort, so no general mountain-versus-coast interpretation follows. For hang gliders only three Alpine cells survive. Logged origin altitude and regional membership describe selection as well as possible environmental constraints.

The larger coastal transport amplitude can be compared qualitatively with the smaller changes of interval-mean velocity measured in Sec. 3.2.4. Greater net progress with smaller velocity changes is compatible with more persistent advancement and less turning, or an added mean motion; it does not identify wind as the cause. The native-grid TA-MSD and the matched-order difference diagnostic use different origins and nearby, rather than identical, reference lags, so this is not an exact paired decomposition. A quantitative mechanism test should align those observations and add independent wind, route, terrain and pilot information.

Reworked in thesis Section 3.2. Original analysis retained below.

4.4.4 Paraglider scaling by altitude and circuit type

For paragliders, we compare the four altitude bands separately within declared open and closed circuits, using the task classes of Sec. 4.1.2. Free-distance tasks and distance tasks with one, two or three points form the open group; triangles, quadrilaterals and out-and-return tasks form the closed group. This is a catalogue declaration of route type, not a test that a retained segment returns exactly to its origin. The eligible sample contains 67,003 open and 87,807 closed flights. The 275 flights whose task labels do not identify either class are excluded from this comparison.

Within each circuit class, we fit all 59 requested lags from 10s to 10,000s, including both endpoints. For each circuit class and altitude band, an unweighted least-squares fit in

logarithmic coordinates gives

$$\log \left[\frac{M_g(\tau)}{1 \text{ m}^2} \right] = a_g + \alpha_g \log \left[\frac{\tau}{1 \text{ s}} \right], \quad H_{\text{eff},g} = \frac{\alpha_g}{2}. \quad (4.35)$$

This is one fit across the full range, rather than an average of the three decade slopes. The notation H_{eff} denotes an effective second-moment scaling exponent. It does not establish self-similarity, stationary increments or an intrinsic Hurst parameter for these flights.

Uncertainty is estimated by 2,000 bootstrap draws of whole site–day groups within each circuit–altitude stratum. Each draw retains all flights in every selected group and uses the same cluster multiplicities at every lag. We recompute the equal-flight mean and refit Eq. (4.35) on the same lag grid. The 2.5th and 97.5th percentiles of the resulting H_{eff} values form the reported interval. This preserves the observed dependence between lags and within a site–day; it does not treat individual displacement origins or fitted lag values as independent observations. Every fit has at least twenty contributing site–day groups at each lag and all 2,000 replicates are finite.

Table 4.8. Paraglider effective exponents H_{eff} from the full 10–10,000-s fit, separately for declared open and closed circuits. Brackets give marginal 95% site–day bootstrap intervals. Flight counts give the fixed cohort, which contributes the same flights at every lag. Weighting matches Fig. 4.17.

Altitude band	Fixed cohort	Flights
<i>Open circuits</i>		
Plains	0.9600 [0.9577, 0.9621]	393
Hills	0.9578 [0.9554, 0.9600]	417
Low mountains	0.9125 [0.9103, 0.9148]	1,564
High mountains	0.9088 [0.9072, 0.9103]	2,209
<i>Closed circuits</i>		
Plains	0.8933 [0.8761, 0.9054]	24
Hills	0.8720 [0.8660, 0.8781]	97
Low mountains	0.8900 [0.8887, 0.8914]	3,887
High mountains	0.8966 [0.8956, 0.8975]	5,770

Table 4.8 and Fig. 4.17 separate the altitude comparison from the mixture of declared task types. Open circuits have larger fitted exponents than closed circuits in every altitude band. The altitude ordering also changes with circuit class: for open circuits Plains and Hills lie about 0.05 above both mountain-labelled groups, whereas the four closed-circuit exponents fall within 0.025 of one another and follow no altitude order, with High mountains highest and Hills lowest. Flight support differs substantially between strata. In particular, the closed-circuit Plains stratum contains only 24 flights in 24 site–day groups. Its interval should be read alongside this small population. The cohort conditions on long continuous records; it does not match the open and closed groups by duration, region, weather or pilot. Differences between their fitted exponents can therefore reflect both route geometry and population composition.

These intervals assume independently resampleable site–day groups. Repeated pilots or weather shared across groups can leave dependence unaccounted for. They quantify sampling uncertainty conditional on the chosen lag range, weighting and population, and do not include uncertainty from cleaning, altitude labels or the adequacy of a power law. In particular, the bending curves and decade slopes above remain relevant even when the full-range interval is narrow. The marginal intervals are not simultaneous tests of differences between altitude bands or circuit classes, and the pooled groups do not isolate a causal altitude effect. Directional moments are needed next because a scalar TA-MSD does not describe the orientation of motion.

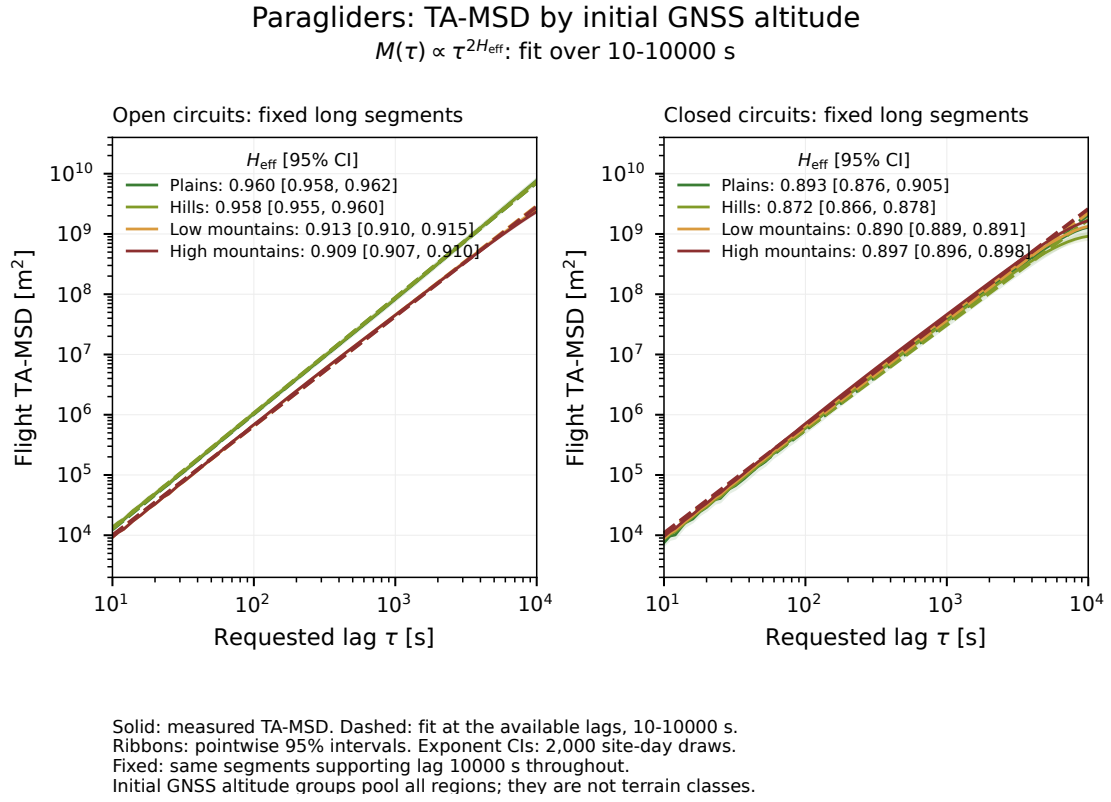


Figure 4.17. Paraglider horizontal flight TA-MSD by initial GNSS altitude, with separate open- and closed-circuit fits over the full 10–10,000-s range. Columns distinguish the declared circuit classes. Every panel uses the fixed cohort of segments supporting the maximum lag, so flight membership does not change with τ . Solid lines show measured means and dashed lines the fitted power laws. Legends give H_{eff} and its marginal 95% site-day bootstrap interval. Ribbons give pointwise 95% intervals for the means. Both use 2,000 complete cluster draws within each circuit-altitude stratum. Flight counts are reported in Table 4.8.

4.5 Flight duration and equipment composition

Longer flights may select different equipment, pilots and conditions. A change in their MSD can therefore reflect a change of population rather than an effect of observation length. We examine both duration bands and their equipment composition, using the identified segment curves from the full archive: 155,085 paraglider and 6,060 hang-glider flights have segments of at least eight fixes with native cadence at most 10 s. Here T_f is the sum of the retained segment spans of flight f ; gaps and discarded time are not part of this observed duration.

First combine the available segment curves within each flight, weighting them by the number of admissible time origins, then give every contributing flight equal weight within a band of observed duration:

$$m_f(\tau) = \frac{\sum_s n_{fs}(\tau) m_{fs}(\tau)}{\sum_s n_{fs}(\tau)}, \quad M_{[T_0, T_1)}(\tau) = \frac{1}{N_{[T_0, T_1)}(\tau)} \sum_{f: T_0 \leq T_f < T_1, n_f(\tau) > 0} m_f(\tau). \quad (4.36)$$

For an eligible segment fs of N_{fs} fixes at interval Δt_{fs} , $n_{fs} = N_{fs} - \text{round}(\tau/\Delta t_{fs})$; it is zero outside that segment's stored lag support, which ends at $\lfloor N_{fs}/2 \rfloor \Delta t_{fs}$. No displacement crosses a segment boundary. This equal-flight average differs from the equal-segment archive average introduced earlier. Requested lags are rounded to each segment's native grid.

The four bands are disjoint, so no flight enters two of them. Nested cohorts $T_f \geq T_0$ answer a different question, namely whether a pooled estimate survives restriction to longer records. They cannot describe how transport depends on duration: consecutive cohorts share most of their flights, since dropping every paraglider flight under one hour removes 6,632 of 155,085, and each cohort is contained in the previous, so only a monotone progression can appear. A band holds its membership fixed out to $\tau = T_0/2$, which is where its curve ends in Fig. 4.18.

At 1,049 s the paraglider RMS displacements $\sqrt{M}$ are 6.00 km below one hour, 5.13 km from one to two hours, 5.47 km from two to four hours and 6.64 km above four hours. Taken as ratios to the all-duration curve these are 1.11, 0.81, 0.92 and 1.35; the hang-glider ratios are 0.86, 0.61, 0.86 and 1.47. The dependence on duration is therefore not monotone in either discipline. Records below one hour displace more per lag than those of one to two hours, and the longest flights displace most. A nested comparison cannot represent this ordering.

One reading consistent with these bands is that short records retain transits and single glides, intermediate durations are dominated by local soaring that returns near its start, and the longest flights are the cross-country ones. The archive carries no declaration of flight intent, so this interprets the ordering rather than measuring it. The bands also differ in equipment and in the sites and days they sample.

Table 4.9 fits every band over one common lag range. The growth exponent is smallest between one and two hours, $H = 0.877$ for paragliders and 0.844 for hang gliders, and largest above four hours, 0.916 and 0.899. It is non-monotone in duration, like the amplitude. Across bands it spans 0.039 and 0.055, against pooled values of 0.895 and 0.877 over the same lags. Observation length therefore moves the exponent by more than the nested cohorts of Fig. 4.19 suggest, which is what a comparison between overlapping populations understates. These remain descriptive slopes of a bending curve over a restricted interval, and the bands stay confounded with equipment, site and day.

Figure 4.19 repeats the comparison within the beginners (EN A/B) and experts (EN C/D/CCC) groups, using nested thresholds $T_f \geq T_0$ rather than disjoint bands. This separates within-class contrasts from changes in their relative abundance.

Without duration conditioning, the fitted slopes over 10 s to 10,000 s are 1.728 for beginners and 1.788 for experts. Table 4.10 compares all eight group-duration cells on 10 s to 1,695 s,

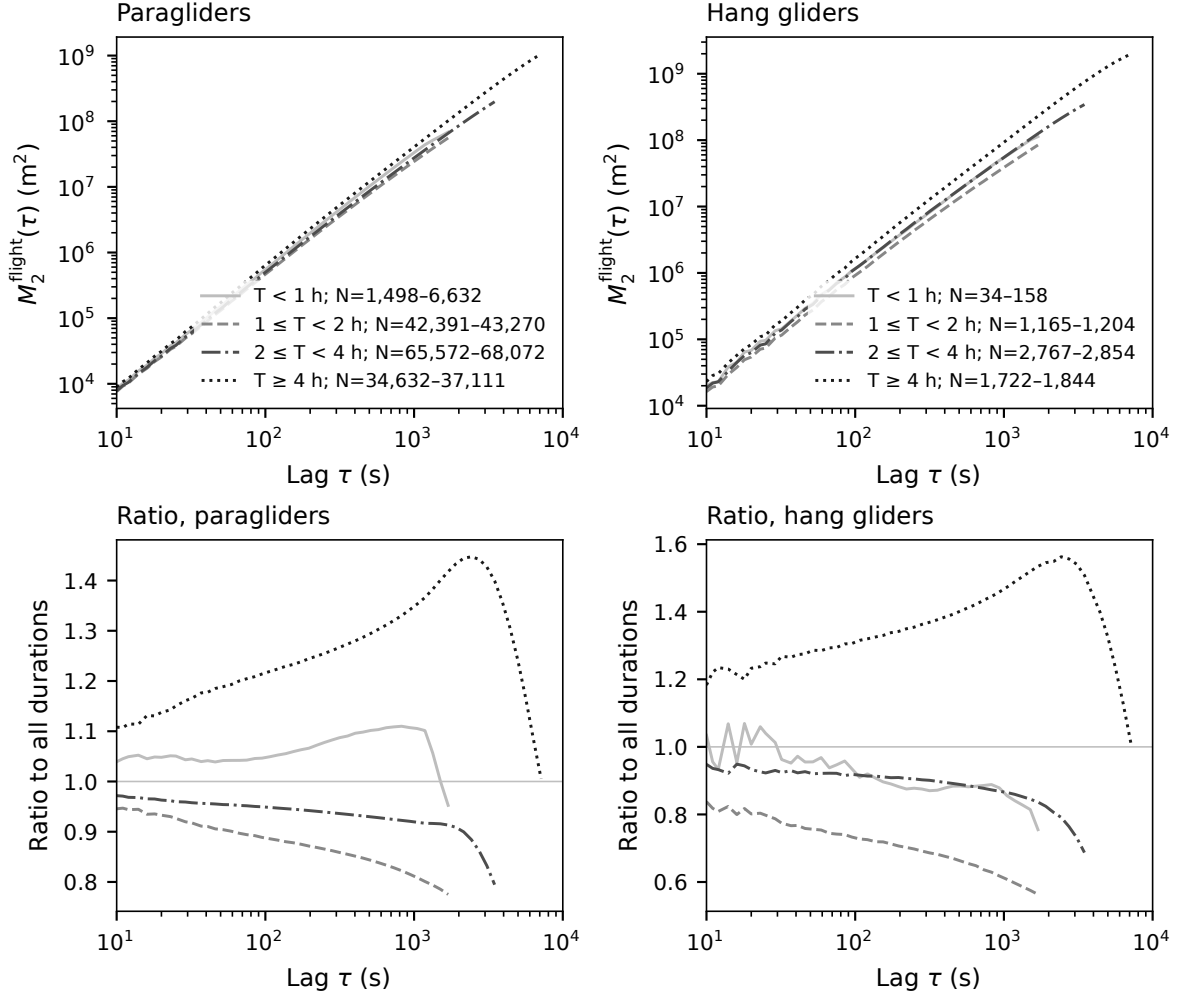


Figure 4.18. Time-averaged squared displacement by retained flight duration, in four disjoint bands. Top: the band curves, with legend counts giving the range of contributing flights. Bottom: each band divided by the all-duration curve at the same lag, which is what separates them on these axes. Each lag requires at least 30 flights. A band keeps its membership out to $\tau = T_0/2$ and its curve ends there. The lowest band has no such range, and for hang gliders it holds 158 flights, so its curve is the noisiest shown.

ending before half the one-hour threshold. Fit residuals measure departures from the line, independently of sampling uncertainty.

Experts have the larger slope in every duration cell. From all durations to $T \geq 4$ h, the slope rises from about 1.755 to 1.806 for beginners and from 1.803 to 1.836 for experts. Duration contrasts thus persist within both classes. The nested cohorts remain dependent and the comparisons descriptive.

To separate within-class behaviour from class proportions, write the exact mixture identity over the two experience proxies,

$$M_{T_0}^{\text{known}}(\tau) = \sum_c p_c(\tau | T_0) M_{T_0,c}(\tau), \quad p_c(\tau | T_0) = \frac{N_{T_0,c}(\tau)}{\sum_j N_{T_0,j}(\tau)}. \quad (4.37)$$

The weights depend on lag as well as duration. A standardized comparison instead holds them fixed at $\pi_c = N_{0,c} / \sum_j N_{0,j}$, the class proportions among all eligible known-equipment flights:

$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau). \quad (4.38)$$

Table 4.9. Effective growth exponents by duration band, all fitted over the same 10 s to 1,695 s, the intersection of the four bands' supports. Here b is the log-log slope of $M_{[T_0, T_1]}$ and $H = b/2$ the exponent of the corresponding magnitude. The residual measures departure from the fitted line, not sampling uncertainty. The all-duration row is refitted over these same lags, so it compares with the bands rather than with the wider ranges quoted elsewhere in this chapter.

Discipline	Duration	Flights	b	$H = b/2$	RMS residual (dex)
Paragliders	All durations	155,085	1.789	0.895	0.018
Paragliders	$T < 1$ h	6,632	1.796	0.898	0.026
Paragliders	$1 \leq T < 2$ h	43,270	1.754	0.877	0.022
Paragliders	$2 \leq T < 4$ h	68,072	1.777	0.889	0.019
Paragliders	$T \geq 4$ h	37,111	1.832	0.916	0.015
Hang gliders	All durations	6,060	1.755	0.877	0.025
Hang gliders	$T < 1$ h	158	1.712	0.856	0.031
Hang gliders	$1 \leq T < 2$ h	1,204	1.688	0.844	0.033
Hang gliders	$2 \leq T < 4$ h	2,854	1.736	0.868	0.030
Hang gliders	$T \geq 4$ h	1,844	1.798	0.899	0.021

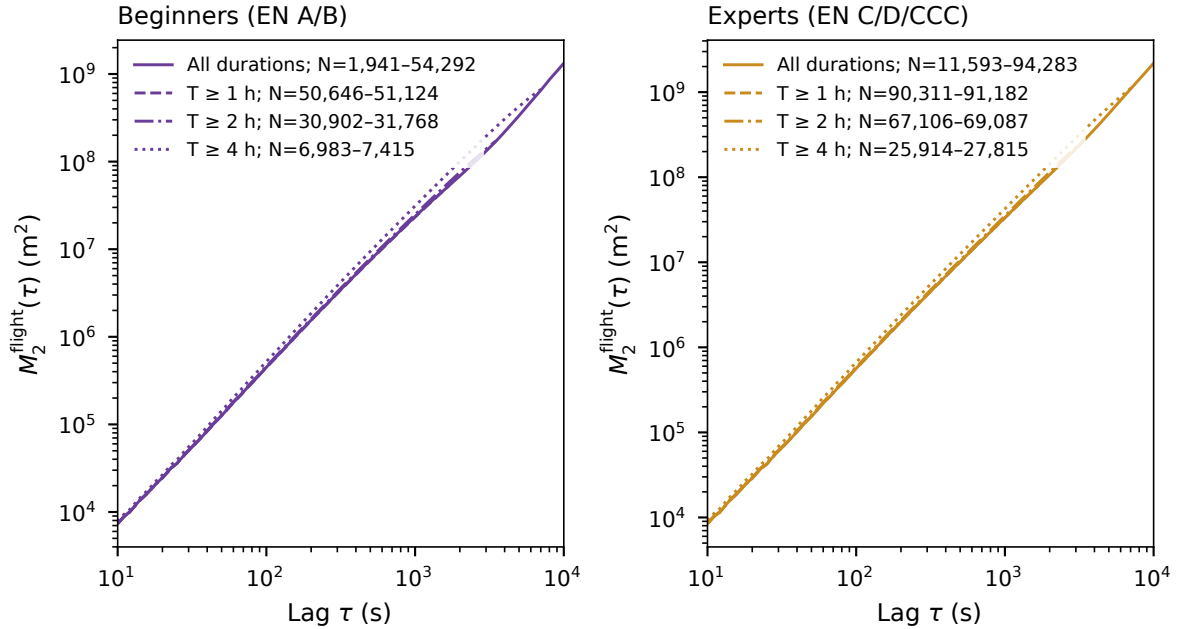


Figure 4.19. The duration comparison repeated for beginners (EN A/B) and experts (EN C/D/CCC), using the same estimator and minimum support as Fig. 4.18, with nested thresholds $T_f \geq T_0$ in place of its disjoint bands, so the cohorts here are not independent of one another. Counts refer to distinct flights, and overlapping groups do not constitute independent samples. Tandem, non-certified and unknown equipment is omitted from this contrast.

Figure 4.20 compares the four-hour/all-duration ratio before and after this adjustment. Both ratios exclude the same unknown or other equipment; otherwise changing the population would be confounded with changing the weights. The adjustment is displayed only where every class has at least 30 contributing flights.

Among the two known classes, the expert share rises from about 63.5% in all durations to 78.9% for $T \geq 4$ h. The latter cohort still includes 7,415 beginner-proxy flights. Fixing the class mixture reduces the MSD amplitude contrast:

Table 4.10. Duration and equipment contrasts on a common supported lag interval, 10 s to 1,695 s. Flight counts include every eligible flight in the cell; the contributing count decreases with lag. The residual is the root-mean-square deviation in $\log_{10} M_2$ from the fitted line.

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

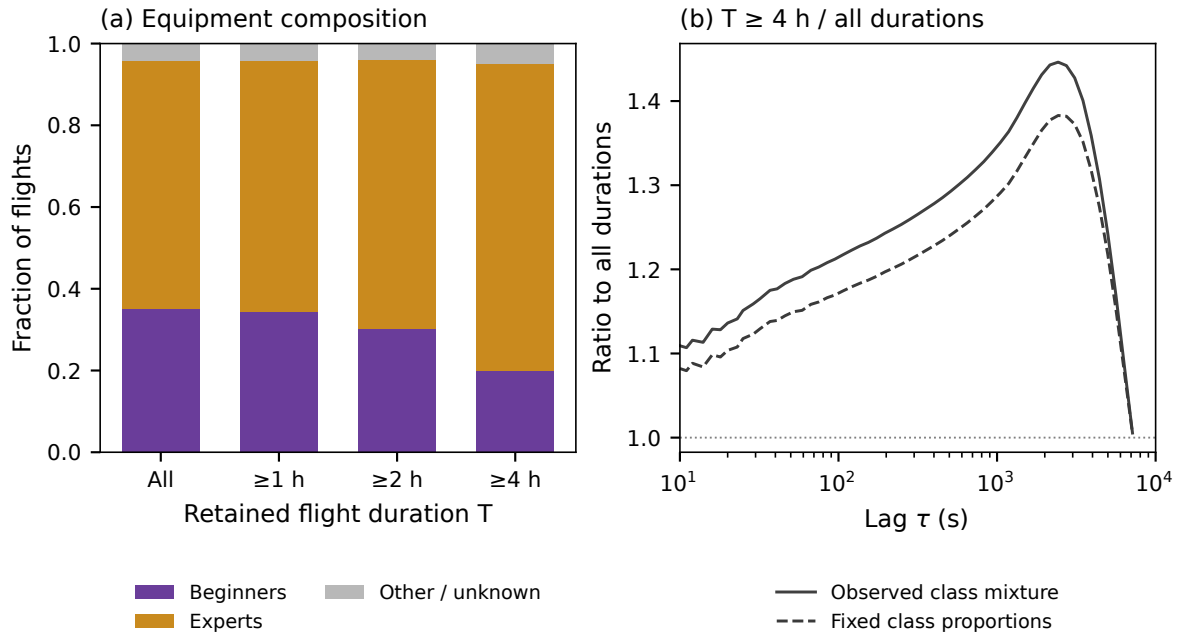


Figure 4.20. Equipment composition and its contribution to the duration comparison. Left: proportions of flights in the nested duration cohorts, including other or unknown equipment (6,510 flights in the full eligible sample). Right: the four-hour/all-duration MSD ratio among known classes, using the observed lag-dependent mixture or fixed class weights. Unity denotes equal MSD amplitudes at that lag, not equal trajectories or established statistical independence.

Lag [s]	Four-hour/all-duration MSD ratio	
	Observed mixture	Fixed class weights
576	1.30	1.25
1695	1.42	1.35

Composition explains part of the contrast under this adjustment; a substantial within-class difference remains. The return towards unity near 7,200 s coincides with shorter flights leaving the reference population and cannot establish duration independence.

Conditions, routes and pilot selection can still vary within each equipment class, so these associations do not isolate pilot skill or a causal effect of duration.

4.6 Constraints for the phase analysis

A model must account jointly for moment growth, changing quantile spread, the regional anisotropy of Chapter 3, velocity memory, and the contrasts by route, duration and equipment. An effective exponent alone cannot describe this set of observations.

The independent terrain comparison strengthens the geographical interpretation of the Pyrenean displacement axis, while the Alpine axis differs from the mean highland orientation. A model of environment-dependent motion should preserve this regional distinction; neither angle comparison supplies a causal separation of terrain, wind and route selection. On the Channel Coast, ERA5 wind at the mean altitude of the contributing flights gives an annual axis about 12–14 degrees from the flight axis, for both all hours and daytime. The warm-season checks retain a broadly similar orientation at this altitude, whereas the surface comparison is more sensitive to season. This supports a descriptive regional resemblance, with a remaining offset; a mean wind direction alone cannot explain centred displacement covariance.

Chapter 5 introduces phase labels. Connecting those phases to global transport requires durations, displacements and dependencies within and between episodes. If c partitions all admissible windows into phase histories, including windows crossing a phase boundary, then

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}, \quad \sum_c w_c = 1, \quad (4.39)$$

for window weighting. With flight weighting, apply the identity within each flight first. Windows wholly inside individual phases omit transitions and cannot generally reconstruct the global moment. Since both weights and conditional amplitudes depend on lag, global exponents are not weighted phase exponents.

Any exhaustive partition satisfies this identity, including one with incorrect labels. Reconstructing the global moments therefore checks the calculation; independent labels are still needed to assess the phase assignment.

Chapter 5

Flight phases

The global observables of Chapters 3 and 4 constrain the motion of a complete flight, but because they mix directed travel, searching and altitude recovery, they do not identify which behaviour contributes to a particular scaling law. Segmentation provides the missing conditional description: given a phase, how does the glider move, how long does that phase last, and what happens next?

The aim here is to assign these phases reproducibly and establish how the labels can be checked, bearing in mind that kinematics provide indirect evidence of behaviour: a coloured climb on a trajectory remains an inference. We first build the probabilistic model and define its measured features, then examine its numerical behaviour and the independent annotations needed to assess its accuracy.

5.1 What the three phases mean

We retain the three-state description used by Vilpellet et al. [16], with labels $\mathcal{S} = \{\text{T}, \text{S}, \text{C}\}$:

Transition (T).

Sustained directed travel between lift regions, usually with limited turning and altitude loss.

Search (S).

Exploratory turns, reversals or recentering without a sustained established climb. Search can be brief or absent between a transition and a climb.

Climb (C).

Sustained exploitation of lift, usually combining positive mean vertical speed and circling.

These definitions concern observable flight geometry rather than a pilot's unrecorded intention, and allow departures from a fixed cycle: a glider may encounter lift directly, leave an unsuccessful search, or interrupt a climb. Ridge soaring and straight flight in rising air may also be ambiguous within this three-state vocabulary. Such cases should remain visible during validation instead of being forced to confirm the proposed interpretation.

5.2 From a Markov chain to hidden-state inference

The phase of a flight is not observed directly: a recorder measures motion, and different behaviours can produce overlapping ranges of speed and turning. A hidden Markov model represents this distinction through two coupled sequences. The latent sequence Q_k describes the state at decision time k , while the observed vector $\mathbf{Y}_k$ carries the measurements associated with



Figure 5.1. Schematic plan views of the three phase interpretations. The drawings are illustrations, not measured trajectories or decision rules. Climbing circles can drift, and searching can cover substantial ground; neither phase is assumed stationary.

that state. The construction and inference below follow the standard HMM formulation [17, Secs. II–III]; its use for behavioural interpretation requires additional care, as discussed by McClintock et al. [18].

5.2.1 Constructing the probability law

Begin with a chain on a finite state space $\mathcal{S}$. Its initial distribution and transition probabilities satisfy

$$\begin{aligned} \Pr(Q_1 = i) &= \pi_i, \quad \sum_i \pi_i = 1, \\ \Pr(Q_{k+1} = j \mid Q_{1:k-1}, Q_k = i) &= a_{ij}, \quad \sum_j a_{ij} = 1. \end{aligned} \quad (5.1)$$

Here $a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i)$ and all probabilities are nonnegative. The Markov assumption concerns the hidden state: its present value contains the information about its history used to predict the next state. A constant matrix A adds time homogeneity; it is a modelling choice, not a consequence of the observations being sampled regularly.

Now attach an emission density $b_i(\mathbf{y})$ to each state, with $\int b_i(\mathbf{y}) d\mathbf{y} = 1$. Conditional on the complete state sequence, the emissions are assumed independent and the density at time k depends only on Q_k . The chain rule then gives

$$p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta) = \pi_{q_1} b_{q_1}(\mathbf{y}_1) \prod_{k=1}^{K-1} a_{q_k q_{k+1}} b_{q_{k+1}}(\mathbf{y}_{k+1}), \quad (5.2)$$

where θ collects π , A and the emission parameters. This is also a generative recipe: draw the first state, generate its observation, then alternate a state transition with a new observation. Although emissions are independent conditional on the states, the observed sequence is generally correlated and need not itself be first-order Markov.

For continuous features, a common emission model is a multivariate Gaussian,

$$b_i(\mathbf{y}) = \frac{\exp[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu}_i)^\top \Sigma_i^{-1}(\mathbf{y} - \boldsymbol{\mu}_i)]}{(2\pi)^{d/2} |\Sigma_i|^{1/2}}, \quad \Sigma_i \succ 0. \quad (5.3)$$

Its full covariance describes dependence among simultaneous features within a state. Gaussianity is an additional assumption about the emission shape; it is not part of the definition of an HMM. Nor does a state number determine a behavioural name: permuting all state labels consistently leaves the observation law unchanged.

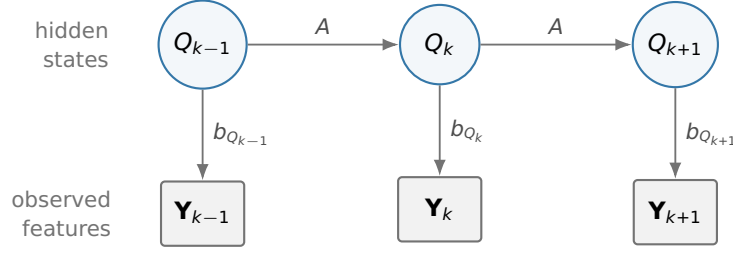


Figure 5.2. Conditional structure of an HMM. Horizontal arrows carry state persistence and transitions; vertical arrows describe how a state generates a measurement. Absence of an arrow between observations represents conditional independence given the hidden sequence, not independence of the observed features over time.

5.2.2 Likelihood and uncertainty about the states

Three tasks must be distinguished: evaluating how well given parameters explain an observed sequence, inferring its hidden states, and estimating the parameters themselves. For the first, the states must be summed out,

$$\mathcal{L}(\theta) = p(\mathbf{y}_{1:K} \mid \theta) = \sum_{q_{1:K} \in \mathcal{S}^K} p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta). \quad (5.4)$$

With continuous observations this is a density evaluated at the recorded vectors, rather than the probability of those exact vectors. Direct evaluation would enumerate $|\mathcal{S}|^K$ paths, but the Markov factorization allows their shared prefixes and suffixes to be accumulated recursively [17, Sec. III-A].

For one valid block of K observations, define forward and backward quantities

$$\alpha_1(i) = \pi_i b_i(\mathbf{y}_1), \quad \alpha_{k+1}(j) = b_j(\mathbf{y}_{k+1}) \sum_i \alpha_k(i) a_{ij}, \quad \mathcal{L} = \sum_i \alpha_K(i), \quad (5.5)$$

$$\beta_K(i) = 1, \quad \beta_k(i) = \sum_j a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j). \quad (5.6)$$

Their cost is $O(K|\mathcal{S}|^2)$, in contrast to enumerating $|\mathcal{S}|^K$ paths. The posteriors used in Baum–Welch are

$$\gamma_k(i) = \frac{\alpha_k(i) \beta_k(i)}{\mathcal{L}}, \quad \xi_k(i, j) = \frac{\alpha_k(i) a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j)}{\mathcal{L}}. \quad (5.7)$$

Here $\alpha_k(i)$ is the joint density of the observations up to k and state i , whereas $\beta_k(i)$ is the conditional density of the remaining observations given that state. Their product combines evidence on both sides of the decision. Thus $\gamma_k(i)$ is a smoothed state probability, while $\xi_k(i, j)$ gives the expected weight of a transition $i \rightarrow j$. Summing γ_k over states, or ξ_k over both indices, gives one; no most-probable path needs to be selected to calculate them.

5.2.3 Decoding a whole sequence

Choosing $\arg \max_i \gamma_k(i)$ separately at every time maximizes the expected number of correct individual states, but those choices need not form the most probable joint sequence and can even violate a forbidden transition. Viterbi instead retains the best path ending in each state [17, 19]. In logarithmic form,

$$d_1(j) = \log \pi_j + \log b_j(\mathbf{y}_1), \quad (5.8)$$

$$d_{k+1}(j) = \log b_j(\mathbf{y}_{k+1}) + \max_i [d_k(i) + \log a_{ij}], \quad (5.9)$$

$$\psi_{k+1}(j) = \arg \max_i [d_k(i) + \log a_{ij}]. \quad (5.10)$$

After choosing $\hat{q}_K = \arg \max_j d_K(j)$, backtracking through $\hat{q}_k = \psi_{k+1}(\hat{q}_{k+1})$ recovers the optimal sequence. Zero probabilities have log score $-\infty$; ties require a consistent convention. The forward recursion adds competing paths, while Viterbi takes their maximum, which explains why a decoded path and posterior uncertainty are different outputs. Both are conditional on the adopted model, so a high posterior does not independently establish that a behavioural interpretation is correct.

5.2.4 Learning from unlabelled observations

If states were observed, transitions could be counted and each Gaussian fitted to its assigned measurements. With hidden states, Baum–Welch replaces those unknown assignments by their posterior weights and iterates the calculation [20]. This is an expectation–maximization procedure: at iteration r , compute

$$\mathcal{Q}(\theta \mid \theta^{(r)}) = \mathbb{E}_{Q \mid \mathbf{y}, \theta^{(r)}} [\log p(Q, \mathbf{y} \mid \theta)], \quad \theta^{(r+1)} \in \arg \max_{\theta} \mathcal{Q}(\theta \mid \theta^{(r)}). \quad (5.11)$$

The E step uses forward–backward probabilities; the M step maximizes the resulting expected complete-data log likelihood. Its monotonicity follows from the EM lower-bound argument under an exact M step [21], but does not imply a global optimum. For B observation blocks treated as independent in the likelihood and indexed by b , the unregularized Gaussian M step gives the following updates, provided each denominator is positive:

$$\pi_i^{\text{new}} = \frac{1}{B} \sum_{b=1}^B \gamma_{b,1}(i), \quad a_{ij}^{\text{new}} = \frac{\sum_b \sum_{k=1}^{K_b-1} \xi_{b,k}(i, j)}{\sum_b \sum_{k=1}^{K_b-1} \gamma_{b,k}(i)}, \quad (5.12)$$

$$\boldsymbol{\mu}_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) \mathbf{y}_{b,k}}{\sum_{b,k} \gamma_{b,k}(i)}, \quad \boldsymbol{\Sigma}_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^{\top}}{\sum_{b,k} \gamma_{b,k}(i)}. \quad (5.13)$$

Initial-state counts use only the first observation of each block, and transition counts stop at its penultimate observation, so no transition is invented across a gap. These formulas explain the statistical fit; numerical scaling, regularization, initialization and stopping rules must also be specified for the actual estimator. The next sections make those choices for flight observations, rather than treating the general HMM theory as evidence that this particular three-state interpretation is correct.

5.3 From the cleaned flight to observations

The input is the final trajectory of Chapter 2, with local position (E, N, z) and the smoothed velocity and acceleration components. Each preprocessing segment defines a separate observation sequence, within which feature windows and hidden-state transitions cannot cross an acquisition gap or an excluded quality interval.

A common decision interval $\Delta = 10$ s gives the transition matrix the same physical meaning for every flight, restricting the analysis to native cadences no slower than this interval. At each decision time t_k , we first compute each scalar feature at native fixes and linearly interpolate those values. Denoting this interpolant by $\mathcal{I}[f]$, we average it over a centred window of width $W = 30$ s:

$$\bar{f}(t_k) = \frac{1}{W} \int_{t_k - W/2}^{t_k + W/2} \mathcal{I}[f](u) du, \quad \mathbf{X}_k = (\bar{v}_z, \bar{v}_h, |\bar{\omega}|, C_{\omega})(t_k). \quad (5.14)$$

The window suppresses fix-scale fluctuations while preserving manoeuvres over tens of seconds; its width is a chosen measurement scale rather than a measured universal thermalling period. Although boundaries lie on a 10-s grid, the overlapping 30-s windows prevent a claim of independent 10-s temporal resolution.

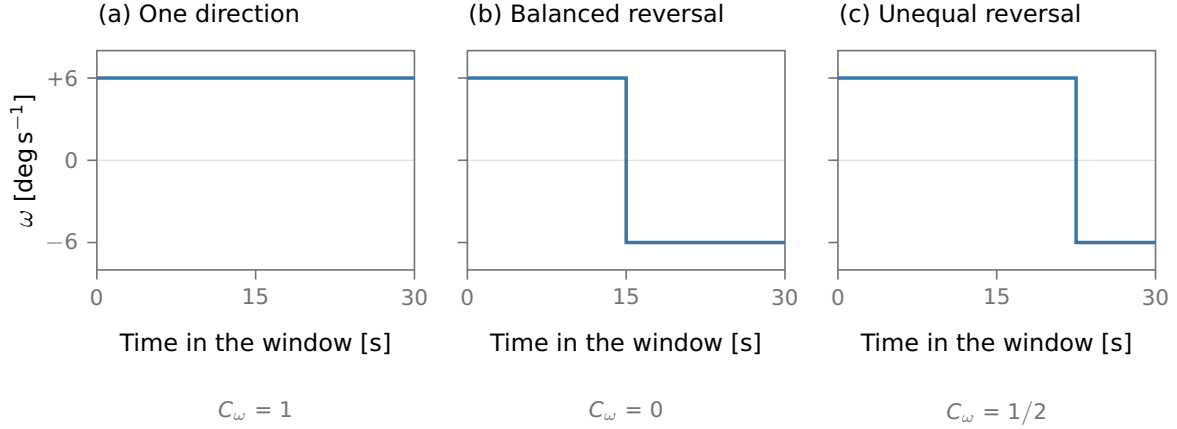


Figure 5.3. Turning amount and directional coherence measure different properties. All three idealized windows have $|\omega| = 6$ degrees per second, but their coherence differs because opposite turns cancel in the signed integral. The jumps illustrate limiting waveforms; the data calculation integrates piecewise-linear interpolants of native-fix values as specified in Eqs. (5.14) and (5.16). No phase label follows from coherence alone.

The horizontal speed and signed turn rate are

$$v_h = \sqrt{v_E^2 + v_N^2}, \quad \omega = \frac{v_E a_N - v_N a_E}{v_E^2 + v_N^2} \quad (v_h \geq 1 \text{ m s}^{-1}). \quad (5.15)$$

Below the stated speed floor the implementation sets $\omega = 0$, because numerical heading is unreliable when the horizontal velocity nearly vanishes. The fourth feature measures whether turning retains its sign:

$$C_\omega(t_k) = \frac{\left| \int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[\omega](u) du \right|}{\int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[|\omega|](u) du}, \quad 0 \leq C_\omega \leq 1, \quad (5.16)$$

The denominator is an accumulated angle in radians. The code sets $C_\omega = 0$ if it is at most machine epsilon (2^{-52} in float64), and clips floating-point round-off to $[0, 1]$. Turning in one direction throughout the window gives $C_\omega = 1$ even when its rate varies: coherence is not a measure of circularity or of lift by itself. At a reversal between two fixes, $\mathcal{I}[|\omega|]$ differs from $|\mathcal{I}[\omega]|$: the former is the quantity the code integrates. The pointwise inequality $|\mathcal{I}[\omega]| \leq \mathcal{I}[|\omega|]$ proves the stated bound on C_ω .

For example, equal-duration turns at $+6$ and -6 degrees per second cancel in the numerator and give $C_\omega = 0$, despite a large mean absolute turn rate. Keeping the positive sign throughout gives $C_\omega = 1$; spending three quarters of a window at $+6$ and one quarter at -6 gives $C_\omega = 1/2$ in the ideal piecewise-constant limit. Thus two different questions are measured: how much the heading turns, and how much opposite turns cancel. Figure 5.3 makes this distinction explicit.

A complete safe window is required. Windows touching reconstructed altitude, a derivative influenced by that reconstruction, or an edge derivative from preprocessing are unclassified, as are unsupported feature edges. Influence includes the entire Savitzky–Golay vertical fitting window, including its off-centre boundary fits. The support includes the native fixes bracketing interpolated window endpoints. Consequently even an unsafe fix just outside the nominal integration interval can invalidate it. Grey in the figures means unavailable support; there is no posterior confidence threshold that turns uncertain classifications grey.

Table 5.1. Quantities shown in the phase time-series figures. Feature windows are centred and 30 s wide; the plotted quantities retain physical units, before standardization.

Axis	Mathematical quantity	Unit / range
Altitude	$z(t_k)$ from the cleaned trajectory	m
Vertical speed	$\bar{v}_z(t_k)$, Eq. (5.14)	m s^{-1} , signed
Horizontal speed	$\sqrt{v_E^2 + v_N^2}(t_k)$	m s^{-1} , ≥ 0
Absolute turn rate	$(180/\pi) \omega (t_k)$	deg s^{-1} , ≥ 0
Coherence	$C_\omega(t_k)$, Eq. (5.16)	dimensionless, $[0, 1]$
Posterior	$\gamma_k(i)$ for $i \in \{\text{T, S, C}\}$	$[0, 1]$, sum = 1
Phase strip	$\hat{q}_k$ from the joint Viterbi path	T, S, C or unavailable

5.4 A probabilistic segmentation, and its assumptions

5.4.1 The fitted hidden Markov model

A separate continuous Gaussian HMM is fitted for each discipline, since speed and turning scale differ between paragliders and hang gliders. Write $Q_k \in \mathcal{S}$ for the hidden phase and $\mathbf{Y}_k$ for the features standardized by training-set means and standard deviations. More explicitly, for the n observations actually selected for fitting and feature j ,

$$\bar{x}_j = \frac{1}{n} \sum_{k=1}^n x_{kj}, \quad s_j^2 = \frac{1}{n} \sum_{k=1}^n (x_{kj} - \bar{x}_j)^2, \quad Y_{kj} = \frac{x_{kj} - \bar{x}_j}{s_j}. \quad (5.17)$$

A numerically constant feature is rejected; the population divisor n , rather than $n - 1$, matches the stored standardizer. The same $\bar{x}_j, s_j$ are used on held-out observations. The model parameters are an initial distribution π , a transition matrix A , and a full Gaussian covariance and mean for each component:

$$a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i), \quad \sum_j a_{ij} = 1, \quad b_i(\mathbf{y}) = \mathcal{N}_d(\mathbf{y}; \boldsymbol{\mu}_i, \Sigma_i). \quad (5.18)$$

Full covariance allows simultaneous vertical speed, horizontal speed and turning to be correlated within a phase. It does not model residual correlation *between* successive observations. The factorization in Eq. (5.2) assumes first-order phase memory, time-homogeneous parameters and conditionally independent emissions. The last is only approximate for overlapping windows; posterior confidence may therefore be sharper than independent observations would justify. Gaussian emissions also place some probability on physically impossible feature values, such as negative speed. They are a classification approximation, not a generative model of flight physics.

Baum–Welch applies expectation–maximization to the unlabelled training likelihood [20, 21]. Ten fixed-seed initializations are compared by training likelihood. Whole valid blocks enter the fit; the final observation of one block never contributes a transition to the next. The fitted scaler is frozen for validation and test flights.

A numerical component has no behavioural name. At present the component with greatest mean vertical speed is provisionally called climb. Among the other two components, transition maximizes $\mu_{v_h} - \mu_{|\omega|} - \frac{1}{2}\mu_{C_\omega}$ in standardized feature coordinates; the remaining component is called search. Human train labels will resolve this permutation. In particular, “search” currently also collects observations unexplained by the other two components: its behavioural interpretation requires a direct check.

5.4.2 Numerical estimation in the fitted model

Equations (5.12)–(5.13) describe unregularized maximum likelihood. The stored models were fitted with `hmmlearn` 0.3.3 [22]. Its default initial/transition Dirichlet parameters are one, and its mean weight is zero, so those updates reduce to the expressions above. For full covariance the default scalar `covars_prior` is 0.01, broadcast to every matrix entry. With $N_i = \sum_{b,k} \gamma_{b,k}(i)$, the implemented covariance update is therefore

$$\Sigma_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^\top + 0.01 \mathbf{1} \mathbf{1}^\top}{N_i}. \quad (5.19)$$

The added matrix is rank one: it does not impose a positive lower bound on every covariance eigenvalue. In particular, `min_covar` = 10^{-3} is added to the initial covariance diagonal; it is not enforced as an eigenvalue floor after every M step. This distinction matters for the very narrow coherence component.

Exact unregularized EM cannot decrease its likelihood, but convergence of the parameters to a stationary point requires further regularity conditions; a mixture can approach a singular covariance. The software’s stopping flag is also true when the iteration cap is reached or the last likelihood change is negative. That flag in an old artifact is not evidence of convergence. New fits record convergence only for a finite, non-negative last gain below the tolerance, reject non-finite final scores, and still retain all restart scores for inspection. For restart r , write $\ell_r = \sum_b \log \mathcal{L}_b(\hat{\theta}_r)$ on the same training observations. Its reported relative score is $(\ell_r - \max_u \ell_u)/n \leq 0$, in natural-log units per fitted observation, whereas the final EM gain is $\ell^{(m)} - \ell^{(m-1)}$ before normalization. These are numerical fit diagnostics, not accuracy scores. Ten fixed-seed restarts are ranked by finite training likelihood; a restart can be selected without satisfying the stated convergence criterion. Training uses scaled forward–backward arithmetic; current Viterbi and posterior decoding use logarithms to avoid underflow.

5.4.3 Why the current decoder differs from the original fit

The original fit used all four features. Its climb component concentrates coherence extremely close to one: the physical standard deviations are approximately 0.0000889 for paragliders and 0.0000604 for hang gliders (Table 5.3). A circling climb with a small turn reversal can therefore receive negligible climb likelihood. More persistent transitions alone cannot repair this emission problem.

The current decoder integrates coherence out of the original four-dimensional Gaussian. If $\mathbf{U} = (Y_1, Y_2, Y_3)$ and $V = Y_4$, then

$$b_i^{(3)}(\mathbf{u}) = \int_{\mathbb{R}} b_i^{(4)}(\mathbf{u}, v) dv = \mathcal{N}_3(\mathbf{u}; \boldsymbol{\mu}_{i,U}, \Sigma_{i,UU}). \quad (5.20)$$

Thus the retained mean entries and principal covariance submatrix define the new likelihood. This is marginalization, not conditioning on perfect coherence, and not a new three-dimensional fit. Coherence remains recorded and plotted for inspection. A separately refitted three-dimensional model remains a validation comparison: the original parameters were selected using the problematic fourth feature.

The coherence mean and width in Table 5.3 undo the training standardization: $\mu_{C,i}^{\text{phys}} = \bar{x}_4 + s_4 \mu_{i,4}$ and $\sigma_{C,i}^{\text{phys}} = s_4 \sqrt{\Sigma_{i,44}}$. They describe a fitted Gaussian component, not the sample mean and standard deviation of decisions labelled climb.

A second, independent choice softens the phase sequence. With $s_i = a_{ii} < 1$ and $q_{ij} = a_{ij}/(1 - s_i)$ for $j \neq i$, the decoding matrix is

$$\tilde{a}_{ii} = \tilde{s}_i, \quad \tilde{a}_{ij} = (1 - \tilde{s}_i)[(1 - w)q_{ij} + wq_{ij}^*], \quad j \neq i. \quad (5.21)$$

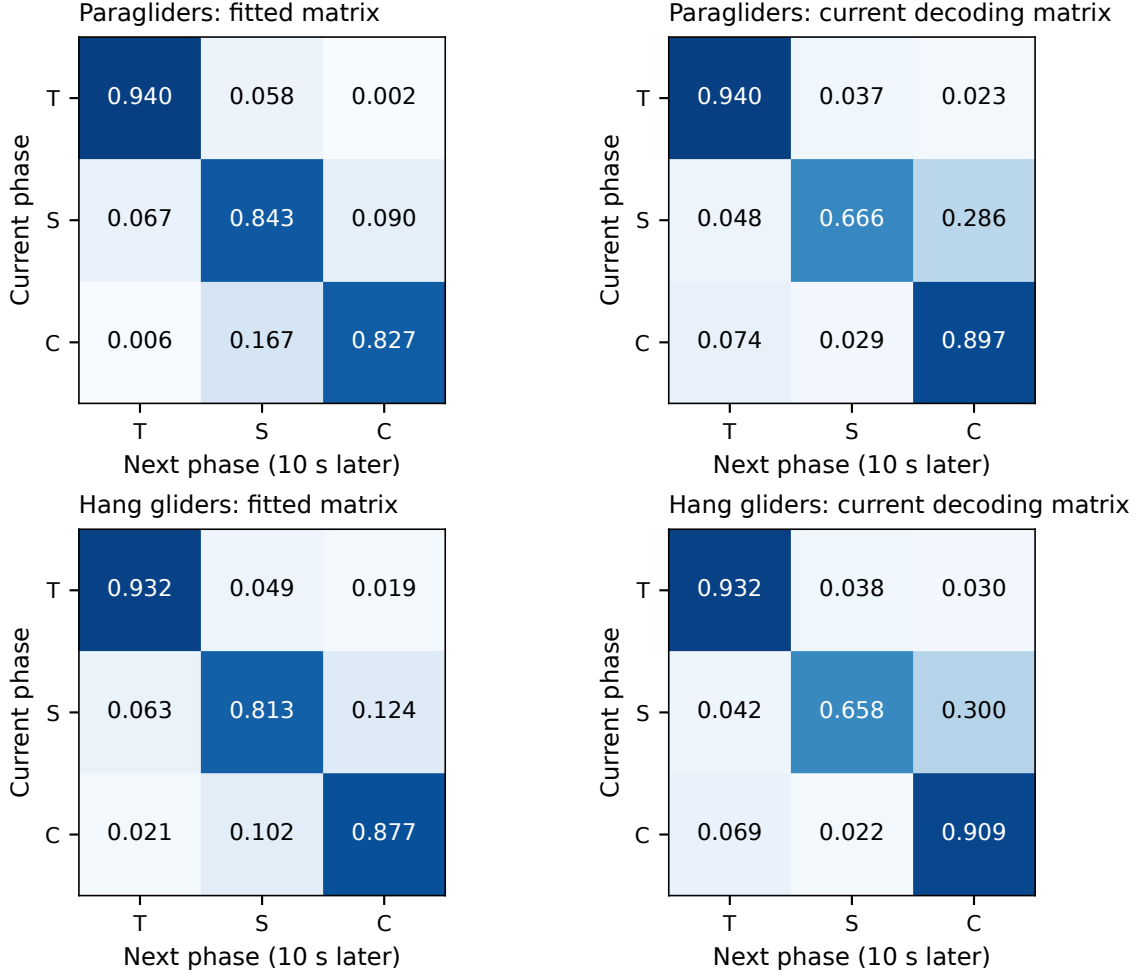


Figure 5.4. Original fitted and current decoding transition matrices, ordered as transition (T), search (S), climb (C). Every entry is a probability per 10-s decision. The right column includes the declared sequence preference and cannot be read as an empirical frequency of pilot behaviour.

Here $w = 0.75$. The target exits from transition split equally between search and climb, allowing direct entry into lift; from search and climb, the next phase in $T \rightarrow S \rightarrow C \rightarrow T$ receives target probability 0.95. The persistence rule is

$$\tilde{s}_i = \begin{cases} s_i + w(e^{-\Delta/\theta_S} - s_i), & i = S, \\ s_i + w \max(0, e^{-\Delta/\theta} - s_i), & i = T, C, \end{cases} \quad \theta_S = 20 \text{ s}, \quad \theta = 120 \text{ s}. \quad (5.22)$$

For an absorbing fitted row ($s_i = 1$), the code starts from equal exit weights and caps the new persistence at $1 - \epsilon_{\text{mach}}$ to retain possible exits. These scales and weights are provisional decoder settings. They are neither fitted behavioural durations nor minimum run lengths. Figure 5.4 shows both matrices so that learned structure and imposed preference can be distinguished. Once a semantic prior is active, changing component names requires decoding again, rather than merely recolouring an existing path.

For both the original and modified model, Viterbi chooses the entire most probable path rather than the most likely phase at each isolated time [19]:

$$\hat{q}_{1:K} = \arg \max_{q_{1:K}} p(q_{1:K} \mid \mathbf{y}_{1:K}), \quad \gamma_k(i) = \Pr(Q_k = i \mid \mathbf{y}_{1:K}). \quad (5.23)$$

The hard path and posterior γ use the same effective transition matrix and marginal emission.

Table 5.2. Archive-scale HMM fit and decoding census. A sequence is a contiguous classifiable feature block; restart numbers are one-based.

Discipline	Fit flights	Fit sequences	Fit points	Decoded points	Restart
Paragliders	972	11,598	952,274	171,663,465	6/10
Hang gliders	874	8,775	1,000,000	7,301,634	8/10

Table 5.3. Support and numerical diagnostics of the segmentation. Excluded time uses all retained cleaned-segment duration as its denominator. Coherence widths belong to the original four-dimensional climb component.

Quantity	Paragliders	Hang gliders
Emitted archive decisions	171,663,465	7,301,634
Excluded cleaned duration	8.24%	5.88%
Climb C_ω mean (original fit)	0.9999986	0.9999996
Climb C_ω standard deviation	0.0000889	0.0000604
Final EM log-likelihood change	-0.0755	-2.15
Annotated intervals supplied	0	0

“Smoothed posterior” refers to inference using observations before and after k , not to an extra filter applied to probabilities.

5.5 What has been checked

5.5.1 Archive support and reproducible examples

The original models were fitted and applied to both archives. Table 5.2 records the actual fitting and decoding counts. Fitting uses a deterministic subset of whole training blocks up to the memory cap; longer selected blocks contribute more observations, so this is not equal weighting of flights. The saved archive uses the current decoder. The four-dimensional path is reconstructed from the same fitted parameters as a comparison. The figures in this chapter use the current configuration and saved features, with complete valid blocks decoded before an eight-minute display excerpt is taken. A shortened figure therefore does not shorten the sequence used for inference.

The selected fits do not both satisfy a finite, non-negative last gain below tolerance (Table 5.3). The software stopping flags therefore cannot establish convergence of these fits; covariance and refitting sensitivity remain to be checked.

Coverage has a separate denominator from classification accuracy:

$$f_{\text{unclassified}} = 1 - \frac{\text{duration covered by classified decision cells}}{\text{total retained cleaned-segment duration}}. \quad (5.24)$$

Decision cells $[t_k - \Delta/2, t_k + \Delta/2)$ are clipped at segment boundaries; acquisition gaps contribute to neither duration. For exclusion reason r , the plotted percentage is $100 \sum_k |I_k| \mathbf{1}\{r_k = r\} / D_{\text{clean}}$, with I_k the clipped decision cell and D_{clean} the denominator in Eq. (5.24). Feature-edge status takes precedence over a quality mask if both apply. Segments with native cadence above 10 s contribute their full duration to the cadence category; remaining tails outside any cell form the final category. The categories are disjoint. The denominator does include retained segments too sparsely sampled for classification. The excluded fractions are 8.24% for paragliders and 5.88% for hang gliders, chiefly from unsafe feature support (Fig. 5.5). These masks are shared by the two decoders compared on that snapshot. The feature masks include the propagated reconstruction support of the vertical Savitzky–Golay derivatives. Their

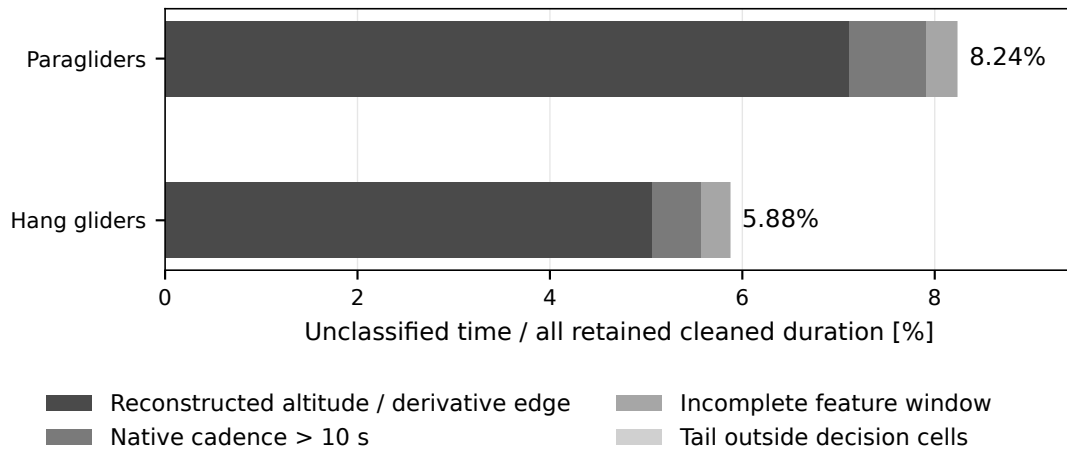


Figure 5.5. Why retained cleaned time lacks a phase label. Fractions use the same duration denominator in Eq. (5.24); they are not fractions of flights or fixes. Windows touching reconstructed altitude or edge derivatives dominate the excluded time.

coverage says how much time is available for subsequent phase statistics, not whether any phase label is correct or whether exclusion is behaviourally unbiased.

The two examples are selected by a fixed priority from validation flights, independently of their apparent phase composition. Recomputing labels with the stored model reproduces the archived paths. Their current labels and posterior curves then use Eqs. (5.20)–(5.22). The audit records the selected identities, model hashes, configuration, and changed-label counts. Thus each plotted label can be traced to its observations and decoder settings.

The paraglider excerpt shows sustained turning and altitude recovery after an initial transition, with a climb label that persists through brief changes in vertical speed and coherence. The hang-glider excerpt is less clear-cut: part of a sustained ascent is assigned to search, and the posterior divides between search and climb. Positive vertical speed is therefore not equivalent to the fitted climb state. This could reflect a difference in turning behaviour, but the figure alone cannot resolve the behavioural name of that component. It gives a concrete interval for human validation rather than an example selected to display apparently perfect classification.

The development example in Fig. 5.8 separates the two decoder changes. On its full preprocessing segment, the four-dimensional reference assigns 6 decisions to climb, the sequence-only variant assigns 6, and the current three-dimensional marginal assigns 32. The comparison shows how the coherence coordinate and sequence preference affect the same observations. The example helped motivate the change, so the comparison is a regression diagnostic, not independent evidence of improved accuracy. Neither a longer green interval nor a smaller number of switches is, by itself, a success criterion.

5.5.2 The manual check that is still needed

The prepared blinded pack contains 40 candidate flights: per discipline, four training candidates, eight validation candidates and eight test candidates. Human reference intervals are needed before reporting accuracy, confusion matrices or a definitive physical meaning for the search state. No human reference intervals were supplied to this audit, so it reports no independent classification scores. The interactive tool is available: select an altitude interval and mark transition, search or climb; edits are saved automatically on the 10-s decision grid. The plan view and kinematics are visible, while model predictions are hidden. The pack must match

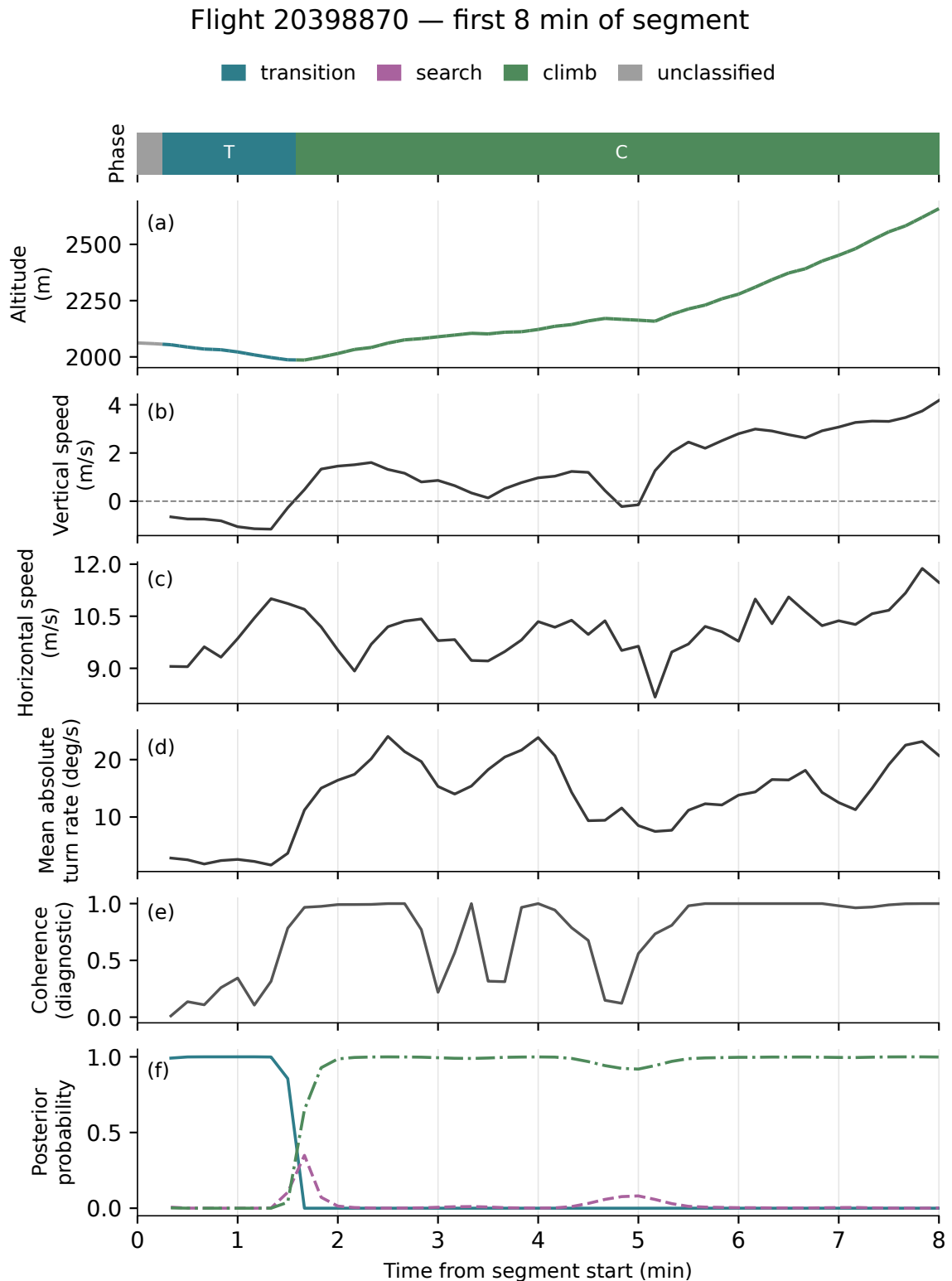


Figure 5.6. Eight minutes of a validation paraglider segment, decoded using the current three-feature marginal and sequence policy. The phase strip, altitude, separate feature axes and posterior probabilities share a time axis marked at one-minute intervals; decisions remain 10 s apart. Table 5.1 defines every displayed quantity. Coherence is shown as a diagnostic and does not enter the current emission likelihood. Grey indicates unavailable support. Phase names remain provisional.

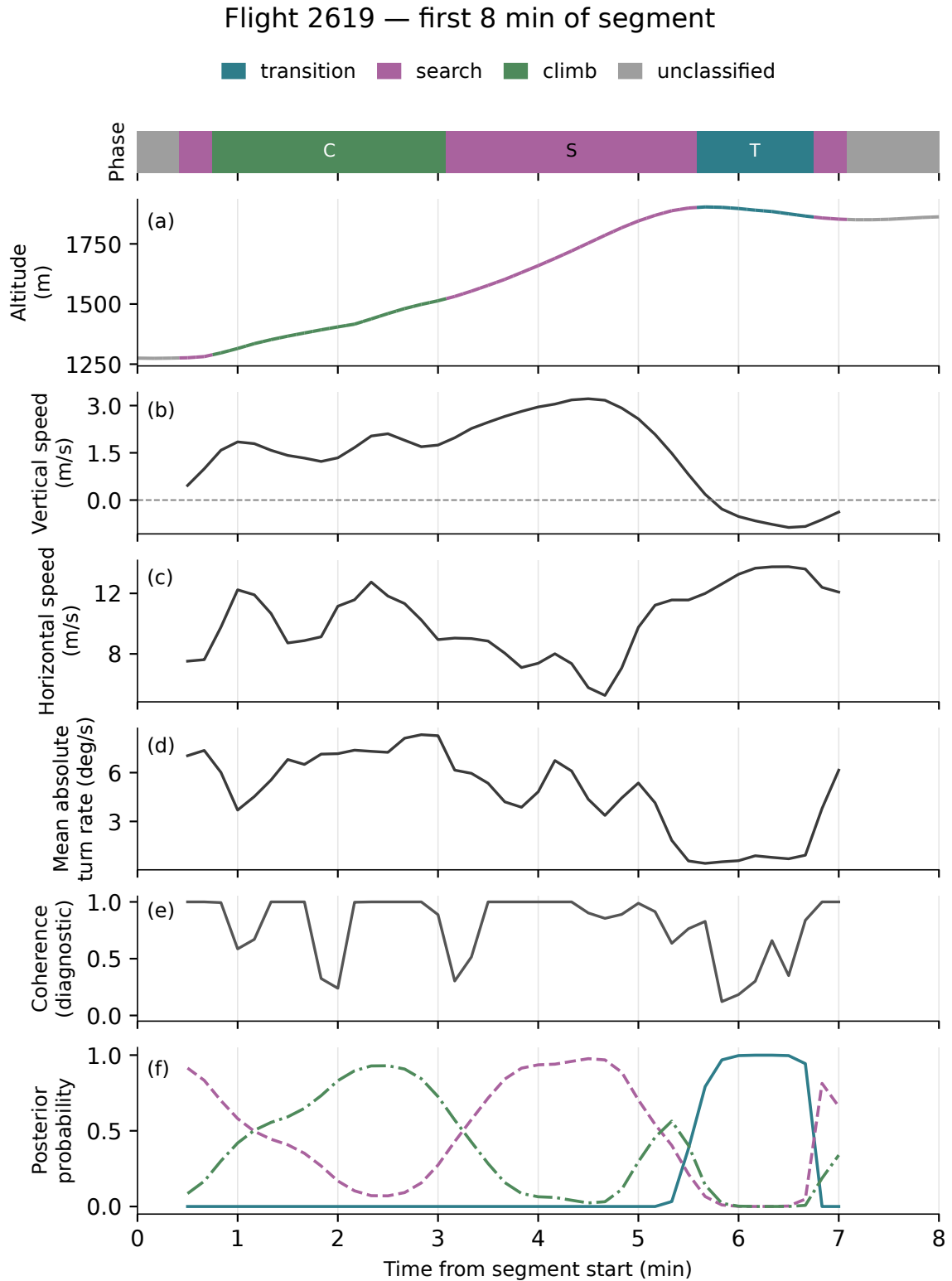


Figure 5.7. The corresponding eight-minute hang-glider validation excerpt. It uses its own fitted model and standardizer, with the same feature definitions and display convention as Fig. 5.6. A high posterior expresses certainty within the adopted model; it is not an independently measured probability of correct behaviour.

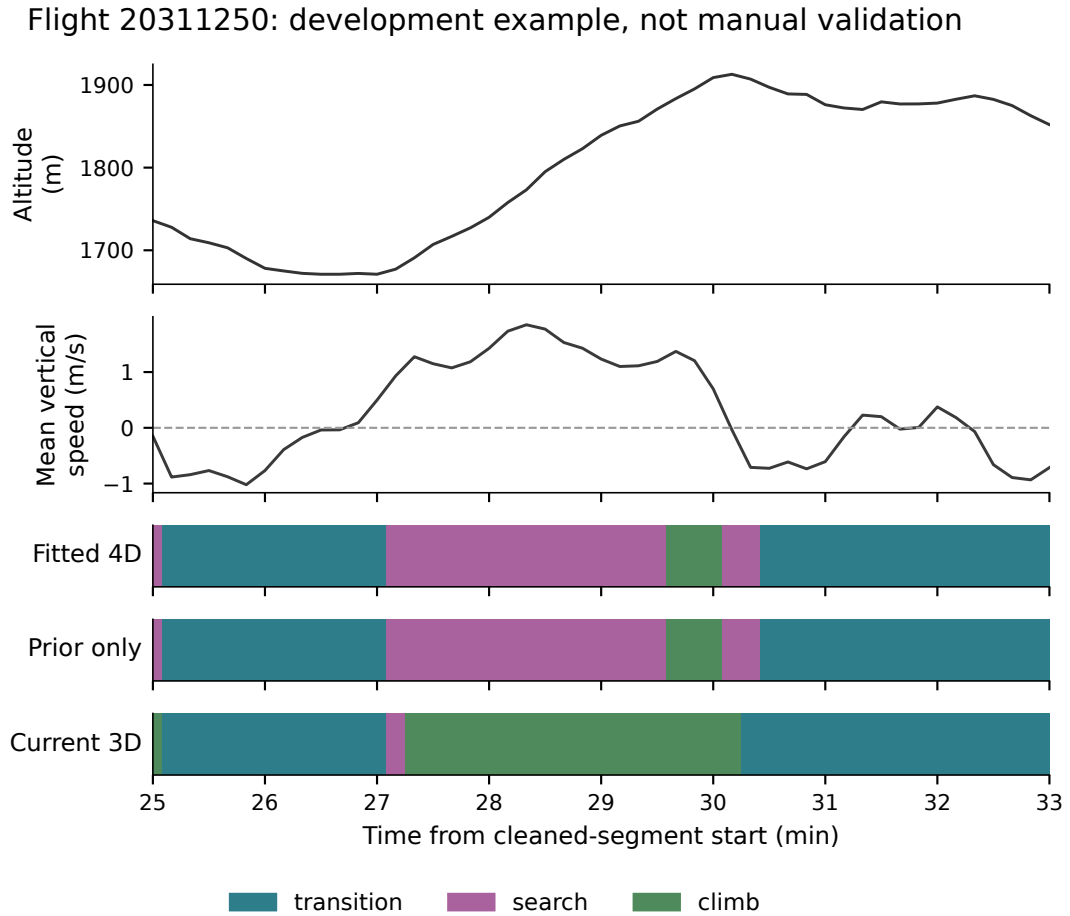


Figure 5.8. An eight-minute development excerpt from flight 20311250. The aligned phase strips compare the fitted four-dimensional reference, a change of sequence policy alone, and the current three-dimensional marginal with that policy. The altitude and vertical speed provide physical context. No human reference labels are present.

the cleaned archive used for evaluation; the regeneration procedure creates a new dated pack while preserving the previous pack and any labels already written.

Clearly interpretable intervals should be labelled across the whole candidate window; ambiguous boundaries can remain unlabelled. The fraction left unlabelled must accompany the scores, since evaluating only easy interiors would otherwise conceal the difficult part of segmentation. A second independent annotation of a subset will measure human agreement and boundary ambiguity. That agreement study has not yet been performed.

The deterministic 70/15/15% split operates at flight level, so no flight is shared between fitting and nominal holdout sets. This alone does not guarantee an untouched test: development flights 20275040 and 20279877 were inspected despite belonging to the nominal test partition. They, and hang-glider example flight 975, are excluded from the prepared annotation test pack and must remain outside final accuracy claims. Repeated pilots, days or sites can also make flights statistically related; generalization to a new region or season requires an additional grouped holdout, rather than being inferred from a random flight split.

Train annotations resolve state naming; validation annotations compare the original four-dimensional model, the marginal decoder with and without the sequence preference, and a refitted three-dimensional model. Model choices are then frozen before test scores are opened. For human label i and inferred label j , let n_{ij} count classified, annotated decisions. The core

metrics are

$$\begin{aligned} \text{Precision}_i &= \frac{n_{ii}}{\sum_j n_{ji}}, & \text{Recall}_i &= \frac{n_{ii}}{\sum_j n_{ij}}, \\ F_{1,i} &= \frac{2n_{ii}}{\sum_j n_{ji} + \sum_j n_{ij}}, & F_{1,\text{macro}} &= \frac{1}{3} \sum_i F_{1,i}. \end{aligned} \quad (5.25)$$

If a precision, recall or F1 denominator is zero, the evaluator assigns that metric zero and retains the phase in the three-state macro average. The corresponding accuracy is $\text{Acc} = \sum_i n_{ii} / \sum_{ij} n_{ij}$ on the same annotated, classified decisions; the evaluator requires at least one annotated, classified decision. On that nonempty set all these classification scores lie in $[0, 1]$. The confusion matrix reveals whether search is systematically confused with climbing or transition; macro-F1 gives a rare phase the same weight as a frequent one. Accuracy alone can be dominated by the most occupied phase. Bootstrap intervals [1] obtained by resampling whole flights preserve within-flight dependence, although they do not remove dependence between related flights. The existing evaluator provides these classification metrics. Boundary errors, annotation coverage and between-observer agreement remain additional validation measurements to complete; no values are supplied before labels exist.

5.6 What the segmentation will let us measure

The next empirical chapter will characterize all three phases before selecting a stochastic transport model. For a phase occurrence $I = [a, b)$, useful primitive quantities are its duration $D = b - a$, horizontal path length L , net displacement R , and altitude change:

$$L = \int_a^b v_h(t) dt, \quad R = \|\mathbf{r}(b) - \mathbf{r}(a)\|, \quad \Delta z = z(b) - z(a), \quad \mathcal{P} = R/L \in [0, 1] \quad (L > 0). \quad (5.26)$$

Here $\mathbf{r} = (E, N)$ and $v_h = \|\dot{\mathbf{r}}\|$ refer to the same horizontal curve; this consistency is needed for the triangle inequality $R \leq L$. The discrete implementation must use a matching position-based path length rather than assume that separately smoothed derivatives satisfy the inequality exactly. The ratio $\mathcal{P}$ distinguishes local circling from directed progress without assuming that either occurs in a particular phase. The ground displacement of a climb can be substantial in wind; search is not automatically an immobile waiting state. Distributions of $(D, L, R, \Delta z)$ and phase-conditioned velocity correlation will quantify those differences.

A phase-conditioned MSD requires an explicit conditioning convention. One useful choice keeps the *entire* displacement interval inside a single occurrence of phase i :

$$M_i(\tau) = \frac{\sum_I \int_{a_I}^{b_I - \tau} \|\mathbf{r}(t + \tau) - \mathbf{r}(t)\|^2 dt}{\sum_I (D_I - \tau)_+}, \quad I \text{ an occurrence of phase } i. \quad (5.27)$$

Integrals for $D_I \leq \tau$ contribute zero, and the observable is defined only when its denominator is positive. This weights eligible time origins; long occurrences contribute more, and short occurrences disappear at large lags. A plot must therefore show the contributing occurrences and durations, and be checked against equal-occurrence or equal-flight weighting. Conditioning only the two endpoints would define a different observable because it could cross other phases.

Prior phase-resolved results [16] supply comparisons, not values that the segmenter must be tuned to recover. Agreement in phase MSDs or time budgets cannot independently validate a classifier designed around the same kinematic expectations. Differences in sampling, geography, wind and the phase definitions must be considered before attributing a disagreement to the dynamics.

5.6.1 Keeping model assumptions separate from measured physics

An HMM imposes a geometric latent dwell-time law. With decoding persistence $\tilde{a}_{ii}$, its model prediction is

$$\Pr(D_i = m\Delta) = \tilde{a}_{ii}^{m-1}(1 - \tilde{a}_{ii}), \quad \mathbb{E}[D_i] = \frac{\Delta}{1 - \tilde{a}_{ii}}. \quad (5.28)$$

for $m = 1, 2, \dots$ and $0 \leq \tilde{a}_{ii} < 1$. A Viterbi run-length histogram is not a sample of observed physical durations independent of this assumption. It depends on the emissions and the decoder as well as on behaviour. Runs cut by a gap, quality mask or feature boundary are censored; merely discarding them can bias the surviving duration distribution. Testing non-geometric phase durations requires human reference intervals and a sensitivity analysis against a model with explicit duration distributions, such as a hidden semi-Markov model [23]. The imposed exit preferences likewise prevent interpreting the current decoded phase order as unqualified evidence of physical memory.

The later transport model must reproduce both these phase observables and the global constraints of Chapter 4. In particular, the duration of a moving leg and its displacement should be studied jointly. A constant-speed Lévy walk couples a leg's length to that *same leg's travel time*; dependence between its length and a *subsequent* climb or search is a separate question. A stereotyped phase cycle by itself also does not disprove renewal at the level of complete cycles. Renewal requires independence of successive cycle variables, which must be tested at the chosen event scale.

5.7 Chapter summary

The chapter has built the hidden-state model from its probability law, connected each feature to a measured kinematic quantity, and separated parameter fitting from sequence decoding. The current labels use a three-feature marginal of a fitted four-feature Gaussian HMM and an explicit sequence preference. Coverage, numerical diagnostics, short trajectory excerpts and a decoder comparison make those choices inspectable, without supplying an independent estimate of behavioural accuracy. The current archive leaves 8.24% of retained paraglider time and 5.88% of hang-glider time unclassified. The narrow coherence components and negative final likelihood changes remain numerical limitations, while the ascent labelled search in the hang-glider example exposes a behavioural ambiguity that posterior probabilities cannot settle.

The next step is the independent phase validation. Once its errors and boundary uncertainties are known, phase durations, displacements and correlations can be used alongside the global observables to test a physical transport model.

Chapter 6

The Vilpellet flight-phase segmentation

This chapter documents the second segmentation currently available in the analysis: the implementation transcribed from Jérémie Vilpellet’s supplied inference code. It provides a reference for reading, running and comparing that method with the Gaussian HMM of Chapter 5. Both methods assign transition, search and climb to a trajectory. A choice between them for subsequent phase-conditioned transport measurements remains open.

The probabilistic foundations are those of Section 5.2: the joint law in Eq. (5.2), Viterbi decoding in Eq. (5.10), and Baum–Welch estimation in Eq. (5.12). Here the observations are binary triples and the emission law is categorical. The present implementation performs inference under supplied parameters; it does not fit them again. The broader phase-resolved study is described by Vilpellet et al. [16], but the operational details below follow the executable implementation and its configuration.

6.1 Input, fitted parameters and admissible trajectories

The supplied inference script, `flight_phase_inference_minimal.py`, contains fixed emission, transition and initial-state probabilities. The parameter provenance recorded with the port identifies a Baum–Welch fit to 2019 paraglider flights from four Alpine take-off sites. Those parameters are transferred unchanged to the present flights, including hang gliders. Only the straightness threshold is discipline-specific. This chapter describes that transfer; it does not independently reproduce the original training or establish that its population represents the present archive.

For the default `cleaned` input, the two segmenters receive the same retained (E, N, z, t) trajectory from Chapter 2. Here z is the pipeline’s altitude coordinate. Each preprocessing segment is decoded separately: neither a feature window nor a hidden-state transition crosses its boundary. The optional `raw_gnss` command-line input instead constructs local east, north and up coordinates directly from GNSS positions using the first fix as origin. In that path, vertical differences are differences of local up coordinates. Changing the input path therefore changes the observation construction, not just the file reader.

The protocol is expressed in numbers of fixes. Its eligibility conditions are shown in Table 6.1. The flight-level size check precedes the segment-level checks. The entry point first orders fixes by segment and time; the time conditions refer to those ordered sequences. The cadence check uses the *mean* interval, not a test that all intervals are equal. Uniformity within the default cleaned input comes from preprocessing.

At a one-second cadence the flight guard corresponds approximately to an hour, but it is a fix-count rule, not a one-hour duration threshold. Likewise, a 30-fix window covers a different

Table 6.1. Eligibility in the current implementation. A failed check leaves the affected fixes **unclassified**, with a recorded reason.

Scope	Required condition
Whole flight	At least 3,600 retained fixes.
Each segment	At least 30 fixes.
Ordered times	Unique, strictly increasing timestamps with finite increments.
Mean cadence	$\bar{\Delta t} < 1.2$ s.

physical interval when the cadence differs from one second. The gate rejects slower logging; it does not make all accepted cadences identical. Consequently the two segmenters need not label the same population, even when they start from the same cleaned archive.

6.2 From geometry to three binary observations

Let k index fixes within one admissible segment. With the default `pre_smooth` option enabled, each coordinate receives a centred Gaussian rolling mean of 6 fixes, with Gaussian standard deviation equal to one sixth of the window width. Denote the resulting coordinates by $(\tilde{E}_k, \tilde{N}_k, \tilde{z}_k)$. Full windows are required for this operation, so its incomplete endpoint windows remain missing. On cleaned input this Gaussian averaging follows the pipeline’s Savitzky–Golay smoothing. Keeping it preserves the supplied feature definition, but introduces a second smoothing stage relative to applying the reference code directly to raw GNSS positions.

Define the incoming heading and the signed change towards the outgoing step by

$$\theta_k = \text{atan2}(\tilde{N}_k - \tilde{N}_{k-1}, \tilde{E}_k - \tilde{E}_{k-1}), \quad (6.1)$$

$$r_k = \text{wrap}_{[-\pi, \pi]}(\theta_{k+1} - \theta_k), \quad \bar{r}_k = \mathcal{M}_6[r]_k. \quad (6.2)$$

Here $\mathcal{M}_w$ denotes the centred, unweighted rolling mean over w fixes, using the available nonmissing entries at a boundary. The quantity named `radius_curvature_lat_long_sgn` in the supplied code is this signed *angle increment*, in radians per fix. It is neither a geometric radius nor the angular velocity of Eq. (5.15): it is not divided by elapsed time. For vertical motion the implementation does divide by time,

$$v_{z,k} = \frac{\tilde{z}_k - \tilde{z}_{k-1}}{t_k - t_{k-1}}, \quad \bar{v}_{z,k} = \mathcal{M}_{30}[v_z]_k. \quad (6.3)$$

This finite-difference velocity is computed for the segmenter; it is not taken from the preprocessing pipeline’s stored vertical-velocity column.

The observation at fix k is the ordered triple $\mathbf{y}_k = (B_k^{\text{straight}}, B_k^{\text{up}}, B_k^{\text{sign}})$. With $\beta = 0.9$ and straightness threshold a , its first two bits are

$$I_k = 1\{-a < \bar{r}_k < a\},$$

$$B_k^{\text{straight}} = 1\{\mathcal{M}_{30}[I]_k > \beta\}, \quad (6.4)$$

$$B_k^{\text{up}} = 1\{\bar{v}_{z,k} > 0\}. \quad (6.5)$$

The configured values of a are 0.20960958 for paragliders and 0.17398039 for hang gliders, in radians per fix. The configuration also supplies 0.10000000 for sailplanes, which are not analysed in the present archive. The straightness inequality is strict: with a complete 30-fix window, more than 90% means at least 28 straight fixes. Positive mean ascent alone sets only one bit; it does not assign the climb phase.

For the third bit, take the 30 values of $\bar{r}$ in the window from $k - 15$ through $k + 14$. Select the 15 largest absolute values, retaining their signs. If n_k^+ and n_k^- count the positive and negative members of this selected set, then

$$B_k^{\text{sign}} = 1 \left\{ \max(n_k^+, n_k^-) \geq \lceil \beta 15 \rceil \right\}. \quad (6.6)$$

Thus at least 14 of the selected increments must turn the same way. Keeping the largest turns reduces the contribution of almost straight steps, whose tiny signed increments provide little directional information. A zero increment counts as neither positive nor negative. The code reflects the increment series at the endpoints for this operation; it does not shorten this window.

These boundary conventions are part of the implemented model. Comparisons with missing values evaluate false, and no additional feature-edge mask is applied before decoding. Gaussian-smoothed endpoint gaps therefore do not automatically become unclassified fixes. On the default path the Vilpellet wrapper also does not use `interpolated` or `z_reconstructed` as an additional exclusion mask. That differs from the quality masking described in Chapter 5, even though both methods share the same retained trajectory.

6.3 A categorical HMM and its decoded labels

The 3 binary indicators have 8 possible joint values. Their lookup index is

$$c(\mathbf{y}) = 4B^{\text{straight}} + 2B^{\text{up}} + B^{\text{sign}}, \quad b_i(\mathbf{y}) = p_{i,c(\mathbf{y})}. \quad (6.7)$$

Each component has one probability vector over all eight triples. This is a joint categorical distribution: the implementation does not multiply three independent Bernoulli probabilities. Within-fix dependence between the bits is represented by that table. The usual HMM conditional independence across observations remains an assumption, while the overlapping feature windows induce additional temporal structure, as discussed for the Gaussian features in Chapter 5.

Table 6.2. Supplied joint emission probabilities, in lookup order and rounded to three decimals. Printed zeros can represent small positive probabilities; inference uses the full values in the configuration.

Component	000	001	010	011	100	101	110	111
0 (climb)	0.000	0.001	0.000	0.999	0.000	0.000	0.000	0.000
1 (search)	0.326	0.178	0.496	0.001	0.000	0.000	0.000	0.000
2 (transition)	0.000	0.000	0.000	0.000	0.738	0.014	0.241	0.006

The component order in this chapter is $(0, 1, 2) = (\text{C}, \text{S}, \text{T})$. In that order, the supplied transition matrix and initial distribution are

$$A \simeq \begin{pmatrix} 0.98961 & 0.00346 & 0.00692 \\ 0.00667 & 0.99000 & 0.00333 \\ 0.00100 & 0.00712 & 0.99188 \end{pmatrix}, \quad (6.8)$$

$$\pi \simeq (5.660756 \times 10^{-2}, 9.433924 \times 10^{-1}, 5.887000 \times 10^{-29}). \quad (6.9)$$

Every segment starts afresh from this π . The matrix is applied per fix; its persistence has approximately a one-second interpretation on one-second data. Its nonzero off-diagonal entries allow all phase transitions, rather than enforcing a transition–search–climb cycle. The initial probability of component 2 is very small but positive. No sequence-prior modification from Chapter 5 is applied to this model.

Component 0 places almost all emission mass on 011: not persistently straight, positive mean ascent and persistent turn sign. Component 2 favours 100 and 110: persistently straight motion without a persistent turn sign. Component 1 spreads mass over the remaining turning patterns. These emission profiles motivate the names climb, transition and search respectively. The supplied minimal script has a naming line that exchanges climb and transition; its numeric component paths are unaffected. The present configuration uses the ordering supported by the emission profiles and the accompanying code documentation. Setting `literal_minimal_script` reproduces the exchanged naming line. Reproducing numeric states and choosing their behavioural names are therefore separate checks.

Viterbi computes the most probable component path using the recurrence already derived in Eq. (5.10). The implementation works in probability space and divides all state scores by their sum at each step when that sum is positive. This common rescaling preserves the maximizing path. Ties choose the lowest component index. The optimized three-state recursion preserves the same arithmetic and is tested against the general recursion.

The published phase column is then obtained by a centred majority vote over 90 decoded components, using partial windows at the endpoints. Ties again choose the lowest numeric component. This post-processing can move phase boundaries and erase short runs; the resulting sequence is generally no longer the Viterbi-optimal path of the original HMM. Its nominal per-fix labels do not imply independent one-second temporal resolution. The pipeline does not compute forward-backward posterior probabilities for these final labels.

6.4 Retained runs, saved products and an example

The supplied code recommends discarding the flight tail, concentrating on long runs, and examining searches followed by a climb. The port exposes these as optional run-selection settings. In the present configuration the tail cut and minimum run duration are both zero, and the following-climb condition is disabled. Thus none of these recommendations removes an otherwise decoded fix. They should not be confused with the active eligibility gates of Table 6.1.

A physical track is broken at a preprocessing segment change, a nonpositive time step, or a gap exceeding 1.5 times that segment's median positive step. A phase run also breaks whenever its label changes. The saved run duration is $t_{\text{end}} - t_{\text{start}}$, and censoring flags record whether either endpoint coincides with a physical track boundary. These flags matter when estimating duration distributions, as already explained in Section 5.6.

The default archive export writes phase intervals and segment coverage under `derived/segmentation/vilpellet/`. It omits the optional fix-level table. Intervals supply the classified labels; their complement and the coverage table identify unclassified support. Geometry remains in `fixes.parquet`. By contrast, the Chapter 5 decision table also stores features, quality masks and posterior probabilities. This storage difference is not a missing Vilpellet segmentation: those posterior diagnostics are not produced by this decoder.

Figures 6.1 and 6.2 use the same cleaned paraglider flight, 2021-09-01_20308851, with 21,125 fixes spanning 352.1 minutes. Both methods are run on this geometry. Table 6.3 reports phase-time shares: each fix represents the interval to the next fix in the same physical track, with the terminal fix assigned that track's median step. This terminal-step convention differs from summing endpoint-to-endpoint run durations.

Among the 20,840 fixes both decoders classify, their phase names agree on 77.4%. This measures agreement between two algorithms, not accuracy: neither label set is an independent behavioural reference. The larger or smaller search fraction alone cannot decide which interpretation is more useful for the intended transport observables.

Flight 2021-09-01 20308851, one cleaned trajectory labelled twice

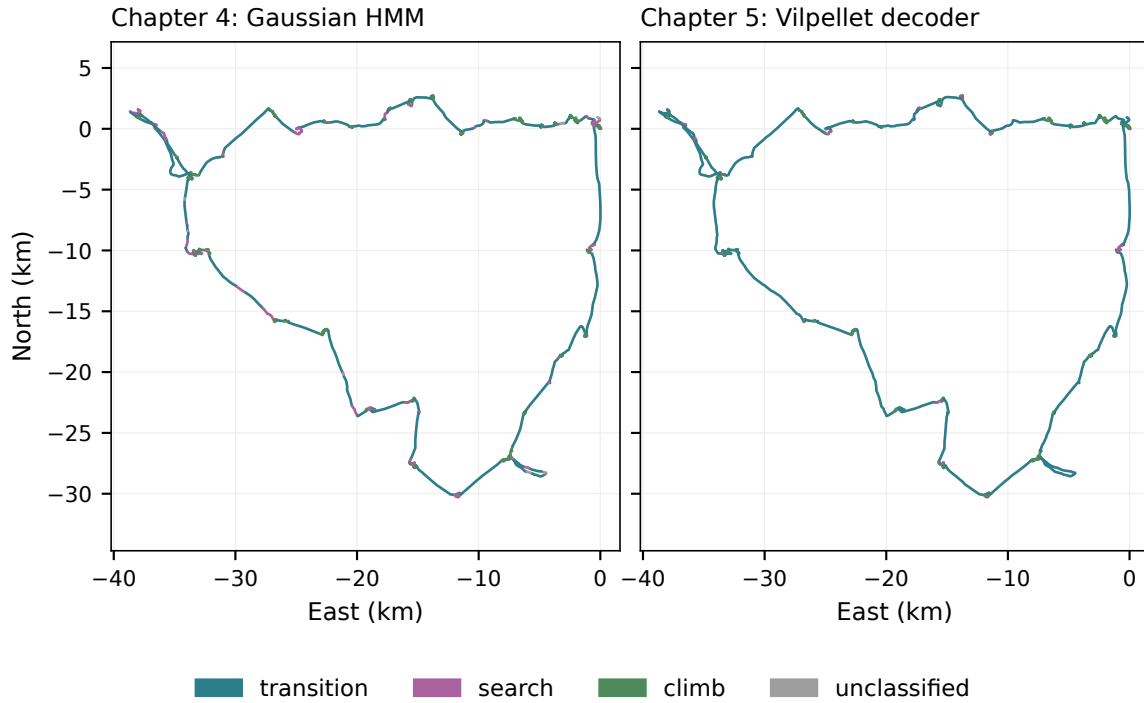


Figure 6.1. The same cleaned trajectory labelled by the current Chapter 4 decoder and the Vilpellet decoder. Coordinates and plotting limits are shared. Colour denotes the inferred phase, not an independently observed thermal or pilot intention.

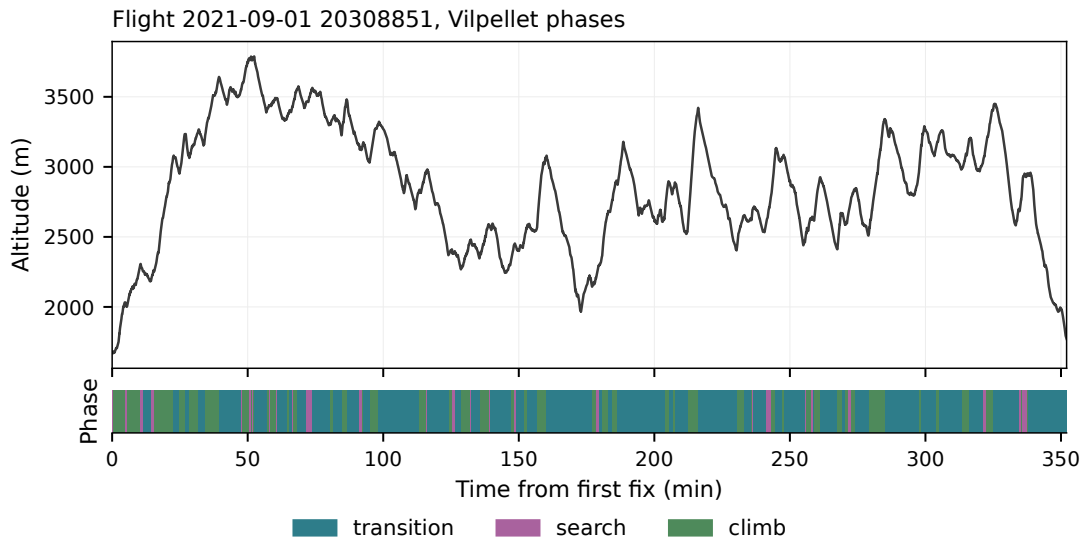


Figure 6.2. Altitude and Vilpellet phase labels for the same flight. The altitude trace supplies physical context; sustained ascent is one input to the model and therefore is not independent evidence of classification accuracy.

Table 6.3. Phase-time shares for the worked example, as percentages of each decoder’s represented track time, including unclassified support. Rounding can prevent an exact sum of 100%. This is one illustrative flight, not an archive-wide estimate.

Phase	Chapter 4 decoder	Vilpellet decoder
Climb	32.6	27.5
Search	17.2	6.4
Transition	48.8	66.1
Unclassified	1.3	0.0

6.5 How to compare the two methods and where to study the code

Several differences can affect a phase-conditioned measurement: the continuous versus binary observation representation; fitted versus transferred parameters; a ten-second decision grid versus per-fix inference; quality masking versus binary boundary conventions; and the additional majority vote. The Vilpellet cadence and flight-size gates also remove support that the Chapter 5 decoder may classify. Comparison therefore needs both the common classified support and each method’s coverage on the original eligible archive.

The reference-equivalence tests compare observation triples, numeric Viterbi paths and majority-smoothed paths on synthetic and real GNSS trajectories. They test the transcription under common inputs and deliberately check numeric components before behavioural renaming. They do not assert that cleaned and raw inputs give the same labels, or that a transferred model is behaviourally accurate. The same limitation applies to the full thermal-viewer cache audit: identical outputs after a code optimization establish computational stability, not a new validation of flight phases.

A choice between the two segmentations should use the independently annotated windows and held-out evaluation protocol of Section 5.5, reporting phase confusion, boundary behaviour and abstention alongside sensitivity of the intended physical observables. It should also examine how smoothing, fixed-length windows and majority voting affect apparent run durations. The geometric sojourn law in Eq. (5.28) belongs to the latent Markov chain; it is not a predicted empirical duration law for the majority-smoothed labels. Neither method is selected as the final segmentation in this chapter.

For a code-first reading, start with `configs/segmentation_vilpellet.yaml` for the exact constants. In `src/soaring/analysis/segmentation/vilpellet/`, read `features.py` for the coordinate operations and binary indicators, `model.py` for the emission lookup, Viterbi recursion and majority vote, then `pipeline.py` for eligibility, separate segment decoding and run construction. `scripts/pipeline/segment_flights_vilpellet.py` defines the archive export. The worked example and all parameter blocks printed here are generated by `scripts/esperimenti/ch06_vilpellet_segmentation/generate_vilpellet_report.py`. The implementation notes in `docs/guide/vilpellet-segmentation.md` provide a longer operational reference. These paths distinguish the implemented protocol from any future change to the segmentation or to its downstream use.

Chapter 7

Comparison with stochastic models

Chapters 3 and 4 established displacement growth, quantile behaviour, directional anisotropy and duration effects without assuming a generating process. This chapter confronts those measurements with explicit stochastic-process predictions: the moment spectrum and a Lévy-walk benchmark, centred kurtosis as a shape diagnostic, velocity memory across averaging resolutions, and a summary of which candidates the measurements challenge or leave open.

7.1 The moment spectrum and the Lévy-walk benchmark

For $q > 0$, define a moment and its growth exponent by

$$M_q(\tau) = \mathbb{E}[R(t, \tau)^q], \quad M_q(\tau) \propto \tau^{\zeta(q)}, \quad \nu(q) = \zeta(q)/q. \quad (7.1)$$

A common scaling law with finite moments gives $\zeta(q) = qH$. Increasing q gives large displacements more weight, so the spectrum tests a different aspect of the distribution from any one quantile. Plain increments are used here: second differences of a straight leg vanish and would change the physical observable being compared.

For an unbiased renewal Lévy walk with independent isotropically chosen headings, constant speed and leg-duration density $\psi(t) \sim t^{-1-\beta}$, the superdiffusive regime $1 < \beta < 2$ predicts

$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & 0 < q < \beta, \\ q + 1 - \beta, & q > \beta, \end{cases} \quad \zeta_{\text{LW}}(2) = 3 - \beta. \quad (7.2)$$

At $q = \beta$ a logarithmic correction accompanies the common power. The bulk and the rare long legs contribute different branches; their meeting produces strong anomalous scaling [24, 25]. This benchmark is an asymptotic model prediction with stated renewal and speed assumptions.

Read per unit order, Eq. (7.2) predicts

$$\nu_{\text{LW}}(q) = \frac{\zeta_{\text{LW}}(q)}{q} = \begin{cases} 1/\beta, & 0 < q < \beta, \\ 1 - \frac{\beta - 1}{q}, & q > \beta, \end{cases} \quad (7.3)$$

a spectrum that is flat below the knee and *rises* above it towards the ballistic value one. The benchmark fixes a level, a knee position and a direction of curvature at once, which is more than the second moment alone can test.

The fixed-population spectrum and its paired bootstrap intervals are awaiting regeneration (Sec. 4.3.2). Its comparison with the benchmark, including any numerical estimate of β and any claimed incompatibility of curvature, remains pending. Figure 7.2 retains the available pooled spectrum, whose contributing population changes with lag; it cannot substitute for that controlled comparison.

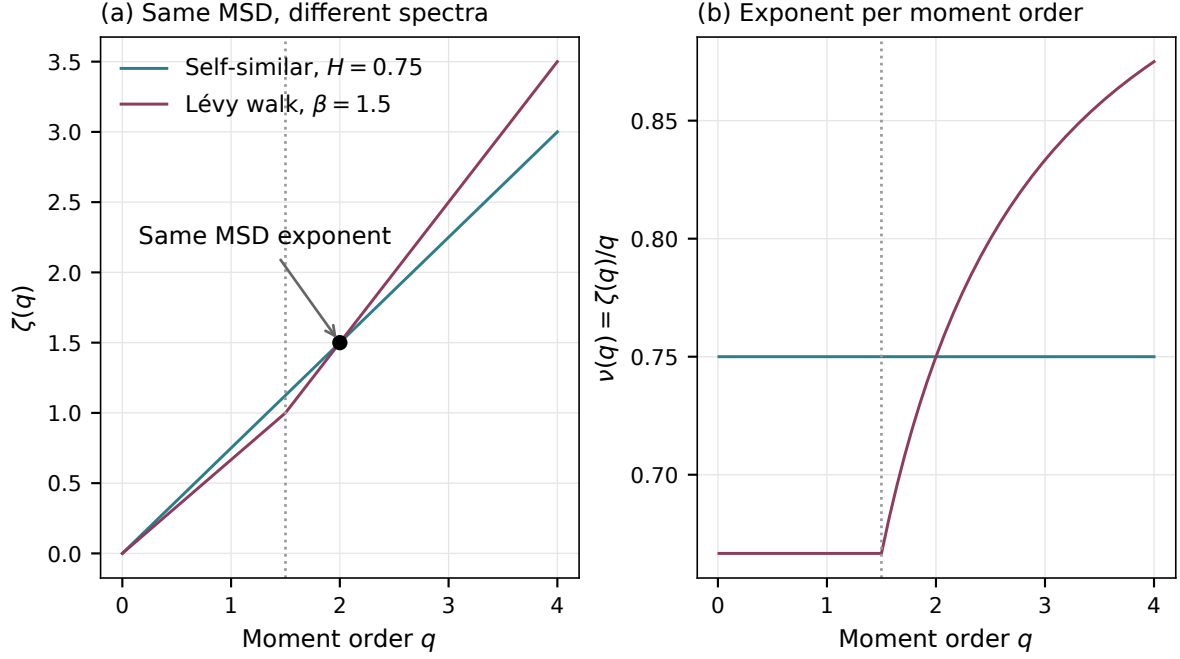


Figure 7.1. Theoretical comparison. A self-similar process with $H = 0.75$ and a standard Lévy walk with $\beta = 1.5$ have the same MSD exponent, $\zeta(2) = 1.5$, but different moment spectra. The dashed vertical line marks $q = \beta$. These are asymptotic predictions, not simulated or empirical estimates.

A spectrum over one window cannot exclude every Lévy walk or every possible crossover. Ballistic Lévy walks, truncated legs, correlated steps and finite-time crossovers need separate predictions. A quantitative test requires simulations with the observed durations, sampling and preprocessing, compared jointly with moments and controlled quantiles. High-order moments also need flight-level sensitivity checks; the 90th percentile alone need not track the ballistic front.

7.2 Centred kurtosis and heterogeneity

At each lag with a finite, positive-definite covariance, centre the increment and standardise by that full covariance:

$$Q = (\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}), \quad K_M(\tau) = \mathbb{E}[Q^2] - d(d+2), \quad d = 2. \quad (7.4)$$

Mardia's kurtosis excess [26] vanishes for a nonsingular Gaussian vector, including an anisotropic one, since $Q \sim \chi_d^2$. Positive excess means a larger fourth moment of standardised radius. It neither requires a power-law tail nor makes zero excess sufficient for Gaussianity.

For the n pooled increments at a lag, the calculation uses

$$\begin{aligned} \hat{\boldsymbol{\mu}} &= \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i, & \widehat{K}_M &= \frac{1}{n} \sum_{i=1}^n \widehat{Q}_i^2 - 8, \\ \widehat{\Sigma} &= \frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i - \hat{\boldsymbol{\mu}})(\mathbf{X}_i - \hat{\boldsymbol{\mu}})^\top. \end{aligned} \quad (7.5)$$

Here $\widehat{Q}_i$ uses these empirical moments. With covariance denominator n , $n^{-1} \sum_i \widehat{Q}_i = 2$, so Jensen's inequality gives $\widehat{K}_M \geq -4$. Negative excess is possible, and even a Gaussian sample need not give exactly zero.

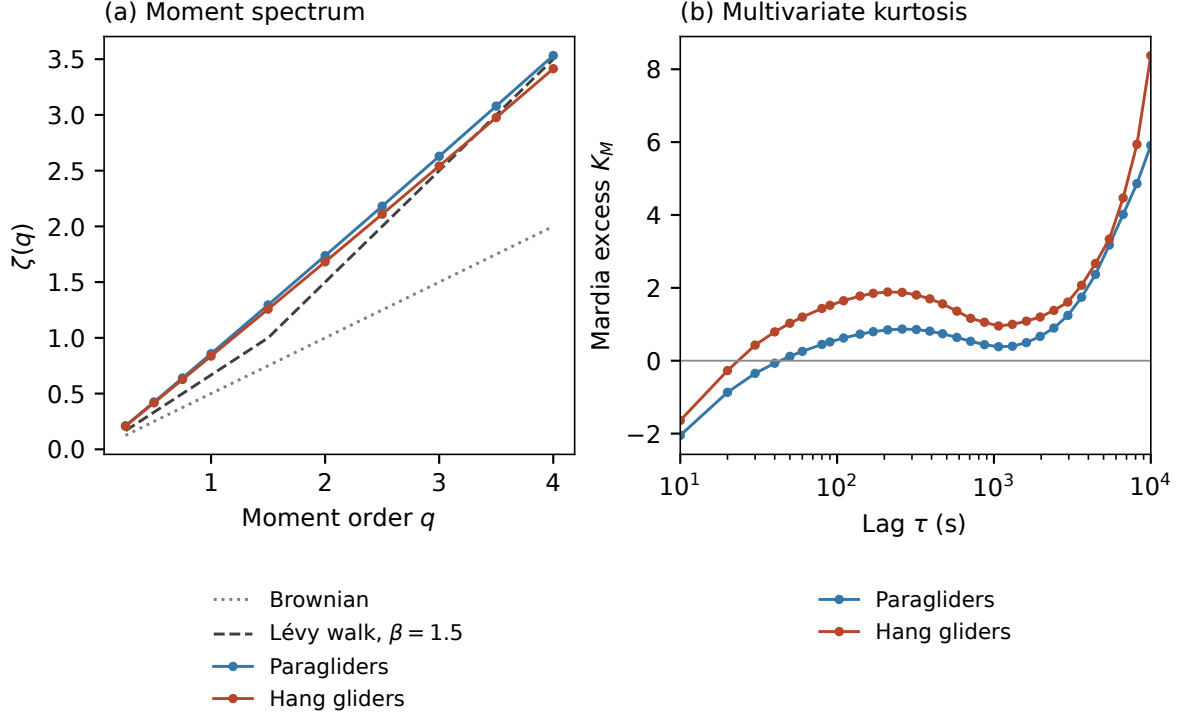


Figure 7.2. Displacement moment spectra and centred multivariate kurtosis in the diagnostic eligible archive, with theoretical moment spectra shown as reference curves. These observables constrain the growth and shape of the increment distribution, respectively, but neither alone identifies the generating process. The contributing flights change with lag here, unlike the fixed cohort of Sec. 4.2.2: short flights leave the pool as τ grows, so part of the change along either curve is a change of population. These spectra are therefore read as archive descriptions; the rank-resolved exponents of Sec. 4.3.1 remain the available controlled measurements; the fixed-population spectrum is pending.

The measured K_M is not constant. Across the displayed interval its range is -2.05 to 5.91 for paragliders and -1.64 to 8.38 for hang gliders. The medians over sampled lags are 0.67 and 1.52, respectively; these lag medians are descriptive summaries, not time averages. The change of sign near the shortest lags and the rise where support is smallest discourage replacing the curve with one claim about a tail.

Excess can also arise from heterogeneity even when the conditional motion is Gaussian. For example, let $\mathbf{X} = \sqrt{A}\mathbf{G}$, where $\mathbf{G}$ is a centred Gaussian vector with fixed covariance, $A > 0$ is independent, and both required moments exist. Then

$$K_M = d(d+2) \left[\frac{\mathbb{E}A^2}{(\mathbb{E}A)^2} - 1 \right]. \quad (7.6)$$

This identity applies to the stated scale mixture; flights with different means and covariance orientations require a fuller calculation. The pooled excess therefore compares the archive with one homogeneous Gaussian law. Conditional Gaussian models need separate centring and covariance estimates within environment or phase groups, with simulations accounting for finite samples and dependence. A peak in pooled kurtosis alone cannot identify a thermal timescale.

7.3 Velocity memory across resolutions

A fix-to-fix velocity mixes recorder cadence, smoothing and physical turning. To examine longer scales, define velocity averaged over a duration h ,

$$\mathbf{v}_h(t) = \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}, \quad C_h(\tau) = \frac{\langle \delta \mathbf{v}_h(t) \cdot \delta \mathbf{v}_h(t+\tau) \rangle}{\langle |\delta \mathbf{v}_h|^2 \rangle}. \quad (7.7)$$

We compare $h = 10, 60$ and 300 s. In each retained segment fs , form n_{fs} adjacent velocity blocks and subtract their segment mean. The segment estimate and archive average are

$$\begin{aligned} \hat{C}_{fs,h}(kh) &= \frac{(n_{fs} - k)^{-1} \sum_{j=0}^{n_{fs}-k-1} \delta \mathbf{v}_{fs,j} \cdot \delta \mathbf{v}_{fs,j+k}}{n_{fs}^{-1} \sum_{j=0}^{n_{fs}-1} |\delta \mathbf{v}_{fs,j}|^2}, \\ \hat{C}_h(kh) &= \frac{1}{|\mathcal{F}_h(k)|} \sum_{f \in \mathcal{F}_h(k)} \frac{1}{|\mathcal{S}_{f,h}(k)|} \sum_{s \in \mathcal{S}_{f,h}(k)} \hat{C}_{fs,h}(kh). \end{aligned} \quad (7.8)$$

Here $\mathcal{S}_{f,h}(k)$ contains the supported segments of flight f :

$$n_{fs} \geq 8, \quad \sum_j |\delta \mathbf{v}_{fs,j}|^2 > 0, \quad 1 \leq k \leq \lfloor n_{fs}/4 \rfloor.$$

Flights with a nonempty set form $\mathcal{F}_h(k)$. Segments receive equal weight within each flight, then flights receive equal weight. This differs from normalising one pooled covariance.

The separation $\tau = kh \geq h$ avoids overlap between the two velocity blocks. Finite-segment centring can nevertheless depress long-lag correlations.

Figure 7.3 places the positive correlations on logarithmic axes, where a power-law interval would satisfy

$$C_h(\tau) \simeq A_h \tau^{-\gamma_h} \implies \log C_h(\tau) \simeq \log A_h - \gamma_h \log \tau. \quad (7.9)$$

The linear-ordinate panels retain zero crossings and negative correlation.

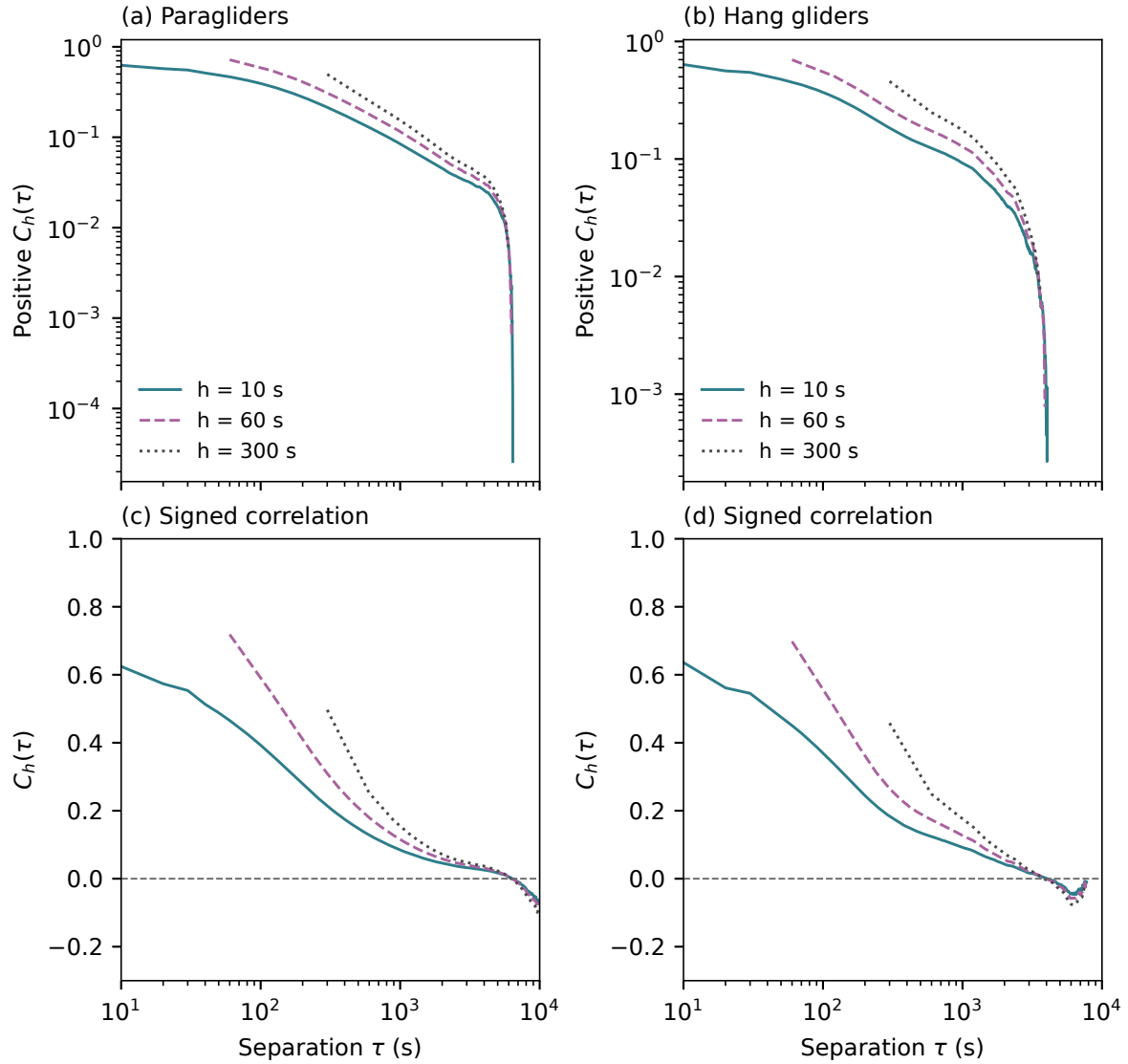


Figure 7.3. Velocity memory at averaging widths $h = 10, 60$ and 300 s. The upper panels use log–log axes for positive correlations; nonpositive values interrupt the curves. The lower panels show the same estimates with a linear vertical axis, preserving zero crossings and anticorrelation. At least 20 contributing flights are required, with separation no shorter than h and no longer than one quarter of each contributing velocity record. The contributing flights change with separation, so a slope read from these curves mixes decaying memory with the loss of short records. A visually straight part of the upper curves is a possible scaling interval; it is not evidence of an asymptotic law.

At a ten-minute separation, all three resolutions retain positive sample means:

Width h [s]	Paraglider $\hat{C}_h(600\text{ s})$	Hang-glider $\hat{C}_h(600\text{ s})$
10	0.128	0.124
60	0.180	0.173
300	0.252	0.249

No calibrated intervals are reported here. Coarse averaging suppresses fast fluctuations and can increase normalized correlation without lengthening the underlying memory. The log–log curves bend, and estimates in both disciplines turn negative at long separation; one decay exponent does not describe the full interval.

Memory and displacement growth. For a stationary centred velocity with variance $\sigma_v^2 = \langle |\delta \mathbf{v}|^2 \rangle$, integrating velocity over the displacement interval and using stationarity gives Taylor's relation [2],

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds. \quad (7.10)$$

The factor σ_v^2 restores the velocity units because C is normalised. Taylor's relation concerns centred displacement in the same stationary population.

For example, an exponentially correlated velocity, $C(s) = e^{-s/t_p}$, gives

$$M_{2,c}(\tau) = 2\sigma_v^2 t_p \left[\tau - t_p(1 - e^{-\tau/t_p}) \right]. \quad (7.11)$$

Its limits are

$$M_{2,c}(\tau) \simeq \begin{cases} \sigma_v^2 \tau^2, & \tau \ll t_p \quad (\text{ballistic}), \\ 2\sigma_v^2 t_p \tau, & \tau \gg t_p \quad (\text{diffusive}). \end{cases}$$

An intermediate superdiffusive slope can therefore arise from finite persistence. Different t_p across flights broaden the crossover. A fitted MSD slope above one does not exclude this model.

An asymptotic tail $C(s) \sim s^{-\gamma}$ with $0 < \gamma < 1$ instead gives $M_{2,c} \propto \tau^{2-\gamma}$. Finite records cannot establish whether the correlation integral diverges. Correlation between successive glides also requires the episode boundaries introduced in Chapter 5.

7.4 Model predictions and remaining tests

Table 7.1 collects the useful exclusions and the hypotheses that remain open. The scope is the pooled data over the observed interval. A candidate can fail there and still describe a restricted phase, environment or limiting time regime.

A glider moves while climbing. Treating climb as a waiting phase in a renewal model therefore needs tests of episode durations, displacement coupling and independence in the segmented analysis.

Table 7.1. Model predictions and the scope of the present comparison. “Challenged” means that a displayed diagnostic disagrees descriptively with the simple prediction; it does not stand for a calibrated statistical rejection.

Candidate	Prediction to confront	Present scope
Homogeneous Brownian motion	$\zeta(q) = q/2$, Gaussian increments, independent disjoint increments	Challenged as a description across the measured interval by displacement growth, shape and memory. A long-time diffusive limit remains untested.
Constant drift plus Brownian motion	$M_2 = \mathbf{v}_d ^2 \tau^2 + 4D\tau$ in 2D; drift-free $V_2 \propto \tau$	A control for apparent superdiffusion. Log-log curvature in V_2 or non-Gaussian residuals require more than a fitted mean velocity; environmental mixtures remain possible.
Homogeneous fractional Brownian motion	Gaussian joint law, common $H_p = H$, $\zeta(q) = qH$	Challenged as a single pooled law when quantile slopes differ and centred kurtosis is nonzero. Conditional Gaussian models are not thereby excluded.
Exponential velocity persistence	Eq. (7.11): ballistic–diffusive crossover	Remains a candidate to test jointly against MSD and coarse velocity correlations; a broad distribution of persistence times is a separate hypothesis.
Standard renewal Lévy walk, $1 < \beta < 2$	Two-branch spectrum in Eq. (7.2)	The controlled spectrum comparison is pending. Finite-record simulations are also needed before claiming exclusion; generalisations require separate predictions.
Unbounded Lévy flight	Instantaneous jumps; divergent second moment for stable index below two	Unsuitable as a literal finite-speed flight mechanism. Truncated or effective coarse models require their own finite-scale predictions.

Bibliography

- [1] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.
- [2] G. I. Taylor. Diffusion by continuous movements. *Proceedings of the London Mathematical Society*, s2-20(1):196–212, 1922.
- [3] Agnan Kessy, Alex Lewin, and Korbinian Strimmer. Optimal whitening and decorrelation. *The American Statistician*, 72(4):309–314, 2018.
- [4] NOAA National Centers for Environmental Information. ETOPO 2022 15 arc-second global relief model, 2022. Elevation subsets accessed through NOAA ERDDAP on 12 September 2026.
- [5] Copernicus Climate Change Service. ERA5 hourly data on pressure levels from 1940 to present, 2023.
- [6] Earthmover. Icechunk ERA5: Registry of open data on AWS, 2026. Pressure-level temporal store; snapshot TGSFKBHQ8D687WS6STMG; source ERA5 values preserved losslessly.
- [7] ECMWF. ERA5: compute pressure and geopotential on model levels, geopotential height and geometric height, 2026.
- [8] Copernicus Climate Change Service. ERA5 hourly data on single levels from 1940 to present, 2023. ECMWF reanalysis; selected wind series accessed through Open-Meteo.
- [9] Open-Meteo. Historical weather api, 2026. ERA5 explicitly selected; nine cells, hourly wind at 10 and 100 m, 2016–2025.
- [10] Deutscher Gleitschirmverband und Drachenflugverband. How are flying devices classified? Official type-inspection guidance. <https://www.dhv.de/en/type-inspection/classification/>. Accessed 12 September 2026. Explanation of type-test flight characteristics and pilot requirements; the paraglider classes on this page use LTF nomenclature.
- [11] DREAL Normandie. Le pays de caux. <https://www.normandie.developpement-durable.gouv.fr/le-pays-de-caux-a1902.html>, February 2018. Published 23 February 2018. Accessed 12 September 2026; regional landscape descriptions including coastal cliffs and valleys.
- [12] Benoit B. Mandelbrot and John W. Van Ness. Fractional brownian motions, fractional noises and applications. *SIAM Review*, 10(4):422–437, 1968.
- [13] Ricardo Fraiman, Leonardo Moreno, and Thomas Ransford. A Cramér–Wold theorem for elliptical distributions, 2022.

- [14] Gustavo Didier and Vlasos Pipiras. Integral representations and properties of operator fractional Brownian motions. *Bernoulli*, 17(1):1–33, 2011.
- [15] Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.
- [16] Jérémie Vilpelle, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. arXiv:2601.01293v1 [cond-mat.stat-mech], 2026. Version 1, 3 January 2026.
- [17] L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.
- [18] Brett T. McClintock, Roland Langrock, Olivier Gimenez, Emmanuelle Cam, David L. Borchers, Richard Glennie, and Toby A. Patterson. Uncovering ecological state dynamics with hidden Markov models. *Ecology Letters*, 23(12):1878–1903, 2020.
- [19] Andrew J. Viterbi. Error bounds for convolutional codes and an asymptotically optimum decoding algorithm. *IEEE Transactions on Information Theory*, 13(2):260–269, 1967.
- [20] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970.
- [21] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–22, 1977.
- [22] hmmlearn developers. hmmlearn documentation and source code. Version 0.3.3. GaussianHMM and ConvergenceMonitor; inspected 10 September 2026. <https://hmmlearn.readthedocs.io/en/0.3.3/>.
- [23] Roland Langrock, Ruth King, Jason Matthiopoulos, Len Thomas, Daniel Fortin, and Juan M. Morales. Flexible and practical modeling of animal telemetry data: Hidden Markov models and extensions. *Ecology*, 93(11):2336–2342, 2012.
- [24] V. Zaburdaev, S. Denisov, and J. Klafter. Lévy walks. *Reviews of Modern Physics*, 87(2):483–530, 2015.
- [25] Adi Rebenstok, Sergey Denisov, Peter Hänggi, and Eli Barkai. Infinite densities for lévy walks. *Physical Review E*, 90:062135, 2014.
- [26] K. V. Mardia. Measures of multivariate skewness and kurtosis with applications. *Biometrika*, 57(3):519–530, 1970.

Appendix 4.A

Squared-displacement distributions

These figures express the test of Section 4.3.1 in squared metres, using the same observations, weights and six lags. Squaring the bin edges preserves probability masses, while the exponent doubles according to Equation (4.30). This representation adds no independent evidence for self-similarity.

The shape of a density also changes under this transformation. If a component magnitude has a finite nonzero density limit $p_Y(0^+) = c$, then $p_{Y^2}(s) \sim c/(2\sqrt{s})$ as $s \downarrow 0$. An approximately straight descending part of a squared-component density on logarithmic axes can therefore come from this Jacobian near zero; it is not by itself evidence of a heavy upper tail. Very small positive bins and exact zero atoms describe numerical observations, without establishing physical displacement resolution at those scales.

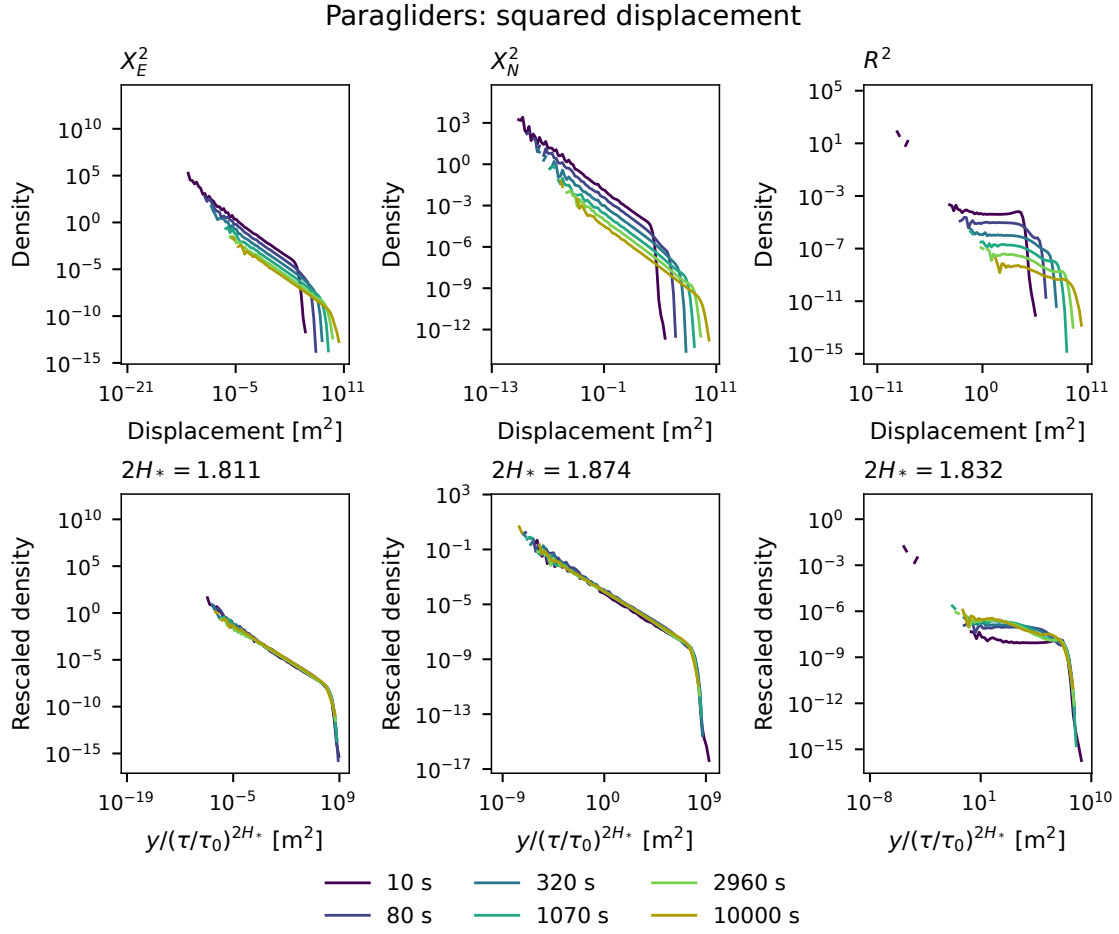


Figure 4.A.1. Paraglider distributions of X_E^2 , X_N^2 and R^2 . Top: raw densities; bottom: rescaling by $(\tau/\tau_0)^{2H_{j,*}}$, with $\tau_0 = 1,000$ s and the doubled exponents from Table 4.4. Zero atoms are recorded separately.

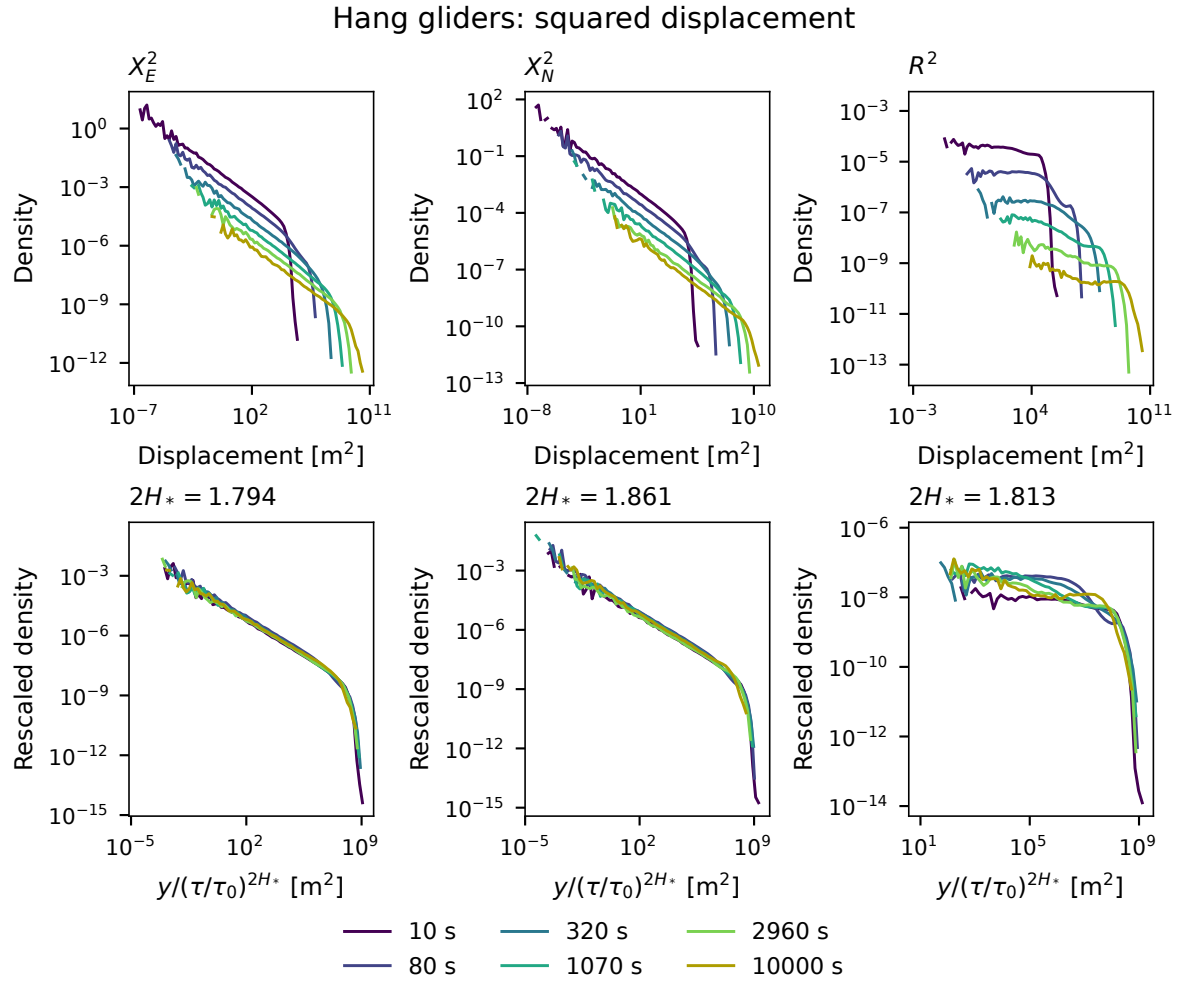


Figure 4.A.2. The same squared-displacement distributions and rescaling for hang gliders. Changes of shape seen for magnitudes remain under the monotone square transformation; the corresponding ECDF separation is exactly unchanged.